Value at Risk

A quantile of tomorrow's profit and loss: how it is computed, what it omits, and the smallest version worth running

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Abstract

Value at risk is a loss level that the portfolio should exceed on a stated small fraction of days, such as one day in 20. It measures an ordinary bad day well. It gives no information about the size of a disaster.

A one-day 95% value at risk of 1,600 dollars means that losses should exceed 1,600 on about one day in 20. It states how often losses cross the threshold. It does not state how large they are. Nothing beyond the threshold changes it. A position that earns a little most days and rarely loses a fortune can report a value at risk of zero.

For an ordinary portfolio, value at risk is roughly current volatility times a fixed multiplier. At 95% the multiplier barely depends on the model, while volatility changes a great deal. A good volatility forecast therefore supplies most of the answer. Every method simulates or replays possible next days for current holdings and reads the loss at the cutoff. On almost a century of US data, filtered historical simulation worked best. It standardises each past return by the volatility at the time, then rescales it by today's volatility. It was the only one of seven methods whose breach rate matched its target at both 95% and 99%. Plain one-year historical simulation reacted late. Losses breached its value at risk three times too often in 2007–2009.

Counting breaches checks value at risk. Over 500 days a 95% model should have about 25 breaches, spread out in time. At 99% the test is weak. A year of data catches a model that breaches twice as often as claimed only about four times in 10. The measure fails in known ways. It ignores losses beyond the threshold, so strategies that sell disaster insurance look safe. Breaches cluster when markets turn. The square-root-of-time rule fails worst in calm markets. From the calmest starting points, losses breached the 10-day 99% figure nearly six times in 100. Once value at risk sets limits and pay, people manage the number.

For an individual, value at risk measures all holdings in one unit of risk. In the worked account, technology was 40% of the capital and 55% of the risk. The chapter recommends a 95% one-day figure built on current volatility, regular breach counts, and stress replays of historical crash days. The worked account had a one-day figure near 1%. It would have lost 18% in the 1987 crash.

What this is. Value at risk is the most widely used number in financial risk management, and the most widely misread. It sets trading limits, determines bank capital, caps the leverage of mutual funds and sizes the margin that a clearing house demands. Critics have blamed it, with some justice, for every market disaster since 1998. This chapter builds it from first principles. It covers what the number is, the dozen ways of computing it, what those ways have in common, and the related measures built on it. It covers how institutions use and display VaR, where it breaks and how to test it. The last third is for a reader who runs their own money. It covers which of this material applies to an individual account and what the smallest worthwhile implementation looks like.

How to read this chapter. §1 and §2 are the conceptual core. They give the definition, the three ideas that the rest of the chapter builds on, and what a VaR number does and does not say. §3 is history and §4 is the bibliography. §5 is the mathematics. §6 and §7 are two surveys, one of ways to compute VaR and one of variations on it. §8 shows that most of §6 and §7 is one calculation with seven choices. §9 and §10 cover practice: who uses VaR for what, and how it is displayed. §11 is the critique and §12 covers testing. §13 and §14 are for the individual investor, and §15 is the synthesis.

Different readers can start in different places.

- Readers who want the idea and nothing else should read §1, §2 and §11.1.
- Readers who run their own account and want something to use this week should read §1, §2, §13 and §14, and treat the rest as reference.
- Readers who already compute VaR and want the parts missing from the textbook chapter should start at §6.8, §11.5 and §12.5.

Appendix A defines the statistical, market and regulatory terms that the main text uses. It presents them in dependency order, for a reader who wants a term explained.

Objectives. After this chapter, you should be able to:

- define VaR as a quantile of a forecast loss distribution, and read a VaR number as a frequency rather than a worst case.
- decompose VaR into a volatility forecast, a tail multiplier and a horizon multiplier, and convert between levels and horizons.
- compute VaR by the main methods, and explain why filtered historical simulation outperformed the alternatives on a century of US data.
- explain expected shortfall, coherence and elicitability, and say when each matters.
- backtest a VaR model for coverage and independence, and judge how much a backtest of a given length can reveal.
- build and maintain a minimal VaR system for a personal account, with a budget, a decomposition, stress replays and written rules.

Relationship to the other chapters. Portfolio Construction and the Covariance Matrix gives the detail behind the covariance matrix that §5.3 and §6.3 take as given. Its section 6.3, on exponentially weighted estimates, matters here. Simple and Log Returns covers the return conventions and the aggregation rules behind §5.4. Trend-Following in Financial Markets treats volatility targeting as a sizing rule, and §13.3 shows that it equals a VaR budget. Implied Volatility gives background for the option examples in §6.10. Each chapter can be read on its own.

Where the numbers come from. Most of the numerical results come from calculations run for this chapter. They are not quotations from published studies, and the text says so where it matters. The calculations use daily US equity returns from the Kenneth R. French data library. The value-weighted market series runs from July 1926 to December 2025. Five of the library's value-weighted industry portfolios represent sector funds in a worked $100,000 account. The script that produces every such number and figure is `figures/var_common.py` in the source repository. Each figure has its own `figures/var_*.py` script. One market and one century do not represent every market. The same exercise on currencies, rates or single stocks would change the numbers. On the reading of the literature taken here, it would not change the conclusions.

A warning about scope. [Practice] Nothing here is investment advice. The worked account is an illustration chosen to make the arithmetic visible, not a recommended portfolio.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but with continuing disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. The evidence may be private or absent. A [Practice] claim has not been debunked.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations or arithmetic. They are true by construction, not by evidence.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in only one section are defined where they appear.

Symbol	Meaning	Defined in
V; ΔV	Value of the portfolio today; its change over the horizon, the **profit and loss** or **P&L**	§1.2
L	**Loss**, $L = -\Delta V$; losses, and so VaR, are positive numbers	§1.2
h	Horizon in trading days	§1.2
α; p	**Confidence level**, 0.95 or 0.99 in nearly all practice; **tail probability** $p = 1 - \alpha$	§1.2
VaR_α; ES_α	Value at risk and expected shortfall at level α	§1.2, §5.6
ρ	Any risk measure, written $\rho(L)$ as a function of the loss or $\rho(x)$ as a function of the positions	§5.5, §5.7
F_L; $q_\alpha(L)$	Distribution function of the loss; its α-quantile	§1.2, §5.1
r_t	Return on day t: a simple return, except in §5.4, which adds up log returns over time	§5.3, §5.4

Symbol	Meaning	Defined in
σ_t	**Conditional volatility**: the standard deviation of r_t as forecast on the evening of day $t-1$	§1.6
σ; $\bar{\sigma}$	Unconditional volatility; the long-run level toward which a mean-reverting model pulls σ_t	§2.2, §5.4
μ; ε	Mean return; **standardised shock**, with mean zero and variance one, so that $r = \mu + \sigma\varepsilon$	§5.2
Φ; φ	Standard normal distribution and density functions	§5.6
z_α	Standard normal quantile $\Phi^{-1}(\alpha)$: $z_{0.95} = 1.645$, $z_{0.99} = 2.326$	§1.5
k_α	**Tail multiplier**: the α-quantile of the standardised loss $-\varepsilon$ under the distribution in use; $k_\alpha = z_\alpha$ is the normal case	§1.6, §5.2
x; N; w	Vector of **dollar positions** in N assets; weights $w = x/V$	§5.3
Σ	$N \times N$ covariance matrix of one-day asset returns	§5.3
σ_p; μ_p	Portfolio one-day volatility **in dollars**, $\sigma_p = \sqrt{x'\Sigma x}$; expected one-day P&L	§1.5, §5.3
λ	Decay of an exponentially weighted moving average (EWMA), $\lambda \in (0,1)$	§6.3
n; T	Number of past observations an estimate uses; number of days in a backtest	§5.8, §12.2
I_t; X	**Hit** indicator, one when the day-t loss exceeded that day's VaR forecast; number of hits, $X = \sum_t I_t$	§1.6, §12.1
$\mathbf{1}\{\cdot\}$; E_t	Indicator, one when the condition holds and zero otherwise; expectation formed with what is known on the evening of day t	§5.4, §5.9
ν; ξ	Degrees of freedom of a Student t distribution; tail index of extreme value theory	§6.4, §6.11
W; B	Account equity, equal to V when nothing is borrowed; a VaR budget as a fraction of it	§13.3

Several conventions carry a qualification:

- Losses are positive numbers, and so is VaR. A negative VaR means that the "bad" outcome is still a gain.
- Hits, exceedances, exceptions, breaches and violations are synonyms. A VaR that rises above a limit or budget is a *limit breach*, a different event.
- A prime marks a transpose. $V = \sum_i x_i$ for an unlevered long-only book.
- In §6.5, γ and κ are the skewness and excess kurtosis. In A.32 and A.39, ϱ is a correlation, distinct from the risk measure ρ.

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1. What value at risk is

1.1 The question it answers

Every risk measure answers a question. Most confusion about value at risk comes from attaching its answer to the wrong question. VaR answers this question:

Over the next h days, with the positions left as they are, what loss will be exceeded only with probability p?

Suppose h = 1 day and p = 5%. A VaR of $1,600 then says that on about one trading day in 20 the loss should exceed $1,600. On the other 19 days the loss should be smaller, or the portfolio should make money. VaR states a *frequency*. It contains only a dollar amount, a time span and a probability.

Banks invented VaR to solve an aggregation problem, and that origin explains its form. In the late 1980s a bank's trading floor held bonds measured in duration, options measured in delta and vega, currencies measured in notional and equities measured in beta. Nobody could add these units together. So nobody could say whether the firm as a whole was taking more risk than it had the week before. A bank can compute a loss quantile for any position in any instrument, and it comes out in money. A bank can also compute it for the sum of the positions. VaR's first use, and still its most defensible one, is as a **common unit**. It measures a Treasury desk and an equity options desk on the same scale.

1.2 The definition

Fix a horizon h and a confidence level α. Let $L = -\Delta V$ be the loss on the current portfolio over the horizon. Treat it as a random variable with distribution function $F_L(\ell) = P(L \le \ell)$. Then

$$\text{VaR}_\alpha(L) \;=\; q_\alpha(L) \;=\; \inf\{\ell : F_L(\ell) \ge \alpha\}.$$

In words, VaR is the smallest loss level ℓ such that the probability of losing no more than ℓ is at least α. When the distribution is continuous, VaR is the number that solves $P(L > \text{VaR}_\alpha) = 1 - \alpha$. The infimum handles distributions with jumps or flat stretches. These cases arise more often than readers might expect. A historical sample has a discrete distribution. So does a bond that either defaults or does not (§5.1, §5.7).

That definition contains three choices. Two VaR numbers can differ at each choice without any visible sign.

The confidence level α. The standard choices are 95% and 99%, with 97.5% and 99.9% in some regulatory uses. The level is a convention. Nobody estimates it from data. It trades relevance against measurability. A higher level reaches further into the tail. The losses that matter most occur there, and almost no data exist there.

The horizon h. Trading desks use one day. The original bank capital rules used 10 days, fund regulation uses 20 days, and credit and insurance use a year. The horizon is also a convention. It should reflect how long exiting the position would take.

The distribution F_L. This is a *forecast*, not a convention. All of the modelling effort goes into it. All of the differences between methods, and nearly all of the failures, arise here. Nobody observes the distribution of tomorrow's P&L. Every VaR number is the output of a model of that distribution.

A fourth assumption is easy to miss. **The portfolio is frozen.** L is the loss on today's positions held unchanged for h days. The same formula applies to a desk that turns its book over three times a day and to a pension fund that never trades. For the desk, the 10-day number describes a portfolio that will not exist.

1.3 The wrong intuition

The most damaging misreading treats VaR as **the most that can be lost**. The phrase "value at risk" invites this reading. So does the usual report, a single dollar figure with the probability in small print.

VaR is closer to a minimum than to a maximum. Take the worst 5% of days. The 95% VaR is the *best* of them: the mildest outcome that still counts as a bad day. A more accurate name would be **the minimum loss on a bad day**. Read "our 95% VaR is $1,600" as "on our bad days we lose *at least* $1,600". VaR does not answer how much more than $1,600 the loss can be.

Three further misreadings follow from the first.

- **"99% confidence means it is almost certain."** If the model is right, losses exceed a 99% daily VaR two or three times a year. Over a 252-day year, the chance of at least one exceedance is $1 - 0.99^{252} = 92\%$. An exceedance does not show that the model has failed. *No* exceedances in three years would show it.

- **"A VaR of $1,600 means risk of $1,600."** The number is meaningless without its level and horizon. Suppose returns were normal and independent from day to day. A portfolio with a one-day 95% VaR of $1,600 would then have a one-day 99% VaR of about $2,300 and a 10-day 99% VaR of about $7,200 (§2.2). All three numbers describe the same risk.
- **"It is a measurement."** No instrument measures VaR. VaR estimates a quantile of a distribution that the analyst must model, from data that contain very few tail observations. Two competent teams given the same portfolio routinely differ by tens of percent (§11.3).

1.4 The right reading

A VaR number is a boundary. Every day's P&L falls on one side of it or the other. Most days fall on the near side, and outcomes there are ordinary. Outcomes on the far side are not ordinary.

VaR is the position of that line. It gives no information about the far side except how often the portfolio will reach it. Used that way, VaR is extremely useful, because a risk model works on the near side. Data about ordinary days are plentiful. Position limits and hedges behave as designed. Markets are liquid enough that "hold the portfolio fixed for a day" is a fair description. [Practice] Aaron Brown ran risk at a large hedge fund and has defended VaR in print more capably than most. In *Red-Blooded Risk* (2011) he argues something close to this. VaR separates the days managed with statistics from the days managed with scenarios, contingency plans and capital. On this view the main output of a VaR system is a daily check. The check shows which kind of day it was, and whether the count of bad days matches the stated probability.

A VaR number therefore does two jobs. First, it gives the **scale of an ordinary bad day**. Sizing positions requires this scale, and it shows whether today's loss was routine. Second, it makes a **claim that anyone can check**. A stated 5% implies about 25 exceedances over 500 days, spread out rather than bunched. No other common risk number is as cheap to falsify, and §12 shows how to do it.

1.5 The master form

Every method of computing VaR follows the same four-stage calculation. The method can be a one-line formula or an overnight simulation on a compute grid.



1. **Exposure.** Express the portfolio as a function of a set of **risk factors**: the things whose movements change its value. For a stock portfolio the factors can be the stock prices themselves. For a bond book they are points on yield curves. For options they also include the underlying price and implied volatility.
2. **Distribution.** Forecast the joint distribution of the factors' changes over the horizon. The diagram draws this stage in rust, because the methods differ here. One method assumes a normal distribution and estimates its covariance matrix. Another replays the last n days of history. A third simulates from a fitted model.
3. **Revaluation.** Compute the portfolio's P&L under each possible factor move. The computation is exact if it reprices every instrument. It is approximate if it uses the portfolio's sensitivities.
4. **Quantile.** Read off the loss exceeded with probability p.

When the portfolio is linear in its factors and the factors are jointly normal, all four stages collapse into one formula. The one-day P&L is then normal with standard deviation σ_p. If successive days are independent and alike, the h-day P&L has standard deviation $\sigma_p\sqrt{h}$ (§5.4). Then

$$\text{VaR}_\alpha \;=\; z_\alpha\,\sigma_p\,\sqrt{h} \;-\; \mu_p h \;\approx\; z_\alpha\,\sigma_p\,\sqrt{h},$$

where μ_p is the expected daily P&L. At short horizons this term is small enough to drop (§5.2). For a single position of value V in an asset with daily volatility σ, $\sigma_p = \sigma V$, and the one-day VaR is $z_\alpha\,\sigma\,V$. VaR is **a multiple of volatility, in dollars.**

1.6 The three central ideas

The rest of this chapter follows from three ideas. They recur throughout the chapter.

Idea 1. VaR is a quantile of a forecast distribution. It is not a property of the portfolio, and it is not a measurement. Every method is the pipeline of §1.5 with different choices at each stage. Every difference between two VaR numbers for the same book comes from a difference in the forecast distribution, almost always in stage 2.

Idea 2. At short horizons VaR is a volatility forecast multiplied by a constant. For anything close to a linear portfolio, $\text{VaR}_\alpha = k_\alpha \, \sigma_t \, V$. The multiplier k_α encodes the shape of the tail. Across every reasonable distribution it varies little: at 95%, between about 1.5 and 1.7. The volatility σ_t varies far more. An EWMA estimate of US equity market volatility, computed for this chapter, was 0.43% a day in February 2007 and 4.80% in October 2008, a factor of 11. An analyst who forecasts volatility well therefore computes VaR well. At 95%, the arguments about distributional shape are second-order by comparison. At 99% and beyond they are no longer second-order, because k_α is the uncertain part (§2.2, §5.8).

Idea 3. A quantile is a threshold, not a tail, and the hit sequence audits it. VaR says how often the loss will cross the line. It gives no information about how far. Expected shortfall, stress tests and the whole of §11 exist to cover what lies beyond the line, which VaR ignores. In exchange, VaR makes the one claim in risk management that anyone can test directly. Define the hit sequence $I_1, I_2, \ldots$ as one on each day the loss exceeded the forecast, and zero otherwise. A correct forecast makes it resemble independent coin flips that come up heads with probability p. The frequency should be right, and the hits should not cluster. Every evaluation method in §12 tests one or both of those properties.

1.7 What VaR is not

The table separates VaR from the measures that people often confuse with it.

It is often confused with	The difference
Volatility	Volatility is a scale. VaR is a scale times a tail multiplier, in money, with a probability attached. For a normal distribution they carry the same information. For an option book or a credit portfolio they do not
Maximum loss	VaR is the *least* lost in the worst p of outcomes. The maximum loss on a long position is its whole value. On a levered long position this is more than the equity behind it. On a short position the maximum loss is unbounded
Expected shortfall	ES is the *average* loss in the worst p of outcomes. It answers "how bad, given bad". VaR does not answer that question (§5.6)
Stress loss	A stress test asks what a named scenario would do, with no probability attached. VaR attaches a probability and names no scenario
Maximum drawdown	Drawdown is a path property over months or years: peak to trough. VaR is a one-step-ahead statement. A year's worst drawdown has typically been about nine daily 95% VaRs (§13.5)
Margin	Margin is what a broker or clearing house demands as collateral. Lenders usually *compute* it from a VaR-like calculation, with add-ons, at the level the lender chooses (§9.4)
Capital	Capital is the buffer held against loss. Regulators set it as a multiple of VaR. The multiple, three or more, shows how far regulators trust the model (§9.1)

1.8 A worked instance

Put $100,000 in the US equity market and look back over the 500 trading days to the end of 2025. The daily returns over that window had a standard deviation of 1.042%, so the P&L had a standard deviation of $1,042.

The formula. Under the normal assumption, the one-day 95% VaR is $1.645 \times 1{,}042$, which is $1,714. The 99% VaR is $2.326 \times 1{,}042$, which is $2,424.

The history. Sort the 500 daily P&Ls from worst to best. The 95% VaR is the loss at the 5% point, with 25 days worse than it. That loss was $1,592. The 99% VaR is the loss with five days worse: $2,881. The average of the worst 2.5% of days, 12 or 13 of them, was $3,051. The single worst day lost $5,900.

The table compares the two methods.

	Normal formula	Historical sample
95% one-day VaR	1,714	1,592
99% one-day VaR	2,424	2,881
97.5% expected shortfall	2,436	3,051
Worst day in the window	—	5,900

Three lessons are already visible. Later sections develop each one.

1. **The two methods disagree, and in opposite directions at the two levels.** The historical 95% VaR is 7% *below* the normal figure, and the historical 99% VaR is 19% *above* it. This pattern is typical of fat tails. Compared with a normal curve, the sample has more days near zero and more days far out, and fewer days in between. At 95% the normal assumption overstates the loss. At 99% it understates the loss (§2.2).
2. **The worst day was more than twice the 99% VaR**, and 3.7 times the 95% VaR. Neither VaR figure gave any hint of it. This illustrates Idea 3.
3. **Neither figure describes today's risk.** Both describe an average over two years. If the last month has been calm, the right number for tomorrow is lower. If it has been violent, the right number is higher. §2.4 and §6 cover how to make the estimate conditional.

§1 Key takeaways

1. VaR is the loss that a frozen portfolio will exceed with a stated probability over a stated horizon. It states how often, never how much.
2. Read it as the *minimum* loss on a bad day. Reading it as "the most that can be lost" reverses its meaning.
3. The confidence level and the horizon are conventions. The distribution is a forecast. Every method differs here, and every failure originates here.
4. Every method is one pipeline: map positions to risk factors, forecast the factors, revalue, take a quantile. For a linear portfolio with normal factors, the pipeline reduces to $z_\alpha \, \sigma_p \sqrt{h}$.
5. At short horizons VaR is a volatility forecast times a tail multiplier. At 95% the multiplier stays in a narrow range, while volatility changed by a factor of 11 between 2007 and 2008. A good volatility forecast does most of the work.
6. Data can falsify VaR, and this is its distinctive virtue. Exceedances should occur at the stated frequency, and they should not cluster.
7. On two recent years of US equity data, the historical 95% VaR was 7% below the normal formula's, and the historical 99% VaR was 19% above it. The worst day was double the historical 99% figure.

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2. Reading a VaR number

§1 defined the object. This section teaches a skill that most users of VaR lack. The skill has three parts: reading a number that someone else supplies, translating it into concrete events, and knowing which questions the number cannot answer.

2.1 Translate the probability into a calendar

Readers misjudge a confidence level more easily than a frequency. The table converts the common conventions into "how often". It assumes 252 trading days a year and a model that is exactly right.

Level and horizon	Exceeded on average	Chance of at least one exceedance
95%, one day	12.6 days a year: one trading day in twenty, about once a month	66% in any 21-day month
99%, one day	2.5 days a year: one day in a hundred, about every five months	92% in any year
99.9%, one day	one day in a thousand, about every four years	22% in any year

Level and horizon	Exceeded on average	Chance of at least one exceedance
99%, ten days	one ten-day period in a hundred: about every four years	—
99%, twenty days	one month in a hundred: about every eight years	—
99.5%, one year	one year in two hundred	—

The last three rows are the regulatory conventions of §9: the original Basel market-risk rule, the European and US fund rules, and European insurance solvency. These levels look extremely conservative. They would be conservative if the models behind them were right. §11 and §12 explain why the models are not right.

Remember the first row for personal use. **A 95% one-day VaR is a monthly event.** Suppose an account's 95% VaR is $1,500. If the account has not lost more than that in three months, the model is probably too pessimistic. If it has lost more than that four times in a month, the model is probably too optimistic, or the world has changed. For practical purposes the two causes are equivalent.

2.2 VaR is a multiple of volatility

Consider a position whose returns follow a location-scale distribution. Examples are the normal, the Student t and any other distribution fixed by a shift and a scale. Then

$$\text{VaR}_\alpha \;=\; \left(k_\alpha\,\sigma\sqrt{h} - \mu h\right) V.$$

Here σ and μ are the daily volatility and mean, and k_α is the α-quantile of the standardised loss (§5.2). The factor $\sqrt{h}$ is the square-root-of-time rule of §5.4. The rule is exact for independent normal days and approximate otherwise. At daily or weekly horizons the drift term is negligible. An asset earning 8% a year has μh = 0.03% a day, against a $k_{0.95}\,\sigma$ of about 1.6% for equities. A VaR is therefore a volatility multiplied by two numbers, one for the tail and one for the horizon. Every VaR conversion applies those multipliers, as the table shows.

Conversion	Multiply by	Where it comes from
Daily volatility to one-day 95% VaR	1.645	$z_{0.95}$
Daily volatility to one-day 99% VaR	2.326	$z_{0.99}$
95% VaR to 99% VaR	1.414	$z_{0.99}/z_{0.95}$, normal only
One-day to ten-day	3.162	$\sqrt{10}$, only if returns are IID: independent and identically distributed (§5.4)
One-day to twenty-day	4.472	$\sqrt{20}$, same caveat
Annual volatility to daily	0.063	$1/\sqrt{252}$
One-day 95% VaR to the Basel ten-day 99%	4.47	1.414×3.162

An all-equity account with annual volatility of 16% therefore has a daily volatility of about 1%. Its one-day 95% VaR is about 1.65% of its value, and its one-day 99% VaR about 2.3%. A conventional 60/40 mix of stocks and bonds, at about 10% annual volatility, has a one-day 95% VaR of about 1%. Use these numbers to sanity-check any system.

How much the tail multiplier can move. The previous table assumed a normal distribution. Fatter tails change k_α much less than most readers expect at 95%, and much more at 99%. The next table shows a Student t distribution rescaled to unit variance, with the US market's own figures alongside.

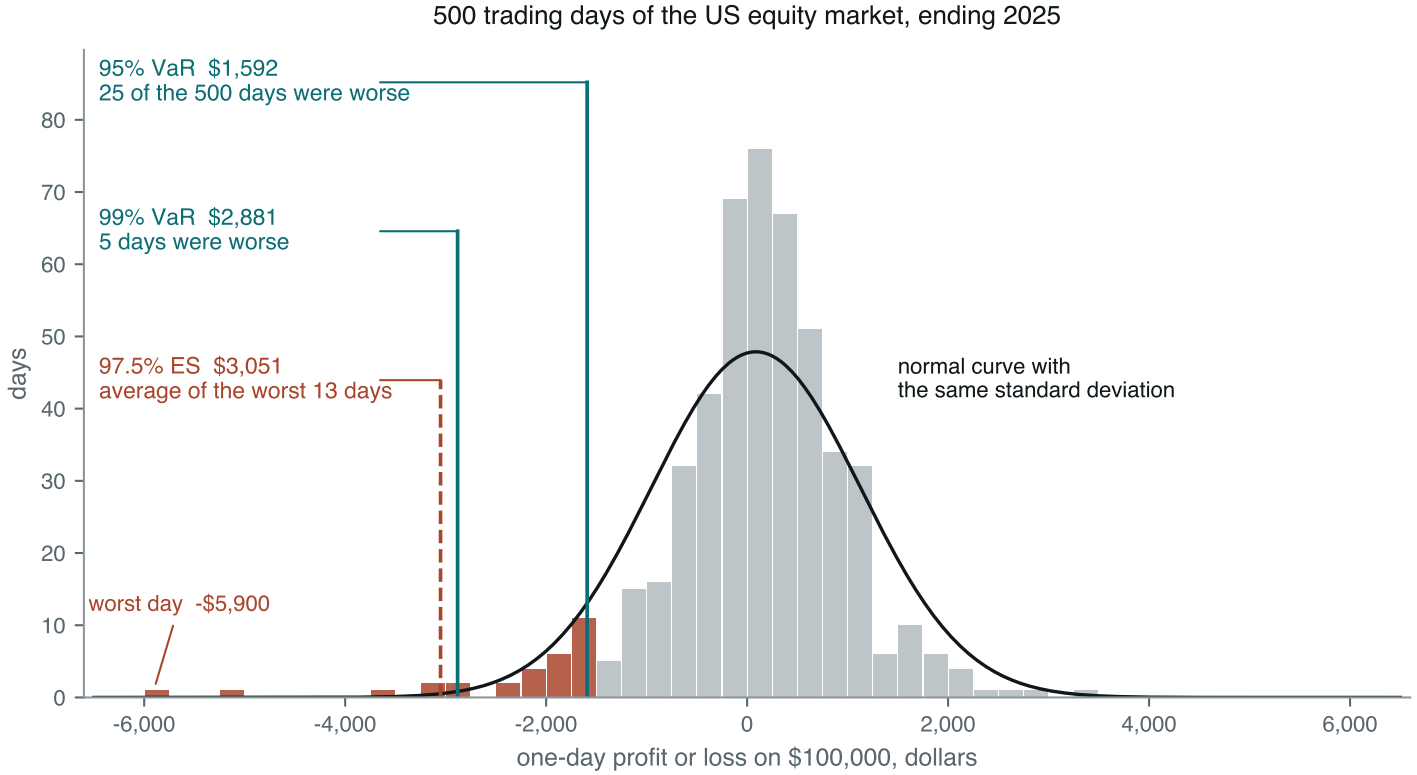
Distribution of the standardised return	$k_{0.95}$	$k_{0.99}$
Normal	1.645	2.326
Student t, 10 degrees of freedom	1.621	2.472
Student t, 5 degrees of freedom	1.561	2.606
Student t, 4 degrees of freedom	1.507	2.649
Student t, 3 degrees of freedom	1.359	2.622

Distribution of the standardised return	$k_{0.95}$	$k_{0.99}$
US market 1929–2025, standardised by that morning's EWMA volatility	1.71	2.90
US market 1926–2025, standardised by full-sample volatility	1.46	2.85

Two facts matter here. First, at 95% every plausible choice lies within about 15% of the normal value, or 17% for the extreme t with three degrees of freedom. Fattening a symmetric tail, as the t rows do, *lowers* the multiplier. A fat-tailed distribution with the same variance moves probability out of the intermediate range, where the 95% point sits, into the centre and the far tail. Real returns need not follow the t here, and the US market's conditional figure is above the normal one (§6.4). Second, at 99% the plausible range is 2.3 to 2.9. This is a 25% spread, and all of it lies on the high side of the normal. Meanwhile the volatility that multiplies these numbers changed elevenfold in 20 months. These figures illustrate Idea 2. At 95% the forecast of σ_t dominates. At 99% the shape of the tail competes with it.

2.3 What the number does not tell you

The figure plots the worked instance of §1.8. Every introduction to VaR contains a figure like this one. Study what it leaves out.



VaR marks where the tail begins. It is silent about how far the tail goes. Daily data from the Kenneth R. French data library.

The 95% VaR is $1,592, inside the body of the histogram. VaR gives no information about the shaded region to its left. That region contains the 99% VaR at $2,881, the average of the worst 2.5% at $3,051, and a worst day of $5,900. Moving that worst day to $50,000 would not change the 95% VaR at all. The black curve is the normal distribution that the parametric method would have used in place of the histogram. It is lower than the histogram in the centre and in the far tail, and higher in between.

VaR can ignore the tail entirely. The table compares two positions.

	Position A	Position B
What it is	A diversified stock portfolio, P&L normal with standard deviation 1,000	A short position in deep out-of-the-money options: gains 200 on 97% of days, loses 20,000 on 3%
95% one-day VaR	1,645	**−200** (a gain)
99% one-day VaR	2,326	20,000
95% expected shortfall	2,063	11,920

Position B has a *negative* 95% VaR. It makes money on 97% of days, which is more than 95%, so the 95% quantile of its loss is a gain of 200. A risk system with a 95% VaR limit would allow any amount of this position. The example is simplified, but real positions of this kind are common. A put option pays its holder the amount by which a stock falls below an agreed strike price. The seller of the put is *short* the option. The seller collects a premium up front and owes that payout (A.35). Consider a sale of 10 put contracts with 30 days to expiry, struck 10% below a $100 stock with 20% implied volatility. The sale collects $71 of premium. The linear (delta-normal) 99% one-day VaR is $91. A day on which the stock falls 10% and implied volatility doubles costs $3,975, which is 44 times the VaR (§6.10).

A better VaR model cannot fix this, because any quantile ignores the tail beyond it. §5.6 and §7.1 introduce the measure that looks past the line. §11.1 discusses how people exploit this gap.

2.4 Today's VaR and the typical VaR

A VaR can answer two different questions. Most disagreements between VaR numbers arise because the two numbers answer different questions. The analysts do not necessarily disagree about the answer to either one.

- **Conditional VaR**, in the sense used here, is the quantile of tomorrow's loss given everything known tonight, including whether markets have recently been calm or violent. Its volatility is a *current* estimate, σ_t.
- **Unconditional VaR** is the quantile of a typical day's loss over a long run of days like the ones in the sample. Its volatility is an average.

Some authors also use the phrase "conditional VaR" as a name for expected shortfall. §8.4 separates the two uses. In this chapter the phrase never means expected shortfall.

The table gives the one-day 95% VaR at the end of 2025 for the worked $100,000 account of §10.4, which holds five sector funds.

Method	Question it answers	95% one-day VaR
Normal, volatility from an EWMA with decay 0.94	Conditional	1,092
Filtered historical simulation (§6.8)	Conditional	1,116
Historical simulation over 500 days	Unconditional, over two years	1,521

The historical figure is 39% higher, and it is not an error. Late 2025 was calm relative to a two-year window that included the tariff shock of April 2025. A two-year average therefore overstates tomorrow's risk. Neither kind of number is *the* VaR. Use the conditional number to size tomorrow's positions. A limit or capital buffer should not change with every shift in market conditions, so a slower number is defensible for that purpose, and regulators deliberately choose one (§9.1). Most practitioners get into trouble by using one kind while believing they are using the other.

2.5 Fat tails, recalibrated

A century of US equity returns appears to reject the normal distribution. From July 1926 to December 2025, over 26,151 trading days, daily returns had a standard deviation of 1.078% and a kurtosis of 19, against 3 for a normal distribution. Kurtosis is the average fourth power of returns measured in standard deviations. It rises with the weight in the tails. The *excess* kurtosis used in §6.5 is the same number minus 3 (A.7).

Loss larger than	Days observed	Days a normal distribution expects
3 standard deviations	232	35
4 standard deviations	105	0.8

Loss larger than	Days observed	Days a normal distribution expects
5 standard deviations	51	0.008
6 standard deviations	29	0.00003

The worst day, 19 October 1987, was a fall of 17.4%: 16.2 standard deviations.

Most of this apparent fat-tailedness comes from time-varying volatility. Volatility clusters: calm periods and violent periods alternate. A sample that mixes them looks fat-tailed even if each day's return is normal *given that day's volatility*. A mixture of normal distributions with different variances always has a kurtosis above 3. Now measure each day's return in units of the volatility that a forecaster could have known that morning, an EWMA estimate (§6.3). The table shows the result.

Day	Return	In full-sample standard deviations	In that morning's volatility
19 October 1987	−17.4%	16.2	10.4
16 March 2020	−12.0%	11.2	2.8
29 October 1929	−11.5%	10.7	3.3
1 December 2008	−9.0%	8.4	2.0

The kurtosis of the standardised series falls from 19 to 8.5. The days in March 2020 and December 2008 look like impossible 11-sigma and 8-sigma events. Given the volatility that everyone could already see, they were three-sigma and two-sigma days. October 1987 remains a genuine outlier on any measure.

[Fact] **Most of the apparent fat-tailedness of daily returns is time-varying volatility. The remainder is real, and it is concentrated in the far tail, from about the 99th percentile out.** The standardised series still has a 99% quantile of 2.90 against the normal's 2.33, and a 99.9% quantile of 5.33 against 3.09. Studies have documented this ordering of causes on many markets since at least the GARCH literature of the 1980s. It is the most practically important fact about VaR methodology. It dictates the ranking of methods in §6.13: model the volatility first, then the shape of the tail.

A remark attributed to the chief financial officer of Goldman Sachs in August 2007 is the classic illustration. He said the firm had seen "25-standard deviation moves, several days in a row". Under a normal distribution, a 25-sigma day should not happen once in the life of the universe. A plainer explanation is that analysts measured the sigma in a calm period and measured the moves in a violent one. A 25-sigma move in yesterday's volatility can be a four-sigma move in today's.

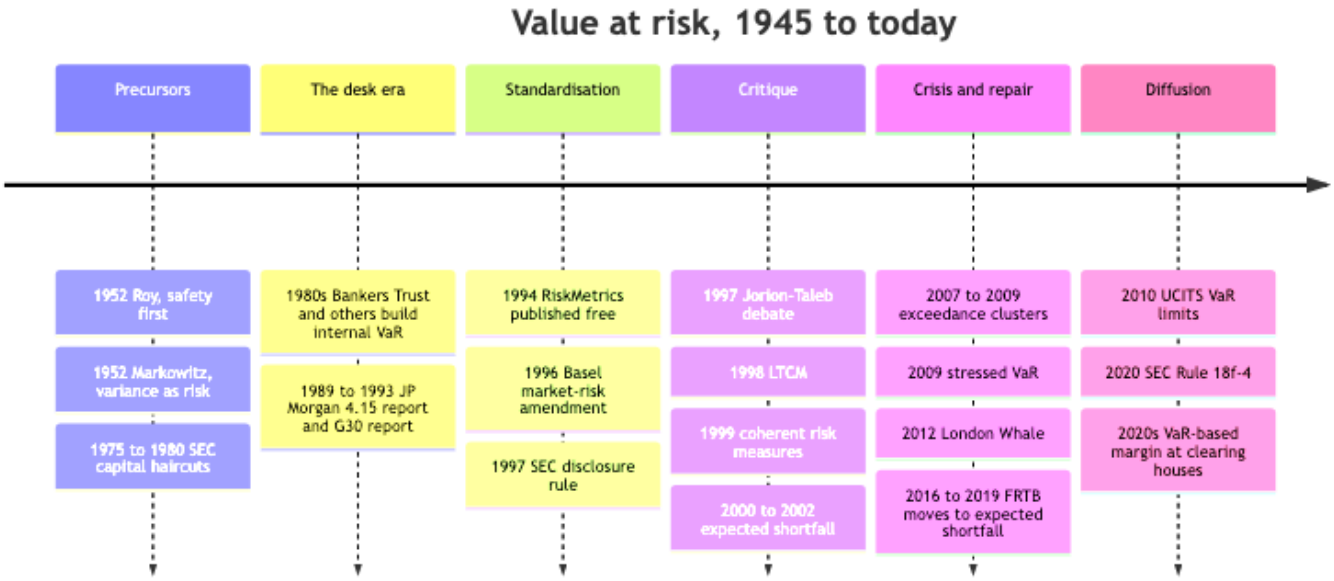
§2 Key takeaways

1. Losses exceed a 95% one-day VaR about once a month, and a 99% one-day VaR two or three times a year. Losses exceed even a correct 99% model's VaR in 92% of years.
2. VaR is volatility times a tail multiplier times a horizon multiplier. An all-equity account has a one-day 95% VaR of about 1.65% of its value, and a 60/40 mix about 1%.
3. At 95% the tail multiplier barely depends on the distribution. Plausible choices lie within 15% of 1.645, and fatter symmetric tails *lower* it. At 99% the plausible range is 2.3 to 2.9, all above the normal value.
4. VaR ignores everything past its own threshold. A short position in deep out-of-the-money options can have a negative 95% VaR and a catastrophic tail.
5. Conditional VaR, given today's volatility, and unconditional VaR, an average over a window, answer different questions. On the worked account they differed by 39%. Check which one a report shows.
6. Standardising by a volatility forecast cuts the kurtosis of US daily returns from 19 to 8.5, and it turns the "11-sigma" day of March 2020 into a three-sigma day. Model volatility first. The shape of the tail is the second-order correction.

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3. How the field evolved

The history of VaR is short, and Holton (2002) documents it unusually well. His working paper is the source for most of the detail below from before 1990. The history has six eras, each with a one-line thesis.



3.1 Precursors (1945–1980): risk as a probability of shortfall

Thesis: the idea of a loss quantile is older than the computers needed to apply it to a trading book.

Holton (2002) identifies a 1945 article by Leavens, on the benefits of diversification, as the earliest published example he found of a VaR-like calculation. The era therefore starts in 1945.

Safety first. Roy (1952), in the same year as Markowitz's portfolio paper, proposed that an investor should minimise the probability of the portfolio falling below a disaster level. *Contribution:* risk as a tail probability rather than a variance. *What changed:* little at the time. Academics adopted mean-variance because it was tractable. *Limitations:* Roy had no way to estimate tail probabilities beyond Chebyshev bounds. *Lasting influence:* VaR inverts Roy's criterion. It fixes the probability and solves for the level.

Markowitz (1952) made variance the measure of portfolio risk. More importantly for VaR, he showed how to aggregate it through a covariance matrix. Every normal parametric VaR equals $z_\alpha \sqrt{x' \Sigma x}$. This is Markowitz's portfolio variance with a quantile multiplier.

Regulatory haircuts. [Fact] According to Holton (2002), the net capital rules of the US Securities and Exchange Commission for broker-dealers moved in 1980 to haircuts on securities positions. The haircuts rested on statistical analysis of historical price moves. They aimed to cover something like a 95th-percentile loss over a one-month liquidation period. *Lasting influence:* the template of a quantile, a horizon tied to liquidation and a capital charge. The 1996 Basel design used exactly this template.

3.2 The desk era (1980–1993): a common unit for a trading floor

Thesis: banks invented VaR for their own use, to answer "how much could we lose tomorrow?" across instruments whose risk measures were incomparable.

Through the 1980s several dealers built internal firm-wide measures of this kind. Bankers Trust is the best-documented case, associated with its risk-adjusted return on capital (RAROC) system. The best-known case is J.P. Morgan's. Its chairman, Dennis Weatherstone, asked for a single-page report delivered at 4:15 each afternoon that summarised the firm's market risk over the next day. The "4:15 report" became the template for the daily VaR report. The methodology behind it was a variance-covariance model over a few hundred risk factors.

The name entered wide use through the Group of Thirty's 1993 report on derivatives practice. The report recommended that dealers measure market risk with a value-at-risk approach and use consistent measures to set limits.

What changed: firms could state firm-wide risk in one number, compare it across desks and over time, and limit it. *Limitations:* the models were parametric and mostly normal. They handled option books through linear sensitivities, and §6.10 shows that these can severely understate the risk of short option positions. *Lasting influence:* the daily VaR report is still the centrepiece of every bank's market-risk function.

3.3 Standardisation (1994–1997): from internal tool to industry and regulatory standard

Thesis: two decisions in three years, one commercial and one regulatory, made VaR universal.

RiskMetrics (1994). In October 1994 J.P. Morgan published its methodology free of charge, together with the volatility and correlation data needed to run it. The fourth edition of the *RiskMetrics Technical Document* (J.P. Morgan/Reuters, 1996) is still the clearest statement of the approach. It maps positions onto a grid of risk factors and models each factor's daily return as conditionally normal. It estimates volatilities and correlations by exponentially weighted moving averages, with decay 0.94 for daily data. *What changed:* any firm could compute a VaR the next morning, and the EWMA estimator became the industry's default volatility model. *Limitations:* conditional normality understates the 99% tail. The estimator with $\lambda = 0.94$ also effectively uses only about 30 observations, which is too few for a large covariance matrix. *Lasting influence:* enormous. The 0.94 decay remains a default everywhere. §6.8 and §6.13 show that, combined with a non-normal tail, it is still close to the best simple method available.

The Basel Market Risk Amendment (1996). The Basel Committee on Banking Supervision (1996a) allowed banks to compute market-risk capital from their own VaR models, subject to standards. The standards required 99% confidence and at least one year of data. The horizon was 10 days, and banks could scale up from one day by $\sqrt{10}$. The capital charge was at least **three times** the average VaR over the previous 60 days. A companion document (Basel Committee, 1996b) set out backtesting against the daily P&L. It also set out the "traffic light", which raises the multiplier when a bank has too many exceedances (§12.4). *What changed:* VaR became the basis of bank capital. Every large bank therefore had to compute, validate and defend it. *Limitations:* nobody derived the multiplier of three. Most observers read it as a safety factor for model error, fat tails and the crude square-root-of-time scaling. *Lasting influence:* every later reform kept the structure of a quantile model, a backtest and a penalty multiplier.

Disclosure. In 1997 the US Securities and Exchange Commission required public companies with material market risk to disclose it, with VaR as one of three permitted formats. [Practice] That rule, and the bank disclosure requirements that followed it, explain why every large US bank's annual report carries a VaR table (§10.3).

3.4 Critique (1997–2002): what a quantile cannot measure

Thesis: within five years of VaR's adoption, the theory of good risk measures showed that VaR is not one and produced a replacement.

The Jorion–Taleb debate (1997). In a pair of articles in *Derivatives Strategy*, Nassim Taleb argued that users should abandon VaR, because it gives false confidence and ignores the events that matter. Philippe Jorion replied that VaR is an imperfect but disciplined improvement on having no firm-wide number at all, and that users should supplement it rather than discard it. The two agreed on more than either side's admirers usually remember. The debate is worth reading because it anticipates nearly every later criticism. §11.9 returns to it.

LTCM (1998). Long-Term Capital Management was a hedge fund run by some of the most sophisticated quantitative traders and economists of the era. It lost most of its capital in a few weeks of August and September 1998. Jorion (2000) reconstructs its risk. The fund's VaR, estimated from a short calm history, was a small fraction of the losses it then suffered. Its positions were highly correlated in a crisis that its data did not include. Its leverage also made liquidation impossible without moving prices. *Lasting influence:* LTCM is the standard case study for model risk, correlation breakdown and liquidity.

Coherent risk measures (1999). Artzner, Delbaen, Eber and Heath (1999) set out four properties that any sensible risk measure should satisfy. They showed that VaR violates one of them. VaR is not **sub-additive**. The VaR of a merged portfolio can exceed the sum of the VaRs of its parts (§5.7). *What changed:* risk measurement gained an axiomatic theory, and VaR's status as the reference measure came under challenge.

Expected shortfall (2000–2002). Rockafellar and Uryasev (2000) showed that linear programming can minimise over a portfolio the average loss beyond VaR, which they called conditional value at risk. Acerbi and Tasche (2002) gave the definition that is coherent for every distribution. *Lasting influence:* expected shortfall is now the regulatory measure for bank trading books.

3.5 Crisis and repair (2007–2019): from one number to a dashboard

Thesis: the financial crisis showed less that VaR was wrong than that institutions had given it jobs it was never designed for.

[Fact] In 2007–2009 bank VaR models produced exceedances far above their stated frequencies, and the exceedances came in clusters. O'Brien and Szerszeń (2017) document both in US bank data. The pattern matches the one that Berkowitz and O'Brien (2002) had found for the 1998 crisis. A backtest of the US market run for this chapter (§6.13, §10.2) reproduces it. Across 2007–2009, losses breached a 99% historical-simulation VaR over a one-year window 24 times, where 7.6 breaches were expected.

The regulatory response kept VaR and added measures around it. **Basel 2.5** (Basel Committee, 2009) added a **stressed VaR**, computed on a 12-month window of significant stress, on top of the ordinary charge. That substantially increased market-risk capital. The **London Whale** episode of 2012 showed a different failure: the firm managed the measure instead of the risk. A US Senate investigation (Permanent Subcommittee on Investigations, 2013) examined the episode. It found that a change of VaR model at JPMorgan's Chief Investment Office cut the reported VaR of its synthetic credit portfolio by about half overnight. Meanwhile the position kept growing and ended in a loss of more than six billion dollars. The **Fundamental Review of the Trading Book** (Basel Committee, 2019) finally replaced 99% VaR with 97.5% expected shortfall. The framework calibrates the new measure to a stress period and uses longer horizons for less liquid risk factors. VaR exceedances remain the backtest. Regulators have postponed implementation repeatedly, and the dates differ by jurisdiction.

3.6 Diffusion (2010–today): VaR outside banking

Thesis: as banks moved to expected shortfall, VaR spread into fund regulation, margin and retail tools, where its simplicity is an advantage.

European fund guidelines (CESR, 2010) cap a UCITS fund's 99% 20-day VaR at 20% of its value, or at twice the VaR of a reference portfolio. For funds using derivatives, the US followed with SEC Rule 18f-4 (Securities and Exchange Commission, 2020). It uses the same 99% 20-day horizon and at least three years of data. Its limits are 20% of net assets or 200% of a reference portfolio's VaR. [Practice] Clearing houses have been moving initial-margin models from fixed scenario grids toward filtered historical-simulation VaR or expected shortfall. Retail brokerage platforms increasingly show a VaR figure for an account. Twenty years after the academic consensus turned against it, VaR is more widely used than ever.

§3 Key takeaways

1. The idea is old. Roy's 1952 safety-first criterion is VaR with the roles of probability and loss level swapped. Parametric VaR is Markowitz's portfolio variance with a quantile multiplier.
2. Banks built VaR to express the risk of desks with incomparable risk measures in a common unit, and that remains its most defensible use.
3. Two events made it universal. J.P. Morgan gave RiskMetrics away in 1994. The 1996 Basel amendment based capital on internal VaR models, with a multiplier of three that nobody derived.
4. The critique began almost immediately: Taleb in 1997, LTCM in 1998, coherence in 1999 and expected shortfall by 2002. None of it displaced VaR, because none of it offered an alternative as cheap to compute and check.
5. The crisis produced clusters of exceedances, not a single bad number. Regulators added stressed VaR, expected shortfall and liquidity horizons to VaR instead of dropping it.
6. The London Whale is the standard case of managing the measure instead of the risk. A model change halved the reported VaR overnight.
7. VaR has since spread to fund regulation and margin, where a simple, auditable number matters more than theoretical elegance.

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4. Foundational references

The references fall into groups by kind, because readers use each kind differently. Each entry says why the work matters. Where a free copy exists, the link points to it. Entries available only behind a paywall carry a mark. Entries without a link have no stable copy.

4.1 Origins and the documents that defined practice

- **Roy, A. D. (1952).** "Safety First and the Holding of Assets." *Econometrica* 20(3), 431–449. — Risk as the probability of falling below a disaster level. VaR inverts Roy's criterion. It fixes the probability and solves for the level.
- **Markowitz, H. (1952).** "Portfolio Selection." *Journal of Finance* 7(1), 77–91. — Portfolio variance through a covariance matrix, which every parametric VaR uses.
- **Holton, G. A. (2002).** "History of Value-at-Risk: 1922–1998." Working paper. — The history, with sources. Most of §3.1–§3.2 relies on it.
- **J.P. Morgan/Reuters (1996).** *RiskMetrics — Technical Document*, 4th ed. New York: Morgan Guaranty Trust. — The document that standardised VaR practice. It covers risk-factor mapping, the conditional-normal model and the EWMA estimator with decay 0.94. It is still clear, and it is the source of more defaults than practitioners remember.
- **Mina, J. & Xiao, J. Y. (2001).** "Return to RiskMetrics: The Evolution of a Standard." RiskMetrics Group. — The update: historical simulation and Monte Carlo alongside the parametric method, and a better treatment of options.
- **Basel Committee on Banking Supervision (1996a).** "Amendment to the Capital Accord to Incorporate Market Risks." — The rule that made VaR the basis of bank capital. It required 99%, 10 days and a multiplier of at least three.
- **Basel Committee on Banking Supervision (1996b).** "Supervisory Framework for the Use of 'Backtesting' in Conjunction with the Internal Models Approach to Market Risk Capital Requirements." — The traffic light (§12.4). The document is short. It shows how to write a statistical rule for non-statisticians, and it includes an honest table of the rule's own error rates.
- **Basel Committee on Banking Supervision (2009).** "Revisions to the Basel II Market Risk Framework." — Basel 2.5: stressed VaR added to the ordinary charge.
- **Basel Committee on Banking Supervision (2019).** "Minimum Capital Requirements for Market Risk." — The Fundamental Review of the Trading Book: 97.5% expected shortfall, liquidity horizons, desk-level backtesting and a P&L attribution test.
- **CESR (2010).** "Guidelines on Risk Measurement and the Calculation of Global Exposure and Counterparty Risk for UCITS." CESR/10-788. — The European fund rules: 99%, 20 days, and absolute and relative VaR limits (§9.2).
- **Securities and Exchange Commission (2020).** "Use of Derivatives by Registered Investment Companies and Business Development Companies." Release IC-34084 (Rule 18f-4). — The US equivalent for funds using derivatives.

4.2 Methods

- **Engle, R. F. (1982).** "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation." *Econometrica* 50(4). — ARCH treats volatility as a quantity to forecast. Every conditional VaR derives from it.
- **Bollerslev, T. (1986).** "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31(3), 307–327. — GARCH(1,1), the model used most often in §6.4.
- **Boudoukh, J., Richardson, M. & Whitelaw, R. F. (1998).** "The Best of Both Worlds." *Risk* 11(5), 64–67. — Age-weighted historical simulation (§6.7), in four pages.
- **Hull, J. & White, A. (1998).** "Incorporating Volatility Updating into the Historical Simulation Method for Value-at-Risk." *Journal of Risk* 1(1), 5–19. [[paywalled]] — Volatility-weighted historical simulation: rescale each past return by the ratio of today's volatility to the volatility at the time.
- **Barone-Adesi, G., Giannopoulos, K. & Vosper, L. (1999).** "VaR without Correlations for Portfolios of Derivative Securities." *Journal of Futures Markets* 19(5), 583–602. — Filtered historical simulation, the method that performs best in §6.13. Together with Hull and White, it forms the most useful pair of papers in this list for an implementer.
- **McNeil, A. J. & Frey, R. (2000).** "Estimation of Tail-Related Risk Measures for Heteroscedastic Financial Time Series: An Extreme Value Approach." *Journal of Empirical Finance* 7(3–4), 271–300. [[paywalled]] — A GARCH filter plus extreme value theory for the residual tail (§6.11). It shows how to "model volatility first, then the tail".
- **Engle, R. F. & Manganelli, S. (2004).** "CAViaR: Conditional Autoregressive Value at Risk by Regression Quantiles." *Journal of Business & Economic Statistics* 22(4), 367–381. — Models the quantile directly, with no distribution at all (§6.12). The link is the NBER working paper.
- **Britten-Jones, M. & Schaefer, S. M. (1999).** "Non-Linear Value-at-Risk." *European Finance Review* 2(2), 161–187. — Delta-gamma VaR for option books, and why the linear approximation fails (§6.10).
- **Glasserman, P., Heidelberger, P. & Shahabuddin, P. (2000).** "Variance Reduction Techniques for Estimating Value-at-Risk." *Management Science* 46(10), 1349–1364. [[paywalled]] — How to make Monte Carlo VaR for option books

fast enough to run overnight.

- **Duffie, D. & Pan, J. (1997).** "An Overview of Value at Risk." *Journal of Derivatives* 4(3), 7–49. [[paywalled]] — The best early survey, strong on jumps, stochastic volatility and options.
- **Linsmeier, T. J. & Pearson, N. D. (2000).** "Value at Risk." *Financial Analysts Journal* 56(2). — The clearest short introduction to the three classic methods, worked through on one portfolio.
- **Litterman, R. (1996).** "Hot Spots and Hedges." *Journal of Portfolio Management* 23(5), 52–75. [[paywalled]] — Risk decomposition by marginal contribution, written for portfolio managers. It is the origin of the "hot spots" display in §10.4.
- **Garman, M. (1996).** "Improving on VaR." *Risk* 9(5), 61–63. — Marginal and component VaR, and the property that the components sum to the total (§5.5).
- **Tasche, D. (2008).** "Capital Allocation to Business Units and Sub-Portfolios: the Euler Principle." arXiv:0708.2542. — Why Euler (gradient) allocation is the only allocation consistent with performance measurement, for VaR and ES alike.

4.3 Coherence, expected shortfall and scoring

- **Artzner, P., Delbaen, F., Eber, J.-M. & Heath, D. (1999).** "Coherent Measures of Risk." *Mathematical Finance* 9(3), 203–228. [DOI] — The four axioms, and the proof that VaR fails sub-additivity. The paper is foundational, and it reads more easily than its reputation suggests.
- **Rockafellar, R. T. & Uryasev, S. (2000).** "Optimization of Conditional Value-at-Risk." *Journal of Risk* 2(3), 21–41. — The formula that computes ES and VaR in a single minimisation, and turns the minimisation of portfolio ES into a linear program (§5.6).
- **Rockafellar, R. T. & Uryasev, S. (2002).** "Conditional Value-at-Risk for General Loss Distributions." *Journal of Banking & Finance* 26(7), 1443–1471. — The extension to discrete distributions, where VaR and ES behave badly.
- **Acerbi, C. & Tasche, D. (2002).** "On the Coherence of Expected Shortfall." *Journal of Banking & Finance* 26(7), 1487–1503. — The definition of ES that is coherent for every distribution, and why the naive "average beyond VaR" is not coherent.
- **Embrechts, P., McNeil, A. J. & Straumann, D. (2002).** "Correlation and Dependence in Risk Management: Properties and Pitfalls." In *Risk Management: Value at Risk and Beyond*, ed. M. Dempster, Cambridge University Press. — When correlation is the right summary of dependence, and when it misleads. It is right for elliptical distributions, where VaR is also sub-additive.
- **Yamai, Y. & Yoshiba, T. (2005).** "Value-at-Risk versus Expected Shortfall: A Practical Perspective." *Journal of Banking & Finance* 29(4), 997–1015. — The practitioner's case for ES, including its cost. Under fat tails, ES needs more data for the same precision.
- **Gneiting, T. (2011).** "Making and Evaluating Point Forecasts." *Journal of the American Statistical Association* 106(494), 746–762. — Shows that ES is not elicitable. No scoring function ranks ES forecasts correctly on its own (§5.9).
- **Fissler, T. & Ziegel, J. F. (2016).** "Higher Order Elicitability and Osband's Principle." *Annals of Statistics* 44(4), 1680–1707. — The pair (VaR, ES) is jointly elicitable, which is the basis of modern comparison of ES forecasts.
- **Emmer, S., Kratz, M. & Tasche, D. (2015).** "What Is the Best Risk Measure in Practice? A Comparison of Standard Measures." *Journal of Risk* 18(2), 31–60. — Compares coherence, robustness, elicitability and allocation side by side, and concludes in favour of ES on balance. It is the most efficient entry point to the whole debate.
- **Danielsson, J., Jorgensen, B. N., Samorodnitsky, G., Sarma, M. & de Vries, C. G. (2013).** "Fat Tails, VaR and Subadditivity." *Journal of Econometrics* 172(2), 283–291. — VaR is sub-additive in the tail for most fat-tailed asset returns. The failures of §5.7 need extremely heavy tails or lumpy payoffs.

4.4 Backtesting and the empirical record

- **Kupiec, P. H. (1995).** "Techniques for Verifying the Accuracy of Risk Measurement Models." *Journal of Derivatives* 3(2), 73–84. — The unconditional coverage test. Kupiec also warned that the test has almost no power at 99% on one year of data. Practitioners still ignore this warning.
- **Christoffersen, P. F. (1998).** "Evaluating Interval Forecasts." *International Economic Review* 39(4), 841–862. — Independence and conditional coverage. Losses must breach a correct VaR at the right rate, and the breaches must be unpredictable.
- **Christoffersen, P. F. & Pelletier, D. (2004).** "Backtesting Value-at-Risk: A Duration-Based Approach." *Journal of Financial Econometrics* 2(1), 84–108. [[paywalled]] — Tests based on the time between exceedances.
- **Jorion, P. (1996).** "Risk²: Measuring the Risk in Value at Risk." *Financial Analysts Journal* 52(6), 47–56. — Standard errors for VaR, and the argument that analysts should report a VaR with one.
- **Hendricks, D. (1996).** "Evaluation of Value-at-Risk Models Using Historical Data." Federal Reserve Bank of New York *Economic Policy Review* 2(1). — Twelve VaR methods on foreign-exchange portfolios. It was the first large comparison, and its main findings have held up.
- **Berkowitz, J. & O'Brien, J. (2002).** "How Accurate Are Value-at-Risk Models at Commercial Banks?" *Journal of Finance* 57(3), 1093–1111. — Banks' own VaR forecasts against their P&L. The forecasts were conservative on average and

had clustered exceedances in 1998. A simple GARCH model on the P&L itself beat them.

- **O'Brien, J. & Szerszeń, P. (2017).** "An Evaluation of Bank Measures for Market Risk Before, During and After the Financial Crisis." *Journal of Banking & Finance* 80, 215–234. — The same exercise for 2007–2009. Bank VaR was conservative before the crisis. During the crisis its exceedances were excessive and clustered.
- **Berkowitz, J., Christoffersen, P. F. & Pelletier, D. (2011).** "Evaluating Value-at-Risk Models with Desk-Level Data." *Management Science* 57(12), 2213–2227. [[paywalled]] — Backtests on desk data. The duration and regression tests have the most power.
- **Kuester, K., Mittnik, S. & Paolella, M. S. (2006).** "Value-at-Risk Prediction: A Comparison of Alternative Strategies." *Journal of Financial Econometrics* 4(1), 53–89. [[paywalled]] — A large out-of-sample comparison. Volatility-filtered methods with fat-tailed residuals win. Plain historical simulation and normal models do worst.
- **Pérignon, C. & Smith, D. R. (2010).** "The Level and Quality of Value-at-Risk Disclosure by Commercial Banks." *Journal of Banking & Finance* 34(2), 362–377. [[paywalled]] — What banks disclose and which methods they use. Historical simulation is the most used method, by a wide margin.
- **Acerbi, C. & Szekely, B. (2014).** "Back-Testing Expected Shortfall." *Risk*, December, 76–81. — Shows that ES can be backtested despite not being elicitable.
- **Nolde, N. & Ziegel, J. F. (2017).** "Elicitability and Backtesting: Perspectives for Banking Regulation." *Annals of Applied Statistics* 11(4). — The modern synthesis of backtesting and forecast comparison for VaR and ES.

4.5 The critiques

- **Beder, T. S. (1995).** "VAR: Seductive but Dangerous." *Financial Analysts Journal* 51(5), 12–24. [[paywalled]] — Eight reasonable combinations of method and parameters, applied to three portfolios, gave VaR estimates that differed by up to 14 times. This was the first demonstration of model risk, and it is still the starkest.
- **Marshall, C. & Siegel, M. (1997).** "Value at Risk: Implementing a Risk Measurement Standard." *Journal of Derivatives* 4(3), 91–111. The link is the working paper. — Software vendors given the same portfolio and the same RiskMetrics method returned materially different numbers. This is implementation risk.
- **Taleb, N. N. & Jorion, P. (1997).** "The Jorion–Taleb Debate." *Derivatives Strategy*, April. — Both sides, at the moment of VaR's adoption. The link is a reprint.
- **Danielsson, J. (2002).** "The Emperor Has No Clothes: Limits to Risk Modelling." *Journal of Banking & Finance* 26(7), 1273–1296. — Risk is endogenous. A model estimated in calm markets gives little information about crises, and a regulatory VaR can increase the risk it measures.
- **Danielsson, J., Embrechts, P., Goodhart, C., Keating, C., Muennich, F., Renault, O. & Shin, H. S. (2001).** "An Academic Response to Basel II." LSE Financial Markets Group Special Paper 130. — The paper predicted procyclicality and herding from VaR-based regulation before the crisis demonstrated them.
- **Basak, S. & Shapiro, A. (2001).** "Value-at-Risk-Based Risk Management: Optimal Policies and Asset Prices." *Review of Financial Studies* 14(2), 371–405. — A manager facing a VaR limit rationally concentrates losses in the states beyond the VaR, so the worst outcomes get *worse* (§11.1).
- **Adrian, T. & Shin, H. S. (2014).** "Procyclical Leverage and Value-at-Risk." *Review of Financial Studies* 27(2), 373–403. — Dealers manage to a roughly constant VaR, which makes their leverage procyclical (§11.6).
- **Danielsson, J. & Zigrand, J.-P. (2006).** "On Time-Scaling of Risk and the Square-Root-of-Time Rule." *Journal of Banking & Finance* 30(10), 2701–2713. — When $\sqrt{h}$ scaling is wrong. With jumps, it understates long-horizon risk.
- **Diebold, F. X., Hickman, A., Inoue, A. & Schuermann, T. (1997).** "Converting 1-Day Volatility to h-Day Volatility: Scaling by √h Is Worse than You Think." Wharton Financial Institutions Center Working Paper 97-34. — Under GARCH, $\sqrt{h}$ scaling gets the level wrong in a predictable direction (§5.4, §11.5).
- **Pritsker, M. (2006).** "The Hidden Dangers of Historical Simulation." *Journal of Banking & Finance* 30(2), 561–582. — Historical simulation reacts late to rising risk, and asymmetrically to gains and losses. The link is the FEDS working paper.
- **Cont, R., Deguest, R. & Scandolo, G. (2010).** "Robustness and Sensitivity Analysis of Risk Measurement Procedures." *Quantitative Finance* 10(6), 593–606. — The counter-critique. Historical VaR is statistically robust and ES is not. Robustness and coherence conflict.
- **Jorion, P. (2000).** "Risk Management Lessons from Long-Term Capital Management." *European Financial Management* 6(3), 277–300. [[paywalled]] — The LTCM post-mortem in VaR terms.
- **Permanent Subcommittee on Investigations, US Senate (2013).** "JPMorgan Chase Whale Trades: A Case History of Derivatives Risks and Abuses." — The London Whale, including the VaR model change, from the primary documents.
- **Nocera, J. (2009).** "Risk Mismanagement." *New York Times Magazine*, 4 January. — The journalistic case against VaR after 2008, with interviews on both sides. It is not a primary source, but it is the best record of how practitioners talked about VaR at the time.

4.6 Volatility management, drawdowns and forecast comparison

- **Moreira, A. & Muir, T. (2017).** "Volatility-Managed Portfolios." *Journal of Finance* 72(4), 1611–1644. — Scaling exposure inversely to recent variance raised Sharpe ratios for the market and many factors. The paper provides evidence that de-risking as VaR rises need not cost return (§11.6).
- **Cederburg, S., O'Doherty, M. S., Wang, F. & Yan, X. (2020).** "On the Performance of Volatility-Managed Portfolios." *Journal of Financial Economics* 138(1), 95–117. [DOI] — The out-of-sample challenge to Moreira and Muir. For many strategies the gains disappear.
- **Magdon-Ismail, M., Atiya, A. F., Pratap, A. & Abu-Mostafa, Y. S. (2004).** "On the Maximum Drawdown of a Brownian Motion." *Journal of Applied Probability* 41(1), 147–161. [[paywalled]] — The expected maximum drawdown of a random walk, used in §13.5.
- **Chekhlov, A., Uryasev, S. & Zabarankin, M. (2005).** "Drawdown Measure in Portfolio Optimization." *International Journal of Theoretical and Applied Finance* 8(1), 13–58. — Conditional drawdown at risk: ES applied to drawdowns (§7.7).
- **Adrian, T. & Brunnermeier, M. K. (2016).** "CoVaR." *American Economic Review* 106(7), 1705–1741. — The VaR of the system conditional on one institution's distress (§7.6).
- **Kou, S., Peng, X. & Heyde, C. C. (2013).** "External Risk Measures and Basel Accords." *Mathematics of Operations Research* 38(3), 393–417. [[paywalled]] — The case for robustness over coherence in regulatory risk measures.
- **Diebold, F. X. & Mariano, R. S. (1995).** "Comparing Predictive Accuracy." *Journal of Business & Economic Statistics* 13(3), 253–263. [[paywalled]] — The standard test of whether one forecast's average loss is significantly lower than another's (§12.6).

4.7 Books

- **Jorion, P. (2006).** *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed. McGraw-Hill. — The standard practitioner text, by the person who did most to defend VaR. It is strong on regulation and bank practice.
- **McNeil, A. J., Frey, R. & Embrechts, P. (2015).** *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press. — The rigorous treatment: risk measures, extreme value theory, copulas and backtesting. The reference for §5.
- **Christoffersen, P. F. (2012).** *Elements of Financial Risk Management*, 2nd ed. Academic Press. — The best book for an implementer. It centres on filtered historical simulation and backtesting, and it includes exercises.
- **Danielsson, J. (2011).** *Financial Risk Forecasting.* Wiley. — Short and practical, with code, and sceptical in the right places.
- **Dowd, K. (2005).** *Measuring Market Risk*, 2nd ed. Wiley. — Comprehensive on methods, including how to estimate the precision of VaR estimates.
- **Holton, G. A. (2014).** *Value-at-Risk: Theory and Practice*, 2nd ed. Published free online. — Thorough on the engineering: mapping, revaluation, Monte Carlo, and the implementation choices that the academic books skip.
- **Alexander, C. (2008).** *Market Risk Analysis, Volume IV: Value-at-Risk Models.* Wiley. — Detailed, with worked spreadsheet examples.
- **Brown, A. (2011).** *Red-Blooded Risk: The Secret History of Wall Street.* Wiley. — A risk manager's defence of VaR. Brown presents it as a tool for finding the boundary of normal markets, not for predicting the abnormal ones. Opinionated and useful.
- **Taleb, N. N. (2007).** *The Black Swan.* Random House. — The case against relying on any probability model of the tail. Read it for the argument, not for the tone.

4.8 If you only read six things

Read them in this order:

1. **Linsmeier & Pearson (2000)** — the three classic methods on one portfolio, in an afternoon.
2. **Christoffersen (2012)**, the chapters on filtered historical simulation and backtesting — the working method of §14, done properly.
3. **Hull & White (1998)** *then* **Barone-Adesi, Giannopoulos & Vosper (1999)** — the volatility-filtered historical simulation that should be the default everywhere.
4. **Artzner, Delbaen, Eber & Heath (1999)** — what a risk measure should satisfy, and the precise way VaR fails.
5. **Christoffersen (1998)** — how to tell whether a VaR is right, and why the timing of exceedances matters as much as their number.
6. **Danielsson (2002)** *and* **Taleb & Jorion (1997)** — the critique, from inside and outside the discipline.

The list contains nothing about Monte Carlo engineering, copulas or extreme value theory. Those matter for a bank's option book. For nearly everyone else, the hard parts are the volatility forecast and an honest backtest.

5. The mathematics

This section gives the formal backing for §1 and §2. §5.2 makes Idea 2 precise. §5.5 gives the decomposition behind every risk report. §5.6–§5.7 cover the theory of expected shortfall and coherence. §5.8 asks how precisely anyone can estimate a tail quantile. Most practitioners skip it, and they should read it.

5.1 Quantiles, and why the definition has an infimum

The α-quantile of a random variable L with distribution function F_L is the generalised inverse

$$q_\alpha(L) = F_L^{-1}(\alpha) = \inf\{\ell \in \mathbb{R} : F_L(\ell) \geq \alpha\}.$$

When F_L is continuous and strictly increasing, this is the ordinary inverse. When it is not, the infimum picks a definite value. Two cases matter in practice.

- **Jumps.** Suppose L takes the value 100 with probability 4% and 0 otherwise. Then $F_L(\ell) = 0.96$ for $0 \leq \ell < 100$, so the 95% quantile is 0. The 95% VaR of a position with a 4% chance of a total loss is zero. This mechanism explains Position B in §2.3 and the failure of sub-additivity in §5.7.
- **Samples.** A historical sample of n losses is a discrete distribution. Over $n = 500$ days, its 95% quantile is the 25th or 26th largest loss, or something between them. Different software makes different choices. At 99% over $n = 250$ days, the quantile sits between the second and third largest losses, 2.5 observations into the tail. Common interpolation conventions can move a 99% historical VaR by 10% or more on the same data. Know this before comparing one system's number with a broker's.

5.2 Location-scale families: VaR is a scaled volatility

Suppose the one-period return is $r = \mu + \sigma\varepsilon$, where ε has mean zero, variance one and a fixed distribution G with quantile function G^{-1}. The loss on a position of value V is $L = -rV = (-\mu - \sigma\varepsilon)V$. The α-quantile of $-\varepsilon$ is $-G^{-1}(1-\alpha)$, so

$$\text{VaR}_\alpha = (-\mu + \sigma\, k_\alpha)\, V, \qquad k_\alpha = -G^{-1}(1-\alpha).$$

For a symmetric G, $k_\alpha = G^{-1}(\alpha)$. For the normal, $k_\alpha = z_\alpha$. Everything in §2.2 follows. VaR is linear in volatility and in position size. The distribution enters only through the one number k_α.

Whether to subtract the mean. Practitioners distinguish **absolute VaR**, measured from zero as above, from **VaR relative to the mean**. The second drops the $-\mu$ term and measures the loss relative to the expected outcome. At a one-day horizon the difference is a rounding error. At a one-year horizon the difference is large. An asset with μ = 7% and σ = 16% has a 95% relative VaR of 26% and an absolute VaR of 19%. At long horizons, always confirm which one a report uses. Fund regulation uses the same two words for something else (§9.2, §8.4).

When the location-scale assumption fails. The assumption holds for any linear portfolio of assets whose joint returns are **elliptical**. The multivariate normal and Student t are the important cases. Every linear combination of an elliptical vector is again location-scale, with the same standardised shape. The assumption fails for option books, whose P&L is a non-linear function of the factors, and for credit, whose losses are lumpy. Those books need the methods of §6.9 and §6.10.

5.3 Linear portfolios and the delta-normal formula

Take a portfolio of N positions with dollar values x_i. Its one-day P&L is approximately

$$\Delta V \approx \sum_{i=1}^{N} x_i r_i = x'r,$$

The approximation is exact for a buy-and-hold portfolio of simple returns over one period. Its variance is $x'\Sigma x$. Under joint normality, and after dropping the mean as §5.2 allows at one day,

$$\text{VaR}_\alpha = z_\alpha \sqrt{x'\Sigma x} = z_\alpha\, \sigma_p.$$

This is the **variance-covariance** or **delta-normal** VaR. The second name comes from how the method puts non-equity positions into the form $x'r$. It replaces each instrument with its first-order sensitivity to each risk factor. An option on a stock at price S becomes a stock position of δS dollars for each share the option covers. The option's **delta** δ is the change in its price per dollar move in the stock. A bond becomes a position in its yield of $-D^\star$ times its value per unit change in yield. The **modified duration** $D^\star$ is the fractional fall in the bond's price per unit rise in yield. A foreign stock becomes two positions, one in its local price and one in the currency. This step, **risk factor mapping**, takes most of a bank's VaR engineering effort. It is trivial for a portfolio of stocks and funds, because each position is its own factor.

5.4 Time aggregation and the square-root-of-time rule

Let the h-day return be the sum of daily log returns, $R_h = \sum_{j=1}^{h} r_{t+j}$. Log returns add across days, while simple returns compound. Over a few days the two are close enough that the loss is about $-R_h V$. Suppose the daily returns are independent and identically distributed (IID), with mean μ and variance σ^2. Then R_h has mean $h\mu$ and variance $h\sigma^2$, so its standard deviation is $\sigma\sqrt{h}$. If the daily returns are also normal, R_h is normal and

$$\text{VaR}_\alpha^{(h)} = (z_\alpha\,\sigma\sqrt{h} - h\mu)V \approx \sqrt{h}\ \text{VaR}_\alpha^{(1)}\,.$$

This is the **square-root-of-time rule**. Both of its assumptions fail. Daily volatility changes over time, and the shape of the tail differs across horizons.

Volatility changes over time and reverts to a long-run level. In a GARCH(1,1) model (§6.4), the recursion $\sigma_{t+1}^2 = \omega + a\,r_t^2 + b\,\sigma_t^2$ gives tomorrow's variance each evening. When $a + b < 1$, the recursion moves variance toward the long-run level $\bar\sigma^2 = \omega/(1-a-b)$. A day's expected squared return is its variance. Taking expectations of the recursion therefore shows that the gap between the forecast and $\bar\sigma^2$ shrinks by the factor $a + b$ for each day ahead:

$$E_t[\sigma_{t+j}^2] = \bar\sigma^2 + (a+b)^{j-1}(\sigma_{t+1}^2 - \bar\sigma^2).$$

The h-day variance is therefore $\sum_{j=1}^{h} E_t[\sigma_{t+j}^2]$, not $h\,\sigma_{t+1}^2$. When today's volatility is below its long-run level, the future variances drift *up*, and $\sqrt{h}$ scaling understates the risk. When volatility is high, the scaling overstates it. Diebold, Hickman, Inoue and Schuermann (1997) build their critique on this point. §11.5 measures the effect on a century of US data. From the calmest fifth of starting points, losses breached a 10-day 99% VaR built by $\sqrt{10}$ scaling of the RiskMetrics one-day figure 5.7% of the time rather than 1%. Part of that excess comes from the thin tail of the normal distribution. §11.5 discusses both causes.

The tail shape changes with the horizon. Sums of independent fat-tailed variables become more normal as h grows, by the central limit theorem. The h-day multiplier therefore drifts back toward the normal one. At 99%, a fat tail raises the one-day multiplier. That effect alone would make $\sqrt{h}$ scaling of the one-day quantile *overstate* the h-day quantile. But volatility clustering pushes in the opposite direction, because a few consecutive bad days are more likely than independence implies. Jumps also push in the opposite direction, for a different reason. A rare jump almost never falls inside a one-day quantile but often falls inside a multi-day one. Danielsson and Zigrand (2006) show that with jumps the rule understates long-horizon risk. On the US data of §11.5, the unconditional one-day 99% loss quantile times $\sqrt{10}$ was 9.68%, and the actual 10-day quantile was 10.07%. The rule was close on average and wrong in each state. Its error lies in its timing, not its average level.

5.5 Decomposition: marginal, component and incremental VaR

Risk reports give the total and say where it comes from. Euler's theorem on homogeneous functions provides the mathematics.

A risk measure $\rho(x)$ is **positively homogeneous of degree one** if $\rho(cx) = c\,\rho(x)$ for $c > 0$. Doubling every position then doubles the risk. VaR has this property in any model where the P&L is linear in positions, and so does ES. If such a ρ is also differentiable, Euler's theorem says

$$\rho(x) = \sum_{i=1}^{N} x_i \frac{\partial \rho}{\partial x_i}(x).$$

The partial derivative is the **marginal VaR** of position i: the change in portfolio VaR per extra dollar in that position. The product $C_i = x_i\,\partial\rho/\partial x_i$ is the position's **component VaR**, and the components add up exactly to the total. For the delta-normal VaR,

$$\frac{\partial\,\text{VaR}}{\partial x_i} = z_\alpha\,\frac{(\Sigma x)_i}{\sigma_p}, \qquad C_i = z_\alpha\,\frac{x_i(\Sigma x)_i}{\sigma_p} = \text{VaR}\cdot\frac{x_i(\Sigma x)_i}{x'\Sigma x}.$$

The fraction on the right is position i's **share of portfolio variance**. It is the position's dollar size times its covariance with the whole portfolio, divided by the portfolio's variance. Equivalently, it is the position's beta to the portfolio times its weight. A position that is uncorrelated with the rest contributes only through its own variance. A position that hedges the rest has a negative component.

The same logic carries over to historical simulation and to ES, with a cleaner interpretation. The marginal VaR of position i equals minus its expected return *on a day the portfolio loses exactly its VaR*. The marginal ES equals minus its average return *over the days the portfolio loses more than its VaR*. An extra dollar in position i changes the loss in every scenario by $-r_i$. The quantile moves by that change, averaged over the scenarios exactly at the quantile. The tail average moves by that change averaged over the tail. The component ES of a position is therefore **its average dollar loss on the portfolio's bad days**, which a spreadsheet filter can compute. Tasche (2008) shows that the Euler allocation is the only one consistent with measuring risk-adjusted performance, This is why it is used universally.

Incremental VaR is the actual change in portfolio VaR from a finite trade: $\text{VaR}(x+\Delta x) - \text{VaR}(x)$. For small trades it is approximately $\Delta x'\nabla\,\text{VaR}$, the marginal VaRs applied to the trade. For large trades it is not, because the derivative itself changes. On the worked account of §10.4, technology had a component VaR of \$599 out of \$1,092. Removing the technology position entirely lowered VaR by only \$468. Once technology is gone, the remaining positions lose the diversification that technology gave them. Component VaR says where the risk is. Only a full recomputation says what a trade will do to it.

5.6 Expected shortfall

The **expected shortfall** at level α is the average of the VaRs beyond α:

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u(L)\,du.$$

For a continuous loss distribution, this equals the conditional expectation $E[L \mid L \ge \text{VaR}_\alpha]$: the average loss on the days that breach the VaR. The integral form is the general definition. Acerbi and Tasche (2002) show that it remains coherent when the distribution has jumps, where the naive conditional expectation does not.

For a normal loss with mean zero and standard deviation σ, the integral has a closed form. Because $\int_z^\infty u\,\varphi(u)\,du = \varphi(z)$, the average of a standard normal beyond z_α is $\varphi(z_\alpha)/(1-\alpha)$. Hence

$$\text{ES}_\alpha = \sigma\,\frac{\varphi(z_\alpha)}{1-\alpha}.$$

The numbers explain a regulatory choice. Normal ES at 97.5% is 2.338σ, and normal VaR at 99% is 2.326σ, within half a percent of each other. [Fact] When the Basel Committee moved from 99% VaR to ES, it chose the 97.5% level so that the new measure would match the old one for a normal distribution. For fatter tails, ES grows faster than VaR. With Student t tails of four degrees of freedom, ES at 97.5% exceeds VaR at 99% by 6.6%. With three degrees of freedom, it exceeds it by 11%. The switch to ES raises capital exactly where the tails are fat. This is the intended effect.

One formula for both. Rockafellar and Uryasev (2000, 2002) proved that for any loss distribution with a finite mean,

$$\text{ES}_\alpha(L) = \min_{c\,\in\,\mathbb{R}}\Big\{c + \frac{1}{1-\alpha}\,E\big[(L-c)^+\big]\Big\},$$

where $y^+ = \max(y,0)$, and that $c = \text{VaR}_\alpha$ attains the minimum. For a continuous loss the argument is short. The derivative of the bracket in c is $1 - P(L>c)/(1-\alpha)$. It is zero where $P(L>c) = 1-\alpha$, which is at the VaR. There, the bracket equals the VaR plus the average amount by which the worst $1-\alpha$ of losses exceed it, which is the ES. Replace

the expectation with an average over historical or simulated scenarios. The expression becomes piecewise linear in c and in the portfolio weights. With one auxiliary variable per scenario standing in for $(L-c)^+$, *minimising a portfolio's ES subject to constraints is a linear program*. This is the practical reason to optimise ES rather than VaR. VaR computed from the same scenarios is not convex in the portfolio weights, and it has many local minima.

5.7 Coherence, and the failure of sub-additivity

Artzner, Delbaen, Eber and Heath (1999) proposed four properties for a risk measure ρ. The table writes them for losses L_1 and L_2.

Axiom	Statement	Meaning
Monotonicity	If $L_1 \le L_2$ always, then $\rho(L_1) \le \rho(L_2)$	A position that always loses less is less risky
Translation invariance	$\rho(L+m) = \rho(L) + m$ for a constant m	Adding a sure loss of m adds m of risk, and holding m in cash removes m of risk
Positive homogeneity	$\rho(cL) = c\,\rho(L)$ for $c > 0$	Doubling the position doubles the risk
Sub-additivity	$\rho(L_1 + L_2) \le \rho(L_1) + \rho(L_2)$	Merging cannot create risk, so diversification never hurts

A measure with all four properties is **coherent**. VaR satisfies the first three. It fails the fourth in general. The standard counterexample uses two bonds.

Two independent bonds each lose 100 with probability 4% and nothing otherwise. Each has a 95% VaR of zero, because the default probability is below 5%. Now hold both. The probability that at least one defaults is $1 - 0.96^2 = 7.84\%$, which is above 5%, so the portfolio's 95% VaR is 100. The diversified portfolio has more VaR than the sum of its parts, which is zero. A system of VaR-based limits would therefore discourage diversification and reward concentration. VaR does not register the risk while each individual bet has a default probability below 5%.

Expected shortfall is sub-additive in this example. Each bond's worst 5% of outcomes consists of its 4% chance of a loss of 100 plus 1% of outcomes with no loss. Its 95% ES is therefore $0.04 \times 100 / 0.05 = 80$. The portfolio's worst 5% contains both defaults together (probability 0.16%, loss 200), plus enough of the single-default outcomes (loss 100) to fill the remaining 4.84%. That gives an ES of $(0.0016 \times 200 + 0.0484 \times 100)/0.05 = 103.2$, comfortably below $80 + 80$.

Researchers contest how much this matters in practice. The evidence has three parts.

- VaR is provably sub-additive, at any confidence level above 50%, whenever the joint distribution is elliptical (Embrechts, McNeil and Straumann, 2002). That covers the normal and Student t models of most linear portfolios.
- [Fact] Danielsson and co-authors (2013) show that VaR is sub-additive far enough into the tail for fat-tailed returns with a finite mean. This covers essentially all traded asset returns. The counterexamples need either extremely heavy tails, with no finite mean, or lumpy, discrete payoffs.
- [Contested] The failure therefore matters on exactly the books where VaR is most dangerous anyway. These are credit portfolios with default risk, short positions in deep out-of-the-money options, and anything else that earns a small, steady income in exchange for a rare large loss. Whether sub-additivity is a decisive objection or an edge case depends on whether the desk runs such a book. Emmer, Kratz and Tasche (2015) weigh it against VaR's advantages and prefer ES. Cont, Deguest and Scandolo (2010) argue that ES's loss of statistical robustness is the larger cost.

5.8 How precisely a tail quantile can be estimated

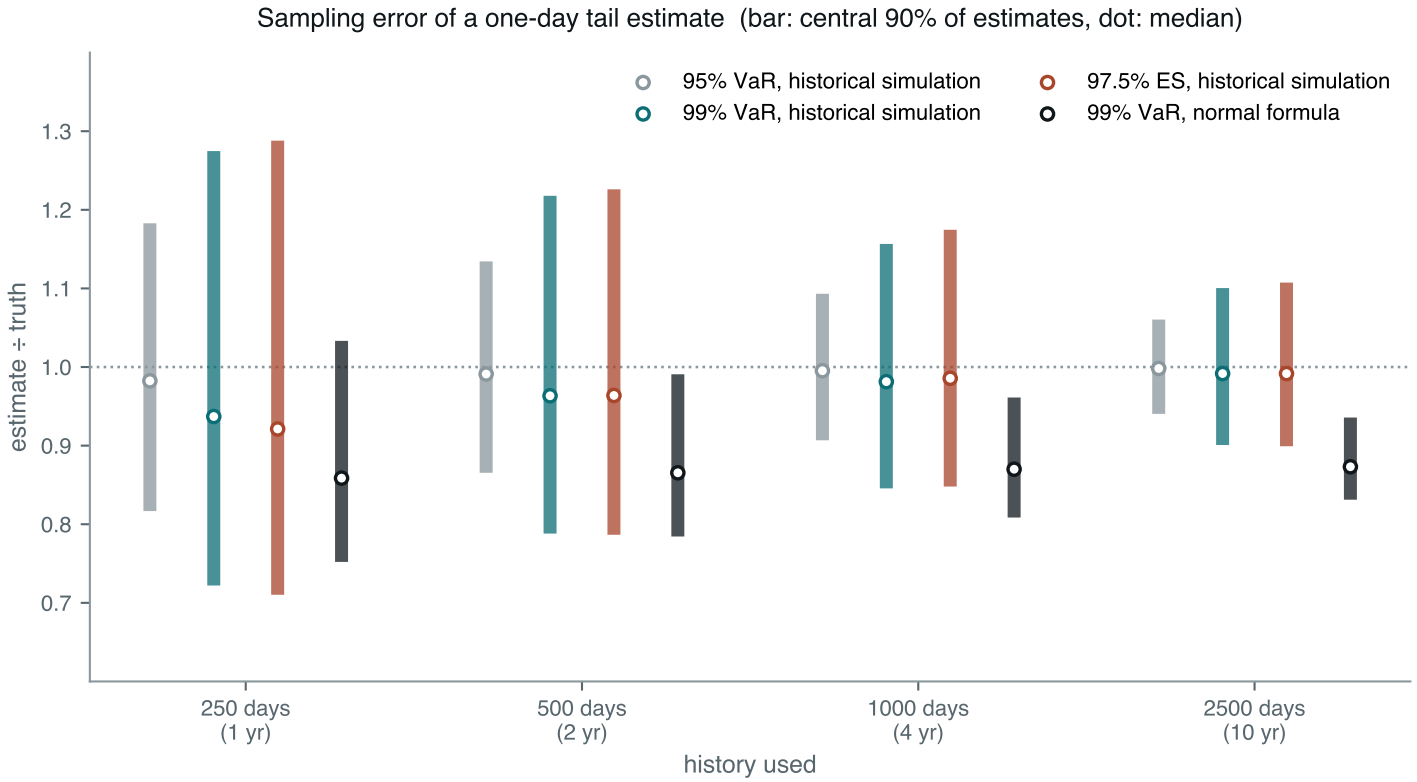
Every VaR is an estimate, and the sampling error is larger than most reports state. Consider an estimate $\hat{q}$ of the quantile q with tail probability p, from n independent observations with density f. Its large-sample standard error is

$$\mathrm{se}(\hat{q}) \approx \frac{1}{f(q)} \sqrt{\frac{p(1-p)}{n}}.$$

The numerator is the binomial uncertainty in the fraction of observations that fall beyond the true quantile. Dividing by the density converts that uncertainty into a distance. The density is small in the tail, which makes the error large. For normal data with n = 250 days, the formula gives a standard error of 10% of the 99% VaR and 8% of the 95% VaR. The normal formula estimates only σ, and under its own assumptions it has a smaller error. When the data really are normal,

se$(\hat{\sigma})/\sigma \approx 1/\sqrt{2n}$, or 4.5% for n = 250. With fat tails, the volatility estimate is itself noisier. At the US market's kurtosis of 19 (A.12), it is about three times as noisy. The formula is then precise, but its model is wrong.

The figure shows the trade-off by simulation. Returns are independent Student t with four degrees of freedom. They are fat-tailed, but nothing changes over time. Every estimator's assumptions therefore hold except the normal formula's, and the only problem is sample size.



Simulated IID Student t(4) returns: fat-tailed, but with nothing changing over time. The normal formula is tighter and reliably wrong.

The table gives the central range of the estimates for four lengths of history.

History	95% VaR, historical	99% VaR, historical	97.5% ES, historical	99% VaR, normal formula
250 days (1 year)	0.82 – 1.18	0.72 – 1.27	0.71 – 1.29	0.75 – 1.03
500 days	0.87 – 1.13	0.79 – 1.22	0.79 – 1.23	0.78 – 0.99
1,000 days	0.91 – 1.09	0.85 – 1.16	0.85 – 1.17	0.81 – 0.96
2,500 days (10 years)	0.94 – 1.06	0.90 – 1.10	0.90 – 1.11	0.83 – 0.94

Central 90% of estimates as a ratio to the true value, 20,000 simulated histories each.

The table supports four conclusions.

1. **A 99% VaR from one year of data is uncertain by about ±28%** at 90% confidence, *with every assumption satisfied*. Two honest analysts with different years of data will often disagree by a fifth, and sometimes by a third.
2. **The 95% VaR is much more precise than the 99% VaR** from the same data. One year gives ±18% against ±28%. The 99% figure needs two to four times the history to match it. One year gives 0.82–1.18 at 95%, while at 99% two years give 0.79–1.22 and four years give 0.85–1.16. This is the main argument for using 95% in personal and internal work, and for backtesting at 99% only on several years of data (§12.5).
3. **The normal formula is tighter and reliably wrong.** Its median estimate is 14% too low at every sample size, because the tails are fat. More data only narrow the interval around the wrong number.
4. **Precision of ±10% at 99% needs about 10 years of data**, under independence. Real volatility changes by a factor of 10 over 10 years. A 10-year window therefore estimates an average volatility that never applies. No window is long enough to estimate the tail and short enough to be current. Filtered historical simulation (§6.8) solves this. It uses a short window for volatility and a long window for the shape of the tail.

Jorion (1996) argued for exactly this reason that analysts should report VaR with a standard error. Almost nobody does so.

5.9 Elicitability, and how to compare forecasts

Choosing between two VaR models requires a scoring function. Counting exceedances (§12) tests each model against its own claim, but it cannot rank two models that both pass. A **scoring function** is a loss $\psi(\text{forecast}, \text{outcome})$ whose expected value the true quantity minimises. A statistic that has such a function is **elicitable**.

The quantile is elicitable. Its scoring function is the **pinball** or **quantile loss**. Write $\hat{q}_t = -\widehat{\text{VaR}}_t$ for the forecast return quantile, with the VaR as a fraction of value, and $p = 1 - \alpha$ for the tail probability. Then

$$\psi(\hat{q}_t, r_t) = \left(p - \mathbf{1}\{r_t < \hat{q}_t\}\right)(r_t - \hat{q}_t),$$

and $E[\psi]$ reaches its minimum at the true p-quantile. The loss charges p per unit of distance for every day the return lands above the forecast quantile, and $1 - p$ per unit for every day it lands below. At 99%, a breach costs 99 times as much per unit as a quiet day. Raising the forecast a little saves p on each day above it and costs $1 - p$ on each day below. The expected score therefore stops improving exactly where a fraction p of returns lie below the forecast, which is the p-quantile. Averaging ψ over a backtest ranks models consistently, and §6.13 uses it for its comparison.

Expected shortfall is not elicitable on its own (Gneiting, 2011): no scoring function has ES as its unique minimiser. Some briefly took this to mean that ES cannot be backtested, but that is wrong. The pair (VaR, ES) is jointly elicitable (Fissler and Ziegel, 2016). Scoring can therefore rank ES forecasts that come with their VaR forecasts, and Acerbi and Szekely (2014) give direct backtests of ES. The real cost of ES appears in §5.8. ES depends on the few largest losses, so it is noisier to estimate and to test than a 95% VaR. It is about as noisy as the 99% VaR it replaced.

§5 Key takeaways

1. VaR is a generalised inverse. With lumpy losses or small samples, the infimum in the definition matters. Conventions for interpolating a sample quantile can move a 99% historical VaR by 10%.
2. For any location-scale return model, VaR equals $(k_\alpha \sigma - \mu)V$. The tail enters through one number. Drop the mean at short horizons, but at long horizons confirm which convention a report uses.
3. Square-root-of-time scaling assumes away the mean reversion of volatility and the change in tail shape. Its error lies in its timing, not its average.
4. Euler's theorem makes component VaRs add up to the total. Component ES is each position's average loss on the portfolio's worst days. Components say where risk is. Only a recomputation says what a trade does.
5. Under normality, ES at 97.5% matches VaR at 99% to within half a percent. Under realistic fat tails it exceeds VaR by roughly 7–11%. ES and VaR come out of one convex minimisation, which makes ES the measure to optimise.
6. VaR is not sub-additive in general. The failure matters on books that sell tail risk, such as credit and short options. For linear portfolios of traded assets it is rarely the binding problem.
7. A 99% VaR from one year of data is uncertain by ±28% with every assumption satisfied, and a 95% VaR by ±18%. No window is both long enough and current. Filter volatility, and use a long window for the shape of the tail.
8. Average pinball loss ranks VaR models, because quantiles are elicitable. ES needs joint scoring with VaR, and it can be backtested.

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6. Computing VaR: the methods

6.1 The two questions every method answers

Stage 2 of the pipeline in §1.5 produces tomorrow's distribution of returns. Every method in this section answers two questions.

- **The source of the scale.** The scale comes from one of two sources. A fixed window of the past weights each day equally (**unconditional**). A model that updates the volatility every day weights recent days more (**conditional**).
- **The source of the shape.** The shape comes from an assumed distribution (normal or Student t). It can also come from the empirical distribution of past returns or from a fitted model of the extreme tail. Methods that model the quantile directly use no distribution.

The table arranges the answers as a grid.

Shape ↓ / Scale →	Unconditional: a window, equal weights	Conditional: volatility updated daily
Normal	Parametric VaR with a rolling window (§6.2)	RiskMetrics: EWMA volatility, normal shocks (§6.3)
Fat-tailed parametric	Cornish–Fisher (§6.5) or Student t on a window	GARCH or EWMA with t shocks (§6.4)
Empirical	Historical simulation (§6.6), age-weighted (§6.7)	Filtered historical simulation (§6.8)
Extreme value tail	Peaks over threshold on raw returns (§6.11)	GARCH filter plus extreme value theory (§6.11)
None: model the quantile	—	CAViaR (§6.12)

Two further choices apply across the whole grid. **Monte Carlo** (§6.9) computes the quantile of any model in the grid by simulating from it, when no closed form exists. **Revaluation** turns the scenarios into P&L, through linear sensitivities, delta-gamma or full repricing (§6.10). It matters only for non-linear positions.

§2.5 already says which axis matters more. Most of the fat tail in daily data is time-varying scale. So expect the right-hand column to beat the left in every row. §6.13 tests two rows in both columns, normal and empirical, and both confirm this.

Each method below uses the same headings, as far as they apply: **intuition**, **definition**, **assumptions**, **strengths**, **weaknesses**, **cost**, **failure modes** and **when preferred**. The backtest numbers quoted come from a backtest run for this chapter on US market daily returns from November 1929 to December 2025. That gives 25,151 forecast days. The sample starts in July 1926. Its first 1,000 days only fill the estimation windows. Each method forecasts one day ahead, using only data available the evening before. The full table is in §6.13.

6.2 Parametric normal (variance–covariance)

Intuition. Estimate the portfolio's volatility, assume the P&L is normal, and multiply.

Definition. $\text{VaR}_\alpha = z_\alpha \sqrt{x' \hat{\Sigma} x}$, with $\hat{\Sigma}$ the sample covariance of the last n days of returns (§5.3). For one position, it is $z_\alpha \hat{\sigma} V$.

Assumptions. Returns are jointly normal and identically distributed over the window, and the portfolio is linear in them. All three are false. §2.5 shows this for the first two, and §6.10 for the third.

Strengths. The method computes instantly. It decomposes analytically into marginal and component VaR (§5.5). It is transparent, because a reader can trace every number to a volatility and a correlation. It is the only method in which a reader can check the answer with a calculator.

Weaknesses. It errs in the tail in a known direction. It is too high at 95% and too low at 99% (§2.2). With an equal-weighted window, it also reacts slowly to changes in volatility, which compounds the tail error with a timing error.

Cost. One covariance matrix: $O(N^2 n)$ to estimate, and negligible to apply.

Failure modes. On this chapter's backtest with a 250-day window, losses breached the 99% VaR on 2.08% of days, more than twice its stated rate. Given a breach yesterday, the chance of another today was 10.3% rather than 1%. Over 2007–2009 losses breached it 34 times where 7.6 breaches were expected. The clustering and the crisis count come from the

equal-weighted window. The volatility it used was an average over the past year. The excess breach rate comes from the normal tail. §6.3 shows that the excess persists with a current volatility estimate.

When preferred. For quick sanity checks, for decomposition, and as the baseline that every other method must beat. It should not be the production number.

6.3 EWMA volatility, the RiskMetrics method

Intuition. Keep the normal distribution, but let volatility respond to recent returns, with the most weight on yesterday's squared return.

Definition. The RiskMetrics recursion (J.P. Morgan/Reuters, 1996) updates the variance forecast each evening:

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2, \qquad \lambda = 0.94 \text{ for daily data.}$$

The covariance updates analogously, $\Sigma_t = \lambda\Sigma_{t-1} + (1-\lambda)\,r_{t-1}r_{t-1}'$, where r_{t-1} here is the vector of asset returns. VaR is then $z_\alpha\sigma_t V$, or $z_\alpha\sqrt{x'\Sigma_t x}$ for a portfolio. Substituting the recursion into itself repeatedly gives

$$\sigma_t^2 = (1-\lambda)(r_{t-1}^2 + \lambda\,r_{t-2}^2 + \lambda^2 r_{t-3}^2 + \cdots).$$

This is a weighted average of past squared returns, with weight $(1-\lambda)\lambda^j$ on the return from $j+1$ days ago. The weights sum to one.

What 0.94 means. The weights halve every $\ln 0.5/\ln 0.94 = 11.2$ days. The weighted average age of the returns used is $1/(1-\lambda) = 16.7$ days. The estimate is as noisy as an equal-weighted average of $(1+\lambda)/(1-\lambda) = 32$ observations. This number is the reciprocal of the sum of the squared weights. For estimation purposes, EWMA therefore behaves like a 32-day window. That is short enough to react within a week to a change of regime. It is too short to estimate a covariance matrix of more than about 30 assets, for the reasons given in Portfolio Construction and the Covariance Matrix.

Assumptions. Conditional normality: returns are normal *given* today's volatility. Volatility follows the recursion, with no long-run level to revert to. EWMA is formally an "integrated" GARCH: a GARCH(1,1) model (§6.4) with a zero constant and a persistence of one.

Strengths. One parameter, no estimation and robust behaviour. It captures the first-order fact, volatility clustering, almost as well as models with more parameters. It is the volatility input to most of the better methods below.

Weaknesses. Conditional normality still understates the 99% tail. Standardised by EWMA volatility, US daily returns have a 99% quantile of 2.90 rather than 2.33 (§2.5). The model has no mean reversion, so its multi-day forecasts are flat, and square-root-of-time scaling repeats the error of §5.4.

Cost. One update per day per covariance entry.

Failure modes. On this chapter's backtest, losses breached the 95% VaR on 5.51% of days, close to the target. They breached the 99% VaR on 2.12% of days. The breaches cluster much less than with a fixed window. The chance of a breach following a breach was 5.6%, against 10.3%. The conditional model causes this improvement. The excess at 99% comes from the tail multiplier. §6.8 keeps this volatility, replaces 2.33 with an empirical multiplier and brings the breach rate to 1.07%.

When preferred. As the volatility model for almost everything. On its own, it serves for the 95% VaR of linear portfolios, where it is about as good as anything.

6.4 GARCH and fat-tailed conditional models

Intuition. Make the volatility model mean-revert, and give the shocks a fat-tailed distribution.

Definition. The GARCH(1,1) model of Bollerslev (1986), building on Engle's (1982) ARCH, is

$$r_t = \sigma_t\varepsilon_t, \qquad \sigma_t^2 = \omega + a\,r_{t-1}^2 + b\,\sigma_{t-1}^2,$$

with $\omega > 0$, $a, b \geq 0$ and $a+b < 1$. Variance therefore reverts to $\bar\sigma^2 = \omega/(1-a-b)$. The sum $a+b$ is the **persistence** (§5.4). EWMA is the case $\omega = 0$, $a = 1-\lambda$, $b = \lambda$. Its persistence is exactly one, so it has no level to revert to. The shocks ε_t are standardised normal variables or, more realistically, standardised Student t variables with ν degrees of freedom. A

standardised t is a t variable multiplied by $\sqrt{(\nu-2)/\nu}$, so that its variance is one. Maximum likelihood estimates the parameter ν jointly with the recursion. VaR is $k_\alpha \sigma_t V$, where k_α comes from the shock distribution. Asymmetric variants (GJR, EGARCH) let negative returns raise volatility more than positive ones. That captures the "leverage effect" observed in equities.

Assumptions. The volatility recursion has the correct form, and the shock distribution is the same every day.

Strengths. Mean reversion gives sensible multi-day forecasts (§5.4). A fitted ν addresses the tail. The model is standard and widely implemented.

Weaknesses. Numerical optimisation estimates three to five parameters. The estimation can be unstable on short samples and across regime changes. A symmetric parametric tail calibrated to fit one quantile can misfit another.

Cost. One likelihood maximisation per refit, and one recursion per day.

Failure modes. A fixed tail shape misfits the data, as a simple test shows. Pair the EWMA volatility with standardised Student t shocks with five degrees of freedom. At 99% this improved the breach rate from 2.12% to 1.44%. At 95% it *worsened* the rate, from 5.51% to 6.22%. This happens because a unit-variance t has a smaller 95% quantile than the normal (1.56 against 1.645), while the actual standardised returns have a larger one (1.71). The data's tail is not a t tail, and any single parametric family forces a trade-off between quantiles. Fitting ν rather than fixing it moves the compromise but cannot remove it. Every unit-variance t has a 95% multiplier below 1.645, and these data have one above it. The family cannot match the data at 99% either. No unit-variance t has a multiplier above about 2.66. The multiplier reaches that value near 3.6 degrees of freedom, against the data's 2.90.

When preferred. When multi-day horizons matter and mean reversion is worth modelling. Also as the volatility filter for §6.8 and §6.11, and when a smooth parametric model is needed, for instance to simulate (§6.9).

6.5 Cornish–Fisher, or "modified VaR"

Intuition. Keep the convenience of a formula, but correct the normal quantile for the skewness and kurtosis of the data.

Definition. The Cornish–Fisher expansion adjusts the normal quantile z using the sample skewness γ and excess kurtosis κ. For the lower tail, $z = -z_\alpha$, and

$$\tilde{z} = z + \frac{(z^2-1)\gamma}{6} + \frac{(z^3-3z)\kappa}{24} - \frac{(2z^3-5z)\gamma^2}{36}.$$

The VaR multiplier is $k_\alpha = -\tilde{z}$. Negative skewness pushes the lower quantile further out at both levels. The kurtosis term changes sign at $|z| = \sqrt{3} \approx 1.73$. Excess kurtosis therefore pulls the 95% quantile in and pushes the 99% one out, which is the pattern of §2.2. Hedge funds and funds of funds commonly report the method as **modified VaR**.

Assumptions. The distribution is close enough to normal for a three-term expansion to be accurate.

Strengths. A closed form, using two moments that everyone already reports.

Weaknesses. The expansion is accurate only for mild departures from normality, and daily returns depart far from normality. It can also fail to be monotone in α, so for large κ the "quantiles" can come out in the wrong order. For symmetric data it stays monotone only while the excess kurtosis is below 8 (A.25), which is half the US market's value.

Failure modes. The US market's full-sample skewness is −0.17 and its excess kurtosis is 16.1. With these values, Cornish–Fisher gives a 95% multiplier of 1.37 against an empirical 1.50, and a 99% multiplier of **6.21** against an empirical 2.89. That is more than double. The empirical figures here are for demeaned returns, which is what the expansion describes. The values 1.46 and 2.85 in §2.2 are measured from zero. A handful of days dominate the sample kurtosis (§2.5), so the correction is both large and unstable.

When preferred. For monthly or quarterly return series of funds, where departures from normality are moderate and only moments are available. Do not use it for daily data, and never for 99% with large kurtosis.

6.6 Historical simulation

Intuition. Historical simulation, HS for short, assumes that tomorrow will look like one of the last n days, chosen at random. Apply each of those days' market moves to today's portfolio, and take the quantile of the resulting P&Ls.

Definition. For each of the past days $s = t - n, \dots, t - 1$, compute the P&L that today's positions would have had. This is either $\Delta V_s = \sum_i x_i r_{i,s}$ or a full repricing of every instrument under day s's factor moves. The VaR is the empirical α-quantile of the losses $-\Delta V_s$ (§5.1).

The phrase "today's positions" is the core of the method. Readers most often misunderstand it. The method is a **replay** of today's book through the markets of past days. It does not use the P&L the portfolio actually earned in the past, because the portfolio was different then.

Assumptions. The last n days are an IID sample from tomorrow's distribution. The window is long enough to contain the tail and short enough to be relevant.

Strengths. The method makes no distributional assumption. Fat tails, skewness and non-linear dependence across assets all come from the data. The method can fully reprice option positions. A risk manager can explain the result to anyone ("on the 13th worst of the last 250 days we would have lost this much"), and an auditor can check it. This is why the method dominates at banks. Pérignon and Smith (2010) found that roughly three-quarters of the banks that disclosed their method used it.

Weaknesses. The window trade-off of §5.8 has no good answer. A one-year window has 2.5 observations beyond the 99% quantile, and it contains no information about anything that did not happen that year. A four-year window estimates an average that does not apply today. Every day in the window also gets equal weight, so the method ignores any difference between current volatility and past volatility.

Cost. One portfolio revaluation per historical day, n in total. This is trivial for linear portfolios and the dominant cost for option books.

Failure modes. The method has three, all visible in the backtest chart of §10.2.

1. **Slow reaction.** In a calm year followed by a crisis, the window is full of calm days. On this chapter's backtest, losses breached the 250-day historical 99% VaR 24 times in 2007–2009, where 7.6 breaches were expected. Ten of them came in September and October 2008, while the VaR climbed in steps.
2. **Stale spikes (the "ghost effect").** A large loss enters the window and VaR jumps. VaR then stays high for exactly n days and drops sharply on the day the loss leaves the window, regardless of what markets are doing. The 250-day 99% VaR reached 7.8% of value in December 2008. It stayed within 10% of that level until late September 2009. By then EWMA volatility had fallen by four-fifths, from its October 2008 peak of 4.8% a day to under 1%.
3. **Clustered breaches.** Because the scale is stale, breaches come together. Given a breach of the 95% VaR yesterday, losses breached the 250-day historical VaR again today 15.1% of the time, three times the stated rate. Pritsker (2006) analyses these effects and shows that they follow from equal weighting.

When preferred. When positions are non-linear and a parametric model of them is unreliable. Also when the explainability of the number to a board or a regulator matters more than its accuracy. Also as an unconditional (§2.4) or stressed (§7.4) measure, where slow reaction is by design.

6.7 Age-weighted historical simulation

Intuition. Keep the empirical distribution, but trust recent days more.

Definition. Boudoukh, Richardson and Whitelaw (1998) give the return from j days ago a probability weight proportional to θ^j, with θ typically 0.97–0.99. Scale the weights to sum to one. Sort the window's losses from largest to smallest, and accumulate their weights until the total reaches $1 - \alpha$. The VaR is the loss at which the total reaches that level.

Assumptions. Recent days are more representative of tomorrow than old ones.

Strengths. Stale spikes fade smoothly rather than dropping out on the day the loss leaves the window, and the effective tail sample adapts to recent events. The method is cheap.

Weaknesses. Weighting does not rescale. The weights decide how much each past P&L counts, not how large it is. Suppose volatility has just doubled. Only the few days since the change carry the new scale. Every other P&L in the window keeps the magnitude it had at the volatility *when it occurred*, however much weight it gets. In a sudden crisis, the method therefore estimates the tail from very few recent observations.

Failure modes. On this chapter's backtest, with θ = 0.98 and a 250-day window, losses breached the 99% VaR on 1.55% of days. Breaches clustered much less than under plain historical simulation, with a 4.4% chance of a breach after a breach against 8.8%. In 2007–2009 the method had 16 breaches against an expected 7.6.

When preferred. As a cheap improvement on plain historical simulation when the infrastructure for volatility filtering is not available.

6.8 Filtered historical simulation

Intuition. Filtered historical simulation (FHS) separates the two jobs. A volatility model gives how *big* tomorrow's moves will be, and history gives what *shape* they will have. Divide every past return by the volatility at the time, so that each return is measured in "standard deviations of that day". Take the empirical quantile of these standardised returns, and multiply it by today's volatility.

Definition. With a conditional volatility estimate $\hat{\sigma}_s$ for each past day, from EWMA or GARCH,

$$\hat{\varepsilon}_s = \frac{r_s}{\hat{\sigma}_s}, \qquad \text{VaR}_{\alpha,t} = \hat{\sigma}_t \cdot \Big(- \hat{q}_{1-\alpha}\big(\{\hat{\varepsilon}_s\}_{s=t-n}^{t-1}\big)\Big) \cdot V.$$

Here $\hat{q}_{1-\alpha}$ is the empirical $(1-\alpha)$-quantile of the window's standardised returns. It is a negative number in the left tail, and the minus sign turns it into a positive multiplier.

The rescaling version of Hull and White (1998) writes the same formula as a replay of each historical return multiplied by $\hat{\sigma}_t/\hat{\sigma}_s$. For a multi-asset book, Barone-Adesi, Giannopoulos and Vosper (1999) filter each risk factor by its own volatility model. They then resample *vectors* of standardised residuals from the same historical day, which keeps that day's cross-sectional dependence. When the positions are fixed, as in §14, the simplest version filters the replayed portfolio P&L directly.

Assumptions. The standardised residuals are roughly IID. The window for the tail shape can therefore be long, years rather than months, while the volatility estimate stays current.

Strengths. The method resolves the window dilemma of §5.8, with a short window for scale and a long window for shape. Fat tails and skew come from the data. It reacts within days to a change in volatility, and the stale-spike effect disappears. Tomorrow's scale comes from current volatility. An old crash is one standardised observation among a thousand, which barely moves a quantile when it enters or leaves the window.

Weaknesses. The method depends on an adequate volatility model. In the multi-asset version, the correlations are those of the residuals in the historical window, not a forecast. It needs slightly more work than the alternatives, though not much. §14 does it in a dozen lines.

Cost. One volatility recursion plus historical simulation.

Failure modes. The volatility filter reacts within days, not instantly, so the first day of a crisis is still a large breach. Over the full backtest, losses breached the EWMA-filtered VaR with a 1,000-day residual window on 1.07% of days at 99% and 5.04% of days at 95%. Of the seven methods in §6.13, only this one passes Kupiec's coverage test (§12.2) at both levels. Its breaches still cluster more than independence allows. The chance of a 99% breach after a breach is 4.5%. No volatility model moves fast enough on the first days of a shock. In 2007–2009 it took 13 breaches where 7.6 were expected, and in 2020 it took four where 2.5 were expected.

When preferred. [Practice] **Recommendation: use it as the default.** [Fact] Large comparisons find volatility-filtered methods with an empirical or extreme-value tail at or near the top, and plain historical simulation and normal models at or near the bottom. Kuester, Mittnik and Paolella (2006) is the most thorough of them. This chapter's backtest agrees.

6.9 Monte Carlo simulation

Intuition. Write down a full statistical model of the risk factors, simulate thousands of scenarios from it, revalue the portfolio in each, and take the quantile.

Definition. Specify the joint dynamics of the factors, for instance GARCH volatilities, a correlation matrix or copula, and fat-tailed shocks. Draw M scenarios for the horizon, reprice every position in each, and take the empirical quantile of the M losses. The sampling error of that quantile follows §5.8 with $n = M$. With 10,000 scenarios, a 99% VaR has 100 of them in the tail.

Assumptions. Monte Carlo assumes whatever the model assumes. It adds no assumption and removes none. It computes a quantile of a model that other methods cannot evaluate.

Strengths. It handles anything: non-linear and path-dependent instruments, multi-day horizons with dynamics, credit migrations and defaults, and assets with no usable history.

Weaknesses. Model risk sits in the specification, and the simulation's precision can mislead readers. A million scenarios give a very precise quantile of a possibly wrong distribution. The method is expensive for large option books.

Cost. M full revaluations. Variance-reduction techniques cut this by one to two orders of magnitude (Glasserman, Heidelberger and Shahabuddin, 2000). They include importance sampling toward the tail and control variates from the delta-gamma approximation.

Failure modes. Mis-specified dependence is the first. A correlation matrix estimated in calm markets understates how assets move together in a crash, and a normal copula has no tail dependence at all. Mis-specified dynamics at long horizons are the second.

When preferred. For option and structured-product books, for horizons beyond a few days, for credit portfolios, and for any instrument without a usable price history.

6.10 Non-linear positions: delta, delta-gamma, full revaluation

The linear approximation $\Delta V \approx x'r$ of §5.3 fails for options. Expand the position's value to second order in the underlying price S, and to first order in implied volatility σ_{imp} and time t:

$$\Delta V \approx \delta\,\Delta S + \tfrac{1}{2}\Gamma\,(\Delta S)^2 + \mathcal{V}\,\Delta\sigma_{\text{imp}} + \Theta\,\Delta t.$$

Here δ and Γ are the position's first and second derivatives in S, its delta and gamma. $\mathcal{V}$ and Θ are its derivatives in implied volatility and in time, its vega and theta. **Delta-normal** VaR keeps only the first term. **Delta-gamma** VaR adds the second. The P&L is then a quadratic function of normal variables, and it is no longer normal. Cornish–Fisher on its moments, Fourier inversion or simulation can find its quantile. Britten-Jones and Schaefer (1999) is the standard treatment. **Full revaluation** reprices every option in every scenario and needs no approximation at all.

The direction of the error is predictable. A long option has positive gamma, so the linear approximation overstates its losses. A short option has negative gamma, so the linear approximation understates its losses, and the understatement grows with the size of the move. Consider a worked example. A $100 stock has 20% implied volatility. That is a daily volatility of $0.20/\sqrt{252} = 1.26\%$ and a one-day 99% move of $2.326 \times 1.26\% = 2.93\%$. Sell 10 put contracts with 30 days to expiry, on 1,000 shares in total. The table gives the losses in dollars.

	At-the-money puts, strike 100	Out-of-the-money puts, strike 90
Premium collected	2,287	71
Option delta	−0.49	−0.03
Delta-normal 99% VaR	1,432	91
Full revaluation at the 99% move	1,695	149
Same, with implied volatility up 5 points	2,191	406
A 10% fall with implied volatility doubled	8,720	3,975
A 20% fall with implied volatility tripled	18,372	11,988

Black–Scholes prices at a zero interest rate. Every revaluation is one day later, so it includes a day of time decay.

The table supports three conclusions. For the at-the-money position, delta-normal VaR is 15% too low at the 99% move. It is 35% too low once implied volatility moves with the price, as it does. For the out-of-the-money position, the linear number is useless. Full revaluation with a volatility shift gives 4.5 times as much. A 10% crash costs 44 times the delta-normal VaR and 56 times the premium collected. The last two rows are not VaR at any confidence level a model would assign. But they are the scenarios in which short-option traders fail, and §7.4 and §13.6 cover how to examine them directly.

When preferred. Use full revaluation whenever the book contains short options. Use delta-gamma when repricing is too expensive. Use delta-normal only for long or small option positions.

6.11 Extreme value theory

Intuition. Model only the tail, and skip the rest of the distribution. Use the one piece of probability theory that says what all tails look like far enough out.

Definition. The Pickands–Balkema–de Haan theorem says that, for a wide class of distributions, the excesses over a high threshold u follow approximately a **generalised Pareto** distribution:

$$P(L - u > y \mid L > u) \approx \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \qquad y \geq 0.$$

Maximum likelihood fits the shape, or tail index, ξ and the scale β to the N_u of the n observations that exceed u. A positive ξ gives a power-law tail. At $\xi = 0$ the formula equals its limit, $e^{-y/\beta}$. Thin-tailed distributions such as the normal fall into this exponential case.

The method estimates the probability of a loss above a level $\ell > u$ in two parts. The first is the fraction of the sample beyond the threshold, N_u/n. The second is the generalised Pareto probability of exceeding the threshold by $\ell - u$. The probability is their product. Setting that probability equal to $1 - \alpha$ and solving for ℓ gives the VaR. Averaging the fitted tail beyond the VaR gives the ES:

$$\text{VaR}_\alpha = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-\alpha)\right)^{-\xi} - 1\right], \qquad \text{ES}_\alpha = \frac{\text{VaR}_\alpha + \beta - \xi u}{1-\xi} \quad (\xi < 1).$$

Both formulas hold for any level high enough that the VaR lies above the threshold, $1 - \alpha < N_u/n$. The ES also needs $\xi < 1$, without which the tail has no finite mean.

[Fact] Estimates of ξ for daily equity returns typically lie between about 0.2 and 0.4. That is a power-law tail with an exponent $1/\xi$ of roughly 2.5 to 5. McNeil and Frey (2000) apply the method to GARCH-standardised residuals rather than to raw returns. This order is correct. Filter the scale first, then model the shape of what remains.

Assumptions. The threshold is high enough for the asymptotic form to hold, and low enough to leave data to fit.

Strengths. A principled extrapolation beyond the data, where historical simulation can say nothing: 99.9% and beyond, or 10-year events from 10 years of data.

Weaknesses. Choosing the threshold takes judgement, and the choice visibly changes the result. The fit uses only the few dozen most extreme observations, so the estimates are noisy.

Failure modes. Unstable estimates of ξ from short samples, and a tail fitted to one regime extrapolated to another.

When preferred. For confidence levels above 99%: economic capital, insurance, and the default funds of clearing houses. At 95% and 99% it adds little to filtered historical simulation.

6.12 Modelling the quantile directly: CAViaR

Intuition. If the quantity of interest is a quantile, model the quantile and skip the distribution.

Definition. Engle and Manganelli (2004) let the VaR follow its own autoregression. One example is the symmetric absolute-value specification

$$\text{VaR}_t = c_0 + c_1\,\text{VaR}_{t-1} + c_2\,|r_{t-1}|.$$

Here a large move yesterday, of either sign, raises today's VaR, and c_1 sets how slowly the effect fades. Minimising the pinball loss of §5.9 over the sample estimates the coefficients. This is a quantile regression in which yesterday's fitted quantile is one of the regressors.

Strengths. No distributional assumption, and the estimation targets exactly the quantity being forecast.

Weaknesses. The objective is not smooth and has local minima, so estimates can be unstable. Each confidence level needs its own model, and the fitted quantiles at different levels can cross.

When preferred. In research and as a benchmark. [Practice] It is not known to be in wide production use.

6.13 Comparison

The table gives the full backtest of the seven methods, run for this chapter on the US market. The horizon is one day, the sample runs from November 1929 to December 2025, and there are 25,151 forecasts.

Method	95% hit rate	99% hit rate	99% hit after a hit	99% hits, 2007–09 (expect 7.6)	99% pinball loss
Normal, 250-day equal-weight volatility	4.91%	2.08%	10.3%	34	4.00
Normal, EWMA 0.94 (RiskMetrics)	5.51%	2.12%	5.6%	18	3.55
Student $t(5)$, EWMA 0.94	6.22%	1.44%	4.4%	12	3.42
Historical simulation, 250 days	5.51%	1.62%	8.8%	24	3.86
Historical simulation, 1,000 days	5.31%	1.35%	9.4%	41	4.13
Age-weighted HS, 0.98, 250 days	5.40%	1.55%	4.4%	16	3.61
Filtered HS, EWMA + 1,000 days	**5.04%**	**1.07%**	4.5%	13	**3.41**

Targets: 5% and 1%. "Hit after a hit" is the probability of a 99% breach on the day after one, which should equal 1% if breaches are unpredictable. Pinball loss is in basis points (§5.9). Lower is better. It rewards both calibration and responsiveness.

The table supports four conclusions.

- **The conditional column wins.** Every method that updates volatility daily has roughly half the breach clustering of its equal-weighted counterpart, and a lower pinball loss.
- **At 95%, almost every method is roughly calibrated on average.** The differences that matter at 95% lie in timing, not level.
- **At 99%, only filtered historical simulation is calibrated.** Normal tails double the breach rate. An assumed t tail helps at 99% and hurts at 95%. Plain historical simulation over a long window is better on average. It is the worst of all in a crisis, with 41 breaches in 2007–2009, because its volatility is four years stale.
- **No method is perfect.** Every method's breaches cluster more than independence allows, and Christoffersen's test (§12.3) rejects all seven. No model anticipates the first day of a volatility shock.

[Fact] The broader evidence agrees. Hendricks (1996) found the 95% VaRs from most methods acceptably calibrated, and the 99% normal VaRs too low. Kuester, Mittnik and Paolella (2006) rank filtered methods with fat-tailed residuals first. Berkowitz and O'Brien (2002) and O'Brien and Szerszeń (2017) found this about a GARCH model fitted to a bank's own P&L. It forecast as well as the banks' elaborate position-level VaR systems. It sometimes forecast better, especially in a crisis.

[Practice] **Recommendation, for a linear book:** estimate volatility with an EWMA or GARCH recursion, and take the tail shape from a long window of standardised residuals. Spend no further effort on distributional sophistication until the backtest shows a need for it. For a book with short options, add full revaluation. For confidence levels above 99%, add extreme value theory. Everything else is refinement.

§6 Key takeaways

1. Every method answers two questions: where the scale comes from (a window, or a volatility updated daily) and where the shape comes from (assumed, empirical, extreme-value, or none). The scale question matters more.
2. The parametric normal method is the baseline. It is transparent and decomposable, and wrong in the tail in a known direction.
3. EWMA with λ = 0.94 behaves like a 32-day window with an 11-day half-life. It is the right volatility model for almost everything. At 95% on linear books it is about as good as anything on its own.
4. Any single parametric tail trades one quantile against another. A $t(5)$ tail improved the 99% VaR and broke the 95% one. Cornish–Fisher on daily data gave a 99% multiplier more than twice the truth.

5. Banks use historical simulation most, for its explainability. It reacts late, leaves stale spikes and clusters its breaches: 24 in 2007–2009 at 99%, where 7.6 were expected.
6. Filtered historical simulation multiplies current volatility by the empirical quantile of standardised history. It was the only method calibrated at both 95% and 99% over 96 years. Use it as the default.
7. Linear VaR badly understates the risk of short options. On out-of-the-money puts, a 10% crash cost 44 times the delta-normal VaR. Reprice options in full whenever the book is short them.
8. Use extreme value theory above 99%, CAViaR mostly in research, and Monte Carlo wherever instruments have no usable history.

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7. Variations: the VaR family

§6 covered how to compute one number. This section covers related numbers that practitioners and researchers built from it. Some replace VaR's statistic, some change what the loss measures, some change the window, and some take the idea outside market risk. Each answers a question that VaR leaves open.

7.1 Expected shortfall

The question. It asks for the average loss, given a bad day.

The measure. ES_α, the average loss beyond the α-quantile (§5.6). It has many names: conditional value at risk (CVaR), average value at risk (AVaR), expected tail loss (ETL), tail VaR and tail conditional expectation. For continuous distributions they coincide. For discrete ones they can differ a great deal, and only the definition of Acerbi and Tasche is coherent. In the two-bond example of §5.7, the naive average of losses at or beyond the VaR is 4 per bond and 102 for the pair. The naive average therefore fails sub-additivity, as VaR does. ES gives 80 per bond and 103.2 for the pair (A.28).

Computing it. Every method of §6 that yields a distribution also yields ES. Average the losses beyond the VaR in a historical or simulated sample, or use a closed form. The normal case is $\sigma V \varphi(z_\alpha)/(1-\alpha)$, and §6.11 gives the generalised Pareto case. On the worked account at the end of 2025, filtered historical simulation gave a 97.5% ES of $2,080 against a 99% VaR of $1,991. Plain historical simulation over 500 days gave an ES of $3,209 against a VaR of $2,849.

Benefits and costs. The table compares the two measures.

	VaR	ES
Sees the size of tail losses	No	Yes
Sub-additive (rewards diversification)	Not always (§5.7)	Always
Convex in positions, so optimisable by linear programming	No	Yes
Elicitable on its own, so forecasts can be ranked	Yes	No, but jointly with VaR (§5.9)
Statistical precision from a given sample	Better	Worse than a 95% VaR; about equal to a 99% VaR (§5.8)
Robust to a few bad data points	More	Less
Easy to explain	"Lose more than this one day in twenty"	"Average of the worst one day in forty"

[Contested] Regulators and most academics have concluded that the first three rows outweigh the last four. Emmer, Kratz and Tasche (2015) give the balanced statement. Critics argue that a capital rule should be robust to data errors and model choices, and that ES is not (Cont, Deguest and Scandolo, 2010; Kou, Peng and Heyde, 2013). This chapter's view is that a single account should compute both. They cost the same, and the ratio of ES to VaR is itself a useful number, which rises when the tail fattens.

7.2 Marginal, component and incremental VaR

The question. It asks where the risk comes from and what a trade would do to it.

The measures. §5.5 defines them. **Marginal VaR** is the sensitivity of total VaR to a dollar more of a position. **Component VaR** is the position times its marginal VaR, and the components add up to the total. **Incremental VaR** is the actual change from a finite trade, which needs a recomputation.

How they are used. [Practice] Component VaR is the basis of the **risk budget**. The firm allots a manager a share of its VaR rather than a dollar amount. The firm judges a position by its return per unit of component VaR. Litterman (1996) introduced the two displays that every risk system now includes. **Hot spots** are the positions whose share of risk most exceeds their share of capital. **Best hedges** are the trades, one per instrument, that would reduce portfolio variance most. For instrument i alone, minimising the variance $x'\Sigma x$ over x_i gives $\Delta x_i = -(\Sigma x)_i/\Sigma_{ii}$.

The worked account. The full table is in §10.4. Technology was 40% of the dollars and 55% of the 95% VaR. A 10% position in autos and durables was 21% of the VaR. Energy was 15% of the dollars and only 5% of the VaR, because it was nearly uncorrelated with the rest. An investor would not have guessed that last fact. It is the reason to compute components.

7.3 Relative VaR and tracking-error VaR

The question. It asks how far the portfolio could fall behind its benchmark.

The measure. The VaR of the *difference* between the portfolio's P&L and a benchmark's. For an active manager judged against an index, the risk that matters is underperformance, not loss. Under normality it is z_α times the **tracking error**, the standard deviation of the active return, times the portfolio value.

The regulatory meaning. Fund rules use the same phrase for a different quantity, a ratio. The ratio is the VaR of the fund divided by the VaR of a reference portfolio without leverage or derivatives, capped at 2 (§9.2). It measures how much extra risk the derivatives add. It does not measure how far the fund could trail.

7.4 Stressed VaR and stress tests

The question. It asks what would happen if today's market were like the worst market in the data.

Stressed VaR recomputes VaR on today's positions using a window drawn from a period of severe stress rather than from the recent past. For most banks that is a 12-month window including the autumn of 2008. The measure is deliberately unconditional and deliberately pessimistic. It does not move when markets calm down. This is intended, because it keeps VaR from falling in calm markets and so limits the procyclicality of §11.6.

Stress tests drop the probability altogether and ask what named scenarios would do. Historical replays apply the factor moves of a specific day or period to today's book. Hypothetical scenarios specify moves that have not happened. The tables apply both kinds of replay to the worked account, holding today's dollar positions fixed.

Replayed day	One-day loss	Multiple of today's 95% VaR
19 October 1987	18,270	16.7
16 March 2020	12,382	11.3
15 October 2008	8,944	8.2
4 April 2025	6,591	6.0
13 September 2022	3,962	3.6

Replayed period, positions held at constant dollar size	Cumulative loss
March 2000 – October 2002	63,837
October 2007 – March 2009	57,720
19 February – 23 March 2020	41,810
Worst twenty-trading-day window since 2000	37,912

Dollars, on the $100,000 account. Multiples are of today's normal EWMA 95% VaR, $1,092 (§2.4). Periods add up daily P&L on positions rebalanced to their current dollar size each day, a simplification.

Nothing in a VaR report of 1.1% of the account says "a bad month can cost 38%". This reflects the division of labour of §1.4. It is not a criticism of VaR. VaR describes ordinary bad days, stress tests describe extraordinary ones, and a risk process needs both.

7.5 Liquidity-adjusted VaR

The question. It asks what it costs to sell the positions.

VaR values positions at mid-market prices, and it assumes that they could be sold at those prices at the end of the horizon. Two adjustments are common.

- **Exogenous spread cost.** Add the cost of crossing half the bid-ask spread on every position, $\frac{1}{2}\sum_i |x_i|\, s_i$, with s_i the proportional spread. Ideally, use a stressed spread rather than today's, because spreads widen exactly when losses breach VaR.
- **Liquidation horizon.** Measure each position's risk over the time it would take to exit without moving the price, rather than over a common horizon. The 2019 Basel rules assign risk factors horizons of 10, 20, 40, 60 or 120 days by category (Basel Committee, 2019).

For an individual in liquid funds and large stocks, neither adjustment is material. Both are material for anyone holding small caps, thinly traded options or a position that is a meaningful fraction of daily volume.

7.6 Beyond market risk

Practitioners have applied the idea of a quantile of loss far beyond trading desks.

- **Credit VaR.** The 99.9% one-year quantile of credit losses from defaults and rating migrations. Banks' own models compute it by Monte Carlo over a portfolio of loans. The Basel internal-ratings capital formula computes it in closed form from a one-factor model. Credit is the setting where VaR's failure of sub-additivity (§5.7) matters most, because credit losses are lumpy.
- **Cash-flow-at-risk and earnings-at-risk.** The versions used by corporate treasurers: the quantile of a quarter's or a year's operating cash flow, given exposures to currencies, commodities and rates.
- **CoVaR.** The VaR of the financial system *conditional on* a given institution being in distress (Adrian and Brunnermeier, 2016). Its excess over the same VaR with the institution in its normal state, called ΔCoVaR, measures how much one firm's distress raises tail risk for everyone else.
- **Growth-at-risk.** Macroeconomists' quantile forecasts of GDP growth, which apply the quantile-regression methods of VaR to economic policy.
- **Margin.** A growing number of clearing houses set initial margin as a VaR or ES of the account. They use a high confidence level and the time needed to close out the account. A retail broker's margin is usually a fixed percentage or a scenario grid (§9.4).

7.7 Drawdown measures

The question. It asks how far below its previous peak the account could fall.

VaR is a one-step measure. It describes the next h days from today's value. Investors experience drawdowns, and drawdowns make investors abandon a strategy. A **drawdown** is the decline from a previous peak, which accumulates over many steps. Chekhlov, Uryasev and Zabarankin (2005) define **conditional drawdown at risk**, the average of the worst p of drawdowns. It has the same formal properties as ES and the same linear-programming optimisation. §13.5 gives the rule of thumb that connects drawdowns to VaR. On US equities, a year's worst drawdown has typically been about nine times that year's average one-day 95% VaR.

§7 Key takeaways

1. Expected shortfall answers "how bad, given bad", and it is coherent and optimisable. It is noisier to estimate and harder to rank. Compute both, and watch their ratio.
2. Component VaR turns the total into a risk budget. Its value lies in what it reveals that weights do not. On the worked account, a 15% energy position carried 5% of the risk, and a 10% autos position carried 21%.
3. "Relative VaR" means the risk of trailing a benchmark to an asset manager, and a ratio to a reference portfolio's VaR to a fund regulator.
4. Stressed VaR and stress tests cover what ordinary VaR ignores by design. Replaying 1987 on the worked account cost 16.7 times its current normal EWMA 95% VaR in one day. A bad month since 2000 cost 38% of the account.
5. Liquidity adjustments matter for anything that cannot be sold in size at the quoted price, and they matter most when VaR is breached.
6. The idea of a quantile of loss applies to credit capital, corporate treasury, systemic risk, macroeconomic forecasting and margin.
7. Drawdown is the path-dependent risk that investors actually experience, and VaR does not describe it directly.

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8. Taxonomy and equivalences

8.1 The master form, with slots

Every number in §6 and §7 is an instance of

$$\rho \;=\; \mathcal{Q}_{\alpha}\Big[\, L(x,\, \Delta f)\Big], \qquad \Delta f \sim \widehat{F}_{t,h}\big(\,\text{scale}_t,\ \text{shape}\,;\ \text{data}\big).$$

The formula applies a statistic $\mathcal{Q}_{\alpha}$, at level α, to the loss L that today's positions x would suffer under factor moves Δf over the horizon h. The moves come from a distribution forecast at time t. A scale model, a shape model and a data window build that forecast. The table lists the seven slots.

Slot	What it decides	Main choices
Statistic $\mathcal{Q}$	Which summary of the loss distribution	Quantile (VaR), tail mean (ES), probability of a given loss (Roy), worst case over a scenario set (stress test, margin grids)
Level α	How far into the tail	95%, 97.5%, 99%, 99.9%
Horizon h	Over what period	1 day, 10 days, 20 days, 1 year; $\sqrt{h}$ scaling or direct
Exposure and revaluation $L(\cdot)$	How positions become P&L	Linear sensitivities, delta-gamma, full repricing; risk-factor mapping
Scale model	How big tomorrow's moves are	Equal-weighted window, EWMA, GARCH, implied volatility
Shape model	What tomorrow's moves look like, scale removed	Normal, Student t, Cornish–Fisher, empirical, generalised Pareto, none (CAViaR)
Data window and weighting	Which past is relevant	Last n days, age weights, a stress period, a hypothetical scenario

8.2 The equivalences

Several things that look different are the same, and several things with the same name are different. The table marks each identity as exact, approximate or heuristic.

Statement	Status	Where
Delta-normal VaR $= z_{\alpha}\times$ portfolio volatility in dollars	Exact under normality, mean dropped	§5.3
A one-day 95% VaR limit of B of equity is a volatility target of $B/1.645$ a day, or $9.65B$ a year	Exact for the daily figure in any location-scale model with the mean dropped, with its own k_{α} in place of 1.645. The annual figure also assumes independent days	§13.3
EWMA volatility is GARCH(1,1) with no constant and persistence one	Exact	§6.3
EWMA with decay λ behaves like an equal-weighted window of $(1+\lambda)/(1-\lambda)$ days	Approximate: matches the variance of the estimate	§6.3
Historical simulation is filtered historical simulation with constant volatility	Exact	§6.8
Historical simulation is age-weighted historical simulation with decay one	Exact	§6.7
Normal ES at 97.5% equals normal VaR at 99%	Approximate: 2.338σ against 2.326σ	§5.6
ES is the average of VaR over all levels beyond α	Exact, and the general definition	§5.6

Statement	Status	Where
VaR and ES are the minimiser and minimum of $c + E[(L-c)^+]/(1-\alpha)$	Exact	§5.6
A position's component ES is its average loss on the portfolio's tail days	Exact for historical and simulated samples	§5.5
A position's share of delta-normal VaR is its share of portfolio variance: weight times beta to the portfolio	Exact	§5.5
VaR is the minimiser of expected pinball loss	Exact	§5.9
Tracking-error VaR $= z_\alpha \times$ tracking error $\times$ value	Exact under normality, mean dropped	§7.3
Ten-day VaR $= \sqrt{10} \times$ one-day VaR	Heuristic: right on average, wrong in each state	§5.4, §11.5
A fixed-fraction stop loss caps the loss at the stop	Heuristic: gaps and halts let losses exceed the stop	§13.3

The second row shows that two communities do the same thing under different names. A trader who sizes positions to a one-day 95% VaR of 1% of the account runs a **volatility target** of about 9.65% a year. A quantitative fund running a 10% volatility target has a one-day 95% VaR budget of about 1.04%. Each community can keep its own terms. Using both does not diversify across two methods, because they are one method.

8.3 Which slot actually matters

For a linear portfolio at one day, the empirical ranking of the slots by importance is as follows.

1. **The scale model.** Volatility moved elevenfold between February 2007 and October 2008, and again between January and March 2020. Moving from an equal-weighted window to EWMA halved the breach clustering in §6.13.
2. **The positions and the data.** A VaR computed on the wrong positions, on stale prices or with a missing account has an error that no model can repair. [Practice] In institutional practice, this is the most common source of large VaR errors. For an individual with several accounts, it is the most likely source.
3. **The level and horizon conventions.** These change the number by factors of 1.4 (95% to 99%) and 3.2 (one day to 10). The analyst chooses them and does not estimate them. Their effect therefore concerns consistency, not accuracy.
4. **The shape model.** It changes the number by about 15% at 95% and by up to 25% at 99% (§2.2). It decides the result only at 99% and beyond.
5. **Everything else** covers interpolation conventions, the exact decay, and the window length within reason. Each changes the number by a few percent at 95%, inside the sampling error of §5.8. Interpolation can change a 99% historical VaR by up to 10% (§5.1).

For option books, revaluation ranks second. For the out-of-the-money puts of §6.10, the linear VaR was under a quarter of the full-revaluation VaR once implied volatility could move.

8.4 Same name, different thing

Several VaR terms carry more than one meaning. The table separates them.

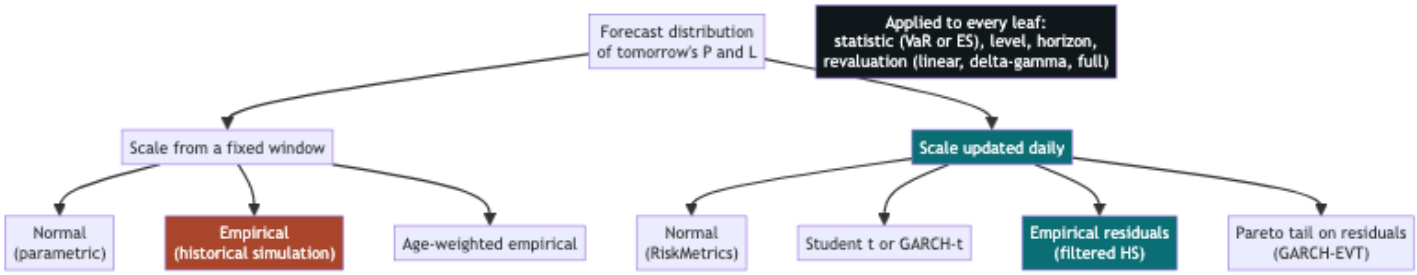
Term	Meaning 1	Meaning 2	Meaning 3
Conditional VaR	VaR given today's information, as opposed to unconditional (§2.4)	Expected shortfall (Rockafellar and Uryasev's CVaR)	—
Absolute VaR	VaR measured from zero rather than from the expected value (§5.2)	VaR as a percentage of net assets, as opposed to relative to a reference portfolio (§9.2)	—
Relative VaR	VaR measured from the expected value rather than from zero (§5.2)	VaR of the active return against a benchmark (§7.3)	Ratio of a fund's VaR to a reference portfolio's VaR (§9.2)

Term	Meaning 1	Meaning 2	Meaning 3
Incremental VaR	The exact change from adding a position (§5.5)	Loosely, marginal VaR times the position, i.e. component VaR	—
Historical VaR	Historical simulation (§6.6)	Any VaR estimated from historical data, including parametric	—
Stressed VaR	VaR on a stress-period window (§7.4)	Loosely, any stress-test loss	—

The first row causes the most confusion. When a paper or a vendor says “CVaR”, check which meaning the author intends.

8.5 The design space

The scale and shape families form the tree in the diagram. Level, horizon, statistic and revaluation apply to every leaf, so the diagram draws them separately.



The teal path is the recommendation of §6.13. The rust leaf is the method most banks use.

§8 Key takeaways

1. Every number in the VaR family fills seven slots: statistic, level, horizon, revaluation, scale model, shape model and data window.
2. The scale model matters most. For option books, revaluation comes next. The shape model matters only at 99% and beyond.
3. A one-day 95% VaR budget and an annual volatility target are the same rule in different units: 1% of equity is a 9.65% volatility target.
4. Historical simulation, age-weighted historical simulation and filtered historical simulation are one method with different weightings and rescalings. EWMA is a special case of GARCH.
5. Normal ES at 97.5% matches normal VaR at 99%, which is why the Basel Committee chose that level.
6. “Conditional VaR” and “relative VaR” each have more than one meaning in common use. Ask which one the author means.

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9. How practitioners use it

[Practice] Regulation and disclosure document much of the institutional practice in this section, but the internal details are private. Where the text describes what firms do rather than what rules require, read it as informed inference.

9.1 Banks: limits, capital and the daily report

A bank trading operation uses VaR in four ways. The list runs roughly in order of how much the bank relies on the number.

Limits. The board approves a firm-wide VaR appetite. The bank divides it among businesses and then among desks, so limits run from the top down. Each desk reports its **limit utilisation**, its current VaR as a percentage of its limit, every day. A breach of limit triggers an escalation. The desk must reduce the position, obtain a temporary increase, or explain. VaR limits sit alongside **sensitivity limits**, on interest-rate duration, option vega, credit spread exposure and the like, and **stop-loss limits** on cumulative P&L. On the usual understanding of practice, the sensitivity limits do most of the day-to-day constraining at the desk level, and VaR does the aggregation. Of the three, VaR is the only one that yields a single measure of the firm's risk across desks. It does not do so by adding up the desks' numbers (§10.3).

The daily report. The daily report is the later form of the 4:15 report. It contains the firm's VaR, VaR by business and risk factor, the change since yesterday, and the attribution of that change to position changes and market changes. It also contains the largest contributors, the limit utilisation, and yesterday's P&L against yesterday's VaR. Risk managers read the *change* more than the level. A VaR that rose 20% overnight with no change in positions shows that the market became more volatile. A VaR that rose with no change in markets shows that somebody traded.

Capital. Under the 1996 rules, the market-risk capital charge is

$$\max\left(\text{VaR}_{t-1},\; m_c \cdot \tfrac{1}{60}\textstyle\sum_{i=1}^{60}\text{VaR}_{t-i}\right),$$

with VaR at 99% over 10 days, and the multiplier m_c at least 3. The backtesting traffic light raises the multiplier toward 4 (§12.4). Basel 2.5 added the same expression for stressed VaR. The 2019 framework replaces both with a stressed expected shortfall at 97.5% with liquidity horizons. It keeps daily one-day VaR at 99% and 97.5% as the backtest, now desk by desk. It adds a test of whether the risk model's P&L tracks the desk's hypothetical P&L (§10.2).

Disclosure. Public banks report VaR quarterly: average, high, low and period-end, by risk category, with a diversification line (§10.3). Many also publish their backtesting chart (§10.2) and their count of exceptions.

9.2 Asset managers and funds

Tracking-error budgets. A benchmarked equity or bond manager works within an ex-ante tracking-error budget, such as "no more than 3% annualised active risk". That is a relative VaR (§7.3) in volatility units. A component analysis of the active risk tells the manager which bets use up the budget.

European funds (UCITS). A fund that uses derivatives in a sophisticated way must compute its global exposure by VaR (CESR, 2010). The VaR uses 99% confidence, a 20-day horizon and at least a year of data. The **absolute VaR** may not exceed 20% of net asset value. The **relative VaR** may not exceed twice the VaR of an unleveraged reference portfolio.

US funds (Rule 18f-4). Since 2022, a US registered fund with more than limited use of derivatives must run a programme of derivatives risk management and a VaR test (Securities and Exchange Commission, 2020). The test uses VaR at 99% over 20 trading days, from at least three years of data. The VaR may be no more than 200% of a designated reference portfolio's VaR, or 20% of net assets where no suitable reference exists. The fund must also backtest it regularly. The rule replaced decades of rules based on asset segregation. It is the clearest case of VaR chosen as a regulatory tool. The reasons are that VaR is simple and auditable, and that regulators can apply it uniformly to very different funds.

Pension and insurance. Asset-liability managers use **surplus at risk**: the VaR of assets minus liabilities over a year. Under Solvency II, European insurers hold capital equal to a 99.5% one-year VaR of their own funds, about a one-in-200-year loss.

9.3 Hedge funds and proprietary trading

[Practice] Hedge funds use VaR in three places. Internally, the fund often sets a portfolio manager's **risk budget** in volatility or VaR terms rather than in capital. A manager running low-volatility relative-value trades then gets more capital than one running directional equity. For investors, monthly letters commonly report VaR alongside exposure and leverage. For lenders, prime brokers set margin and financing terms from risk models of the fund's book.

In the 2010s, multi-manager platforms came to dominate hedge fund assets. These platforms pair a VaR or volatility budget with hard **drawdown rules**. The platform cuts the allocated capital of a manager whose cumulative loss reaches a set fraction of it. At a larger fraction, the platform stops the manager out. Public reports put the thresholds in the single digits of percent. This combination matters for §13. VaR is a forward-looking limit on ordinary risk, and drawdown is a backward-looking limit on realised damage. The platforms rely on neither alone.

9.4 Clearing houses and brokers: margin

Initial margin is collateral posted against the risk that a counterparty defaults and its positions must be closed out at a loss. Setting it is a VaR problem by construction. The margin should cover the loss over the close-out period with high probability.

[Practice] For decades, fixed scenario grids set the margin on exchange-traded derivatives. CME's SPAN system, introduced in 1988, revalues each portfolio under a standard set of price and volatility moves and charges the worst loss. Several large clearing houses now compute initial margin as a filtered historical-simulation VaR or ES over a multi-day close-out horizon, at 99% or higher. They often add a floor based on a stress period, against procyclicality. For uncleared derivatives between dealers, the industry's standard initial margin model (ISDA's SIMM) is a calculation based on sensitivities and calibrated to a 99% 10-day loss.

For an individual, the relevant fact is that **a broker's margin requirement is somebody else's risk model of the account**. A US Regulation T margin account applies fixed percentages. A portfolio-margin account revalues the positions under a grid of price moves and charges the worst. The moves are about ±15% for an individual stock, and narrower for broad index products. Neither margin rule reflects the account holder's risk tolerance. The second can expand sharply on the day the market falls, because the grid revalues options at higher volatility. An investor who knows the account's own VaR and stress losses does not need to find out the broker's figure when the broker acts on it.

9.5 Corporates and the public record

Non-financial companies with material market exposure report it in their annual filings. In the US the formats are tabular exposure, sensitivity to a hypothetical move, or VaR. Treasury functions use cash-flow-at-risk (§7.6) to set hedging policy. [Practice] The VaR format is uncommon outside financial companies. Sensitivities are easier to explain to shareholders whose concern is a move in one specific price.

9.6 What VaR is actually good for

The table gives this chapter's assessment, drawing on §3, §11 and §12.

VaR is good at	VaR is bad at
Putting different instruments and desks on one scale	Saying how much could be lost in a crisis
Limiting ordinary, day-to-day risk	Measuring the risk of anything that sells tail insurance
Detecting change: "risk went up 20% overnight — why?"	Anything at a horizon longer than the time to unwind
Being checked: exceedances are countable	Setting capital against rare events: too little data
Decomposing risk into contributions	Governing a firm as a single number to be managed
Communicating risk to non-specialists, with care	Being trusted when the model's inputs are someone's incentive

The left column explains why the number has survived 30 years of deserved criticism. The right column explains why it has never been enough. The institutions that used VaR well treated it as one measure among several. They read it alongside stress tests, sensitivities, and measures of concentration and liquidity, and they kept looking at the backtest. The institutions that used it badly relied on it alone.

§9 Key takeaways

1. In banks VaR aggregates, while sensitivity and stop-loss limits do most of the day-to-day constraining. Risk managers read the overnight *change* in VaR more than its level.
2. Since 1996, bank capital has been three to four times a 99% 10-day VaR. It is now moving to stressed 97.5% ES, with VaR kept as the backtest.

3. Fund regulation in Europe and the US caps the 99% 20-day VaR at 20% of assets, or at twice a reference portfolio's VaR. Regulators chose VaR for its simplicity and auditability.
4. Hedge-fund platforms pair a VaR or volatility budget with hard drawdown stops. The platforms trust neither alone.
5. Margin is someone else's risk model of the account. A portfolio-margin grid can expand sharply on the day markets fall.
6. VaR works well for aggregation, limits on ordinary risk, detecting change, decomposition and backtesting. It works poorly for crises, books that sell tail risk, long horizons, and use as the only number.

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10. How VaR is presented

A VaR system produces a handful of standard displays. Each answers a different question. Knowing which display answers which question is most of what a reader needs to read a risk report. All the examples below use real data. They use either the US market or the worked $100,000 account of five sector funds. The account holds each fund at its end-2025 dollar size: technology 40,000, health care 20,000, energy 15,000, utilities 15,000, and autos and durables 10,000.

10.1 The distribution, with the line drawn on it

Shows how a bad day compares with a normal day.

The standard display is the histogram of §2.3. It shows P&L outcomes, historical or simulated, with the VaR drawn as a vertical line and the tail beyond it shaded. A well-made version also marks the ES and the worst outcome. It also overlays the normal curve with the same standard deviation, so the reader can see how far the tails depart from it. The version without those three additions is the most common display in risk reporting. It also misleads most often, because it draws the reader's attention to the line and away from the tail.

10.2 The backtest chart

Shows whether the model's forecasts match what happened.

The single most informative VaR display plots the daily P&L against the previous day's VaR forecast, and marks the days on which the loss exceeded it. Banks publish this chart in their regulatory disclosures. Anyone running a VaR should draw it before trusting the number. The figure shows the US market through 2007–2009, under three methods.



Bars: daily return of the US equity market. Line: minus the 99% one-day VaR forecast made the evening before. Dots: days the loss exceeded the forecast. Data from the Kenneth R. French data library.

Look for four things, in order.

1. **The count.** The expected number of exceedances is pT, which is 7.6 here. Historical simulation had 24, RiskMetrics 18 and filtered historical simulation 13.
2. **The clustering.** Historical simulation's breaches cluster in September and October 2008, while its VaR line climbs in steps and lags the market. Bunching shows that the scale estimate is out of date.
3. **The shape of the line.** A step function that stays flat for a year after the crisis (top panel) is the stale-spike effect of historical simulation. A line that jumps and then decays (lower panels) shows a volatility model responding to new data. If the line barely moves while the P&L bars swing widely, the model does not use current data.
4. **The size of the breaches.** On the full backtest run for this chapter, the average breach of a 99% VaR was 1.33–1.43 times the VaR, depending on the method. A model whose breaches are routinely twice the VaR has a tail problem, not only a scale problem.

Which P&L. Banks distinguish **hypothetical** P&L from **actual** P&L. Hypothetical P&L revalues today's positions at tomorrow's prices, which is what the VaR forecasts. Actual P&L also reflects intraday trading and new positions. Backtest against

hypothetical P&L to test the model, and against actual P&L to test whether the model describes the business. An individual who rarely trades intraday can use actual P&L for both.

10.3 The disclosure table

Shows how much risk the book ran over the period, and of what kind.

Every large bank's annual report contains a table in this format. The rows are risk categories, and the columns summarise the daily VaR over the period. The table below shows the format for the worked account over 2025. It uses sectors in place of risk categories and the one-day 95% EWMA VaR. The positions stay at their year-end sizes throughout.

One-day 95% VaR, 2025 (dollars)	Average	High	Low	31 Dec
Technology	1,017	2,719	519	722
Health care	360	700	230	271
Energy	363	920	243	284
Utilities	238	474	150	162
Autos and durables	523	1,030	324	350
Diversification	−630	†	†	−700
Total	**1,871**	**5,474**	**918**	**1,089**

† Not meaningful: each category's high and low fell on a different day. The last column is the VaR forecast for 31 December, made with data to 30 December. §10.4 shows the next day's forecast.

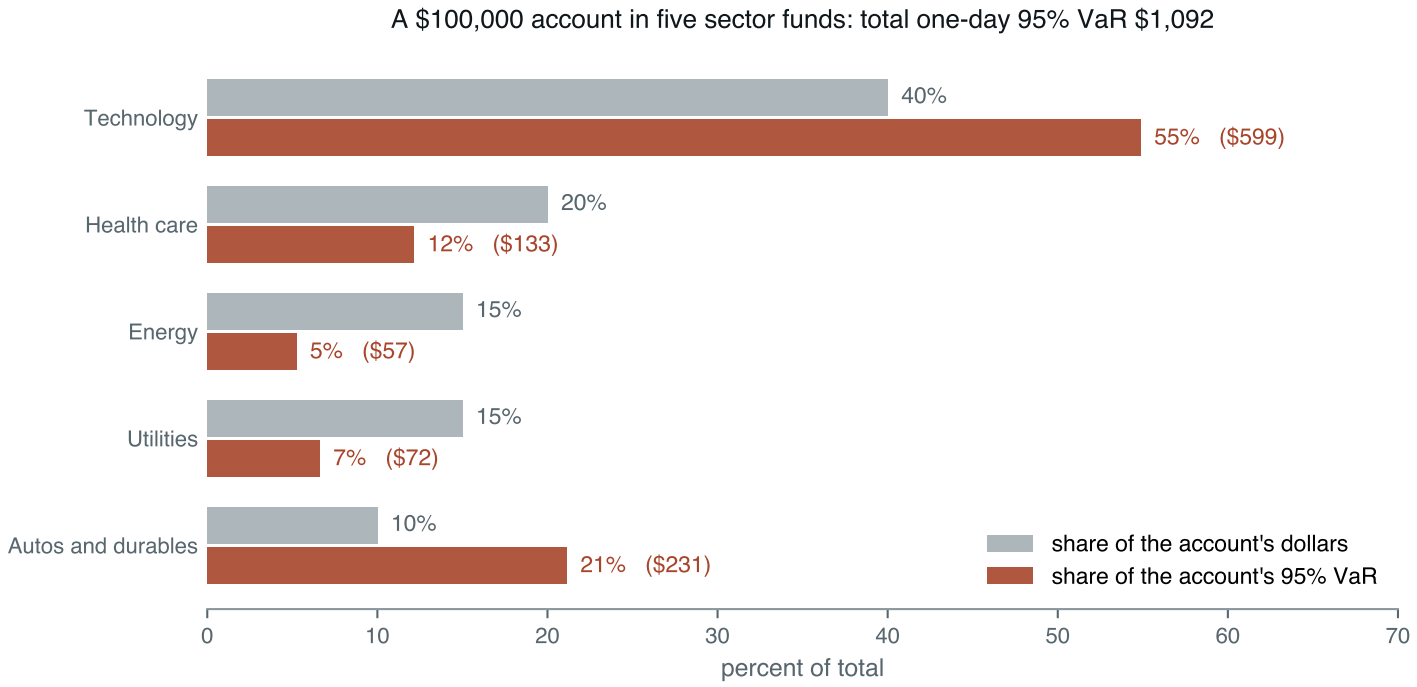
Read the table with four points in mind.

- **The category rows are stand-alone VaRs.** Each row treats the other positions as absent. The rows sum to more than the total.
- **The diversification line is a plug**: the total minus the sum of the categories. It is negative because the positions partly offset one another. Its size measures how diversified the book is, and here it averages a quarter of the stand-alone sum. A diversification benefit that shrinks suddenly means that correlations rose.
- **Highs and lows do not add across rows**, because they happened on different days. The total's high, $5,474, came on 11 April 2025, days after the tariff announcement. Its low, $918, came on 25 September. The ratio of six between them, for a portfolio that never traded, illustrates Idea 2 again. The volatility moved and the positions did not.
- **Only the count of breaches shows whether the model was right.** Losses exceeded this VaR on 14 of 250 days in 2025, against an expected 12.5.

10.4 The decomposition

Shows where the risk is, and whether it matches where the money is.

The decomposition display shows component VaR (§5.5) by position, desk or risk factor, usually beside each position's share of capital.



Component VaR from an EWMA (0.94) covariance matrix as of the end of 2025. Value-weighted US industry portfolios from the Kenneth R. French data library stand in for sector funds.

The table gives the full decomposition at the end of 2025, from the minimal implementation of §14.

Position	Dollars	Daily volatility	Stand-alone 95% VaR	Component VaR (share)	Average loss on the worst 5% of days (share)
Technology	40,000	1.08%	711	599 (55%)	1,299 (51%)
Health care	20,000	0.81%	266	133 (12%)	242 (10%)
Energy	15,000	1.12%	277	57 (5%)	320 (13%)
Utilities	15,000	0.65%	161	72 (7%)	196 (8%)
Autos and durables	10,000	2.08%	342	231 (21%)	475 (19%)
Total	**100,000**		**1,758**	**1,092**	**2,532**

Component VaR comes from the EWMA covariance. Tail losses come from the last 500 days replayed on current positions, and their total is the 95% historical ES.

The table has two readings. First, technology is the **hot spot**. It carries 55% of the risk on 40% of the capital. Autos and durables follow, at twice their capital share. That position looks small in a brokerage statement and is the second-largest risk in the account. Second, the two columns of shares come from different sources. One comes from a covariance model, and the other averages losses on the bad days. They agree to within a few points for four of the five positions. Energy is the exception, at 5% of the covariance VaR and 13% of the tail losses. §10.6 shows the same difference in the crash-day replays. When the two columns disagree, the covariance model misses something about how the positions behave in the tail.

10.5 Over time, against a limit

Shows how close the book is to the limit, and why it moved.

This display is a time series of VaR with the limit drawn on it. It usually shows **utilisation**, which is VaR divided by the limit. The informative version attributes each day's change in VaR to a *position effect* and a *market effect*. The position effect is VaR at today's positions and yesterday's market, minus yesterday's VaR. The market effect is the remainder. For the worked account over 2025, every movement in the total of §10.3 was market effect, because the positions did not change. Take a limit of 1,250, the 1.25% budget of §14.4. Utilisation would have run from 73% in September to 438% in April without a single trade. A VaR limit therefore needs a rule for what happens when the market, not the trader, breaches it (§13.3).

10.6 The scenario matrix

Shows what specific bad days would do, and to which positions.

Stress results usually appear as a matrix. Positions or risk factors form the rows, scenarios form the columns, and the cells hold P&L, often shaded as a heat map. The table gives one-day replays for the worked account, in dollars.

Position	19 Oct 1987	15 Oct 2008	16 Mar 2020	4 Apr 2025	13 Sep 2022
Technology	−7,988	−3,196	−5,256	−2,392	−2,152
Health care	−3,578	−1,182	−1,924	−1,146	−656
Energy	−2,978	−2,293	−2,005	−1,286	−342
Utilities	−1,929	−1,338	−1,748	−926	−412
Autos and durables	−1,798	−934	−1,449	−842	−399
Total	**−18,270**	**−8,944**	**−12,382**	**−6,591**	**−3,962**

Compare the columns with the decomposition. In the replays every position loses. Energy carried 5% of the VaR, yet it lost 2,000–3,000 on three of the five days. The VaR reflects energy's low correlation with the rest of the account in 2025. In a crash, measured correlations rise toward one, and the scenario matrix captures that. The rise is partly a real change in how assets move together. It is partly selection, because the sample contains only violent days (A.32). **The decomposition shows the risk on an ordinary bad day, and the scenario matrix shows it on an extraordinary one. The two rank the positions differently.**

10.7 A one-page report for one account

A single account needs very little. The listing gives the whole report for the worked account on the first trading day of 2026, as the code of §14 produces it.

```
ACCOUNT RISK, 2 Jan 2026                                    value 100,000
------------------------------------------------------------------------
one-day volatility (EWMA 0.94)              664      0.66%
95% VaR, filtered historical simulation    1,116     1.12%   budget 1,250
99% VaR, filtered historical simulation    1,991     1.99%
97.5% expected shortfall                   2,080     2.08%
------------------------------------------------------------------------
backtest, last 500 days    95%: 25 breaches (expect 25)
                           99%: 10 breaches (expect 5)   <- check the tail
------------------------------------------------------------------------
contributors to 95% VaR    technology 55%   autos 21%   health care 12%
                           technology above the 40% cap  <- decide
------------------------------------------------------------------------
replays                    19 Oct 1987  -18,270   16 Mar 2020  -12,382
                           worst 20 days since 2000            -37,912
```

The report holds 12 numbers. A human adds the budget and the two arrows, using the standing rules of §14.4. The budget says whether to act on the level. At 89% utilisation, the answer is no. The contributors' arrow flags that technology carries more than the 40% of the VaR that the rules allow one position. That calls for a decision, written down either way. The backtest arrow flags that losses exceeded the 99% VaR twice as often as expected over the last two years. The 10 breaches spread from January 2024 to October 2025, and two came on consecutive days in April 2025. This is a reason to look before trusting the 99% figure. Ten breaches against five expected sit exactly at the edge of statistical significance for 500 days (§12.5). Over the last 2,500 days, the same method had 27 breaches against 25.

§10 Key takeaways

1. The histogram shows where the VaR line is. Add the ES, the worst day and a normal overlay, or it misleads.
2. The backtest chart is the most important display. Count the breaches, look for bunching, check whether the VaR line moves with the market, and see how far breaches exceed the VaR.
3. In a disclosure table, the category rows are stand-alone, the diversification row is a plug, and highs and lows do not add. On the worked account, the total's high was six times its low in a year without a trade.

4. The decomposition finds hot spots: a 10% position carried 21% of the risk. Compare the covariance-based and tail-based shares. Disagreement means that the tail behaves differently from the average.
5. Utilisation of a fixed VaR limit swung from 73% to 438% with no trades. Decide in advance what a breach driven by the market means.
6. The scenario matrix ranks positions differently from the decomposition, because correlations rise in a crash. Read both.
7. A single account needs about a dozen numbers on one page.

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11. Limitations and failure modes

Earlier sections refer ahead to this section. Each limitation comes with its mechanism and, where possible, a measurement.

11.1 It does not look past its own line

This is the structural limitation, and several others follow from it. VaR is a quantile, and no event beyond the quantile changes it (§2.3). It has two consequences.

It misses the size of the disaster. On the full backtest of §6.13, a breach of a 99% VaR went on average 33–43% beyond the VaR, depending on the method. That is tolerable. The average, however, does not show the cases that matter. The worst day in the sample was 10 times its conditional volatility (§2.5), several times any 99% VaR.

Traders can exploit it. Some strategies earn a small, steady income and occasionally lose a great deal. Examples are selling out-of-the-money options, selling credit protection, carry trades in pegged or managed currencies, and short volatility products. A trader can build such a strategy to have a low or even negative VaR at any chosen level, as long as the probability of disaster is below p. Its ordinary days look excellent and its VaR looks small. That profile attracts exactly the trader with a VaR limit or the investor who screens on VaR. Basak and Shapiro (2001) prove the general version, in a model of a utility-maximising investor. A manager subject to a VaR constraint chooses to take *larger* losses in the worst states, beyond the VaR, than an unconstrained manager. The reason is that the constraint penalises only the probability of a large loss, not its size.

The remedy is to look past the VaR line deliberately, using expected shortfall (§7.1), stress tests (§7.4) and, for option books, the scenario rows of §6.10. A cheap early warning is the ratio of ES to VaR, which rises as the tail gets heavier. On the worked account, filtered historical simulation gave a ratio of 1.045 (ES at 97.5% against VaR at 99%), where a normal distribution gives 1.005 (§5.6). For a book that sells the tail, the ratio has no upper bound. Position B of §2.3 has a 95% VaR that is a gain, and a 95% ES of 11,920.

11.2 It can penalise diversification

VaR is not sub-additive (§5.7). This does practical damage in two ways. First, traders can game limits set desk by desk by spreading lumpy risks across desks. Each desk's chance of a large loss stays within its own p, so its VaR does not register the risk, even though the firm's combined chance is above p. Second, a VaR-based allocation can make a diversified portfolio look riskier than its concentrated parts. In the two-bond example of §5.7, each bond alone has a 95% VaR of zero, and the pair has a VaR of 100. The failure needs lumpy payoffs or extremely heavy tails. It is therefore rare in portfolios of liquid stocks and funds, and common in credit and short-option books (§5.7).

11.3 It is an estimate, with a large error

Every VaR number carries two kinds of error.

Sampling error. As §5.8 showed, a 99% historical VaR from one year of data has an uncertainty of ±28%, even with every assumption satisfied. That range contains the central 90% of estimates. The 95% figure has an uncertainty of ±18%.

Model error. Different reasonable methods give different numbers for the same portfolio on the same day. Beder (1995) found differences of up to 14 times across methods and parameter choices on simple portfolios. Marshall and Siegel (1997) found that vendors implementing the *same* methodology on the same portfolio returned materially different numbers. On the worked account at the end of 2025, three methods gave different one-day 99% VaRs. The normal EWMA method gave $1,545, filtered historical simulation gave $1,991, and plain historical simulation over two years gave $2,849. That is a range of 1.8 to 1, from three methods that a textbook would each call standard.

Responsible practice reports the number with its uncertainty, or at least reports it from two methods. It treats differences smaller than the sampling error as noise. Almost nobody does either.

11.4 It is accurate on average and wrong when it matters

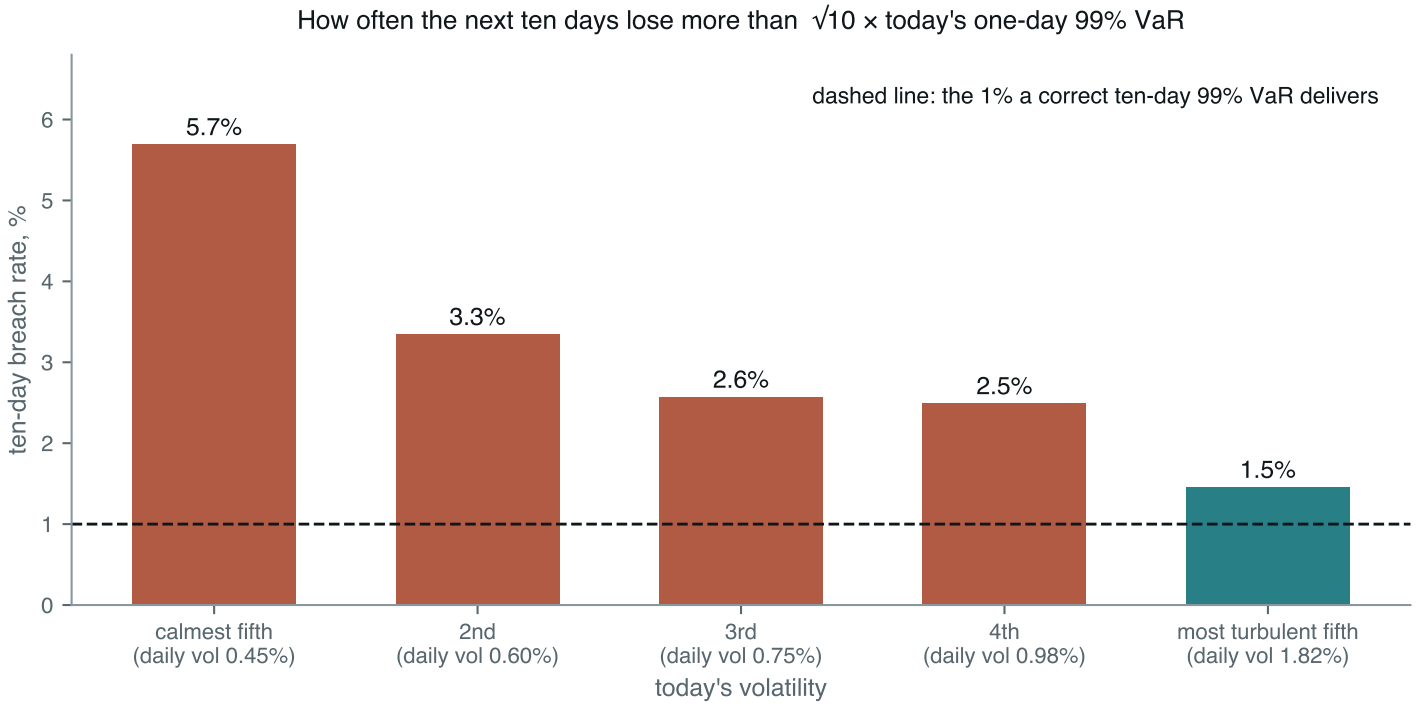
A VaR model can pass a long backtest and still fail exactly when it is needed. Its breaches bunch in the periods of rising volatility and do not spread evenly. On the full backtest of §6.13, every one of the seven methods failed the independence test of §12.3. In 2020, the year of the pandemic crash, losses exceeded the RiskMetrics 99% VaR 12 times, where 2.5 breaches were expected. Even filtered historical simulation, the best method on the full sample, had four breaches. In 2017, a calm year, the seven methods had between zero and four breaches at 99%, against 2.5 expected.

The mechanism is that volatility rises in jumps, while every model estimates it from past data. The first days of a shock are breaches for every model. A model with a slow volatility estimate keeps producing breaches for weeks. The figure in §10.2 shows the pattern for 2007–2009. [Fact] Berkowitz and O'Brien (2002) and O'Brien and Szerszeń (2017) found the same clustering in banks' own VaR models, in 1998 and in 2007–2009 respectively. Before the crisis the models were conservative. During it they had excessive and clustered exceedances.

A VaR is most likely to be wrong on exactly the days when the user most needs to rely on it.

11.5 The horizon is usually wrong

Practitioners compute most VaR at one day and scale it to longer horizons by $\sqrt{h}$ (§5.4). Averaged across a century of US data, the rule gets the level about right. The unconditional 10-day 99% loss quantile was 10.07%, against a scaled one-day figure of 9.68%. The rule gets the level wrong in each state, however, and the error has a predictable direction, as the figure shows.



US equity market, overlapping ten-day windows, 1929–2025; one-day VaR is 2.33 × EWMA (0.94) volatility.
Data from the Kenneth R. French data library.

Consider a 10-day 99% VaR built as $\sqrt{10}$ times the RiskMetrics one-day figure. Starting from the calmest fifth of days, losses exceeded it 5.7% of the time, nearly six times its stated rate. Starting from the most turbulent fifth, losses exceeded it 1.5% of the time. Two effects explain the pattern. First, volatility mean-reverts. From a calm start the next 10 days are likely to be more volatile than today, and from a turbulent start less volatile. Second, the normal tail is too thin, which by itself produced a one-day breach rate of 2.1% (§6.3). From calm starts the two effects add. From turbulent ones they offset, but not fully, so even there the rate stays above 1%.

Scaled VaR errs in the dangerous direction. It is lowest, and most wrong, in exactly the calm periods when adding leverage is most tempting. [Practice] **Recommendation: for any horizon longer than a day, compute the multi-day risk directly rather than scaling.** Use a mean-reverting volatility forecast, or historical or filtered-historical returns over the multi-day horizon.

11.6 It assumes the holder can get out, and that nobody else is trying to

VaR treats the market as exogenous. In this model, prices move, the portfolio's value follows, and the holder's own actions affect nothing. For a small account this holds. For the financial system it does not.

Liquidity. VaR values positions at mid-prices over a fixed horizon. A position that the holder cannot sell within that horizon without moving the price carries more risk than the VaR reports (§7.5). LTCM is the standard case. Jorion (2000) shows that its positions were large relative to their markets, so reducing them itself caused the loss.

Procyclicality. An institution that holds its VaR constant must cut its positions when volatility rises. An EWMA estimate of US market volatility, computed for this chapter, rose from 0.43% a day in February 2007 to 4.80% in October 2008, a factor of 11.2. In 2020 it rose by the same factor in two months, from 0.47% in January to 5.27% in March. VaR is proportional to volatility times position (§1.6). A book run at constant VaR would therefore have had to cut its equity position to 1/11.2, or 9%, of its starting size. In both cases the cut would have come after much of the fall had already happened, and in 2020 within days of the bottom. [Fact] Adrian and Shin (2014) show that dealer leverage does behave this way. It rises in booms and falls in busts, consistent with VaR-type management.

Endogenous risk. When many institutions use similar models, they respond to a rise in VaR in the same way, by selling the same assets at the same time. The selling raises volatility further. Higher volatility raises VaR, which forces more selling. [Hypothesis] Before the crisis, Danielsson (2002) and Danielsson and co-authors (2001) argued that VaR-based regulation would amplify crises through this mechanism. The dynamics of 2008 were consistent with it, though one crisis cannot separate cause from coincidence.

For an individual, the systemic channel does not apply, because the individual's selling moves nothing. The procyclicality does apply. A VaR budget mechanically reduces exposure after volatility rises, which usually means after losses. [Contested] Researchers dispute whether that helps. Moreira and Muir (2017) find that scaling exposure inversely to recent variance raised the Sharpe ratio of the market and of many factor portfolios. Returns have not risen proportionally after volatility spikes. Cederburg and co-authors (2020) find that the gains are fragile out of sample for many of the strategies. For the broad market, the view taken here is that de-risking when volatility rises has not hurt on average and has cut the worst drawdowns. That is all a risk rule needs to do.

11.7 It can be managed instead of the risk

Once an institution uses a number to set limits, capital and bonuses, the number becomes a target. Staff can then meet the incentive to reduce the number without reducing the risk. Documented cases show three routes.

- **Change the model.** In the London Whale case, a new VaR model for JPMorgan's Chief Investment Office cut the reported VaR of its synthetic credit portfolio by about half. According to the Senate investigation (Permanent Subcommittee on Investigations, 2013), the VaR fell from $132 million to $66 million. The change ended a limit breach while the position kept growing. Later review found spreadsheet errors in the model.
- **Hold positions the model measures poorly.** Positions in instruments with short or calm price histories, or mapped to proxies that do not move like them, carry a low measured VaR. Highly rated structured credit before 2007 is the standard example. Its history contained no stress, so its measured risk was small.
- **Sell the tail.** As §11.1 showed, the optimal response to a VaR limit includes shifting losses past the quantile.

The defence is institutional, not statistical. It consists of independent model validation, limits on stress losses and concentrations alongside VaR, and scepticism toward any position whose VaR seems low for its expected return.

11.8 Case studies

The table summarises five episodes.

Episode	What the VaR said	What happened	The limitation
LTCM, 1998	Daily VaR estimated from a short, calm history, small relative to capital	Lost most of its capital in about five weeks	Correlation and liquidity in a stress absent from the data (§11.6). Leverage made the horizon wrong
Banks, 2007–2009	Conservative in 2006. Exceedances ran far above stated rates from mid-2007	Clustered exceedances. Large write-downs on positions the models treated as low-risk	Slow volatility estimates (§11.4). Positions the model could not measure (§11.7)
London Whale, 2012	Halved by a model change	Losses above six billion dollars	The measure was managed instead of the risk (§11.7)
Short-volatility products, February 2018	Low risk on any history that excluded such a day	The VIX index more than doubled in a day, and products that were short volatility lost most of their value	The tail was beyond the line by construction (§11.1)

Episode	What the VaR said	What happened	The limitation
Pandemic crash, March 2020	Calm-period VaR in January	RiskMetrics 99% VaR breached 12 times in 2020 against 2.5 expected	Volatility rose elevenfold in two months (§11.6). Scaled multi-day VaR was lowest just before (§11.5)

11.9 The debate, and the view taken here

The arguments have changed little since Taleb and Jorion exchanged them in 1997.

The case against comes from Taleb and Danielsson, and Nocera's interviews illustrate it. A probability model fitted to the past cannot describe the events that matter, which are rare and different each time. A precise-looking number displaces judgement. A regulatory standard makes everyone use the same flawed model, which synchronises their failures.

The case for comes from Jorion and Brown. On this view, VaR never aimed to describe crises. It describes ordinary risk, gives a common unit, and makes an explicit claim that anyone can check every day. Its failures in crises belong to the people who treated it as a worst-case measure and to the institutions that allowed traders to game it. The alternative is not a better number but no number.

The view taken here is that both sides are right about different questions. As a description of ordinary daily risk, conditional VaR from a reasonable volatility model is accurate, cheap and falsifiable. §6.13 shows that one can calibrate its breach rate to within sampling error over a century. As a measure of capital adequacy or of the possibility of ruin, it is the wrong tool. The tail beyond 99% has too little data, and models there disagree by factors rather than percentages. Errors in practice have come from using the first kind of number to answer the second kind of question. An individual investor can avoid this more easily, because no regulator or incentive scheme forces the confusion. [Practice] **Recommendation: use VaR to size and monitor the ordinary, and use stress scenarios and a written plan for the extraordinary.**

§11 Key takeaways

1. Nothing beyond its threshold changes VaR. Strategies that sell tail risk can therefore show a low or negative VaR, and a VaR constraint pushes a manager to take larger losses beyond the line. Watch the ratio of ES to VaR.
2. The failure of sub-additivity is real for credit and short options, and it rarely binds for liquid long portfolios.
3. Three standard methods put the worked account's 99% VaR between $1,545 and $2,849 on the same day. Report two methods, and ignore differences inside the sampling error.
4. Breaches cluster in exactly the periods when users rely on VaR. Every method tested here fails the independence test.
5. Square-root-of-time scaling is right on average and wrong in each state. From calm starting points, losses exceeded a scaled 10-day 99% VaR 5.7% of the time. Compute multi-day risk directly.
6. A constant-VaR book had to cut equity exposure to 9% of its starting size in both 2008 and 2020. For institutions this may amplify crises. For an individual in the broad market, de-risking as volatility rises has, on contested evidence, done little harm and cut the worst drawdowns.
7. Staff will manage a number that serves as a target. A change of model halved the London Whale's VaR overnight.
8. Use VaR for the ordinary and stress scenarios for the extraordinary. Most VaR disasters came from the first kind of number answering the second kind of question.

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12. Backtesting and evaluation

A VaR model makes a checkable claim. This section covers how to check it, and how much a check can reveal.

12.1 The hit sequence

For each day t in a backtest, compare the realised loss with the VaR forecast made the evening before:

$$I_t = \mathbf{1}\{L_t > \text{VaR}_{\alpha,t}\}.$$

Suppose the model is correct, so that $\text{VaR}_{\alpha,t}$ really is the α-quantile of the loss given everything known at $t-1$. Then each I_t is a Bernoulli variable with probability $p = 1-\alpha$, *and it is independent of everything known at* $t-1$, including all earlier hits. This is Idea 3 of §1.6. It splits into two properties, and every test checks them.

- **Unconditional coverage:** the hit rate is p.
- **Independence:** hits are unpredictable. They do not cluster, and they do not depend on the level of VaR or on anything else observable.

Christoffersen (1998) calls the combination **conditional coverage**. A model can pass the first test and fail the second. In §6.13, filtered historical simulation passes the coverage test at both levels and fails independence. Such a model is right on average and wrong when it matters.

Two technical rules apply. First, use the forecast made *before* the outcome, with no data from day t. A backtest that refits the model on the full sample and then "tests" it proves nothing. Second, backtest at a one-day horizon, even for a 10-day VaR. Overlapping 10-day windows produce hits that autocorrelate by construction, which invalidates every test below.

12.2 Unconditional coverage: Kupiec's test

Suppose there are X hits in T days. The likelihood ratio test of Kupiec (1995) compares the stated rate p with the observed rate $\hat{\pi} = X/T$:

$$\text{LR}_{\text{uc}} = -2\ln\frac{(1-p)^{T-X}\,p^{X}}{(1-\hat{\pi})^{T-X}\,\hat{\pi}^{X}} \;\sim\; \chi^2_1.$$

The distribution holds approximately, in large samples, under the null hypothesis that the stated rate p is the true one. Reject at 5% if $\text{LR}_{\text{uc}} > 3.84$, the 95th percentile of χ^2_1. An exact and more transparent alternative is the binomial probability of seeing X or more hits, or X or fewer, if the model were right. That is a different test, usually run one-sided, so its p-value need not match the likelihood ratio's.

Apply the test to the worked account over the last 2,500 days, with today's positions replayed. The normal EWMA 99% VaR had 52 hits against 25 expected, with a likelihood-ratio p-value of 2×10^{-6}, so the test rejects it. Filtered historical simulation had 27, with a p-value of 0.69, so the test does not reject it.

12.3 Independence and conditional coverage

Christoffersen (1998) models the hit sequence as a two-state Markov chain, with state 1 a hit and state 0 a day without one. He tests whether the probability of a hit today depends on whether yesterday was a hit. Let n_{ij} be the number of days in state j that follow a day in state i. The estimated transition probabilities are $\hat{\pi}_{01} = n_{01}/(n_{00}+n_{01})$, the chance of a hit after a day without one, and $\hat{\pi}_{11} = n_{11}/(n_{10}+n_{11})$, the chance of a hit after a hit. Under independence, both equal the overall hit rate $\hat{\pi}$. Here the same $T-1$ consecutive pairs of days give $\hat{\pi}$, as $(n_{01}+n_{11})/(T-1)$. This differs from X/T only through the first day. The test compares the two fits:

$$\text{LR}_{\text{ind}} = -2\ln\frac{(1-\hat{\pi})^{n_{00}+n_{10}}\,\hat{\pi}^{\,n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}}\,\hat{\pi}_{01}^{\,n_{01}}\,(1-\hat{\pi}_{11})^{n_{10}}\,\hat{\pi}_{11}^{\,n_{11}}} \;\sim\; \chi^2_1.$$

The joint test of conditional coverage is $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}} \sim \chi^2_2$, where both statistics use the same $T-1$ days.

The useful summary statistic is $\hat{\pi}_{11}$, the probability of a hit given a hit yesterday, which should be close to p. In §6.13 it was 8.8–10.3% for the equal-weighted 99% methods and 4.4–5.6% for the conditional and age-weighted ones, against 1%. The

first-order Markov test looks only one day back. **Duration tests** (Christoffersen and Pelletier, 2004) use the whole distribution of the time between hits. That distribution should be geometric with mean $1/p$, or 100 days at 99%. Duration tests are more powerful against the slow clustering of a model whose volatility estimate lags. Regression tests go further and ask whether *anything* known at $t-1$ predicts hits, such as the VaR level, recent returns or volatility. That test covers the full content of the independence property.

12.4 The Basel traffic light

The regulatory backtest (Basel Committee, 1996b) counts 99% exceptions over the last 250 days and assigns a zone. The zone sets the amount added to the capital multiplier of three (§9.1), as the table shows.

Exceptions in 250 days	Zone	Increase in the capital multiplier
0–4	Green	0
5	Yellow	0.40
6	Yellow	0.50
7	Yellow	0.65
8	Yellow	0.75
9	Yellow	0.85
10 or more	Red	1.00, and the model is presumed flawed

The probabilities of each zone show what the test can and cannot do.

True breach rate	P(green)	P(yellow)	P(red)
1% (model correct)	89.2%	10.8%	0.03%
2% (twice too many)	43.9%	53.1%	3.0%
3%	12.8%	65.1%	22.1%
4%	2.7%	42.8%	54.5%

A correct model falls in the yellow zone more than one year in 10. A model with twice as many breaches as it claims stays in green 44% of the time. The committee's own document states these error rates openly. The committee designed the yellow zone as a range in which supervisors use judgement rather than a rule.

12.5 How much a backtest can reveal

The traffic-light table illustrates a general fact: **backtests of 99% VaR have very little power on the sample sizes anyone has.** The table below quantifies it.

Test	Sample	Reject at 5% if hits ≥	Power against a true rate twice the stated one
99% VaR	250 days	6	38%
99% VaR	500 days	10	54%
99% VaR	1,000 days	16	85%
99% VaR	2,500 days	34	99%
95% VaR	250 days	19	92%
95% VaR	500 days	34	99.5%

One-sided binomial tests, exact.

One year of data detects a doubled breach rate at 95% nine times out of 10, and at 99% fewer than four times in 10. Kupiec (1995) made the point 30 years ago. This is the strongest practical argument for backtesting at 95%, even when the number of interest is at 99%. The 95% test can show within a year whether the model's scale is right. The 99% test needs four years. If the 95% VaR matches its stated breach rate and the 99% VaR does not, the problem is the shape of the tail. If neither matches, the problem is the volatility.

Equally, a model that produces too *few* hits is failing. Zero exceedances of a 95% VaR in a year mean that the model overstates risk by a wide margin. For a trader that means positions that are too small. For a bank it means capital that it did not need. Berkowitz and O'Brien (2002) found that pattern in banks' VaR outside crises.

12.6 Comparing models

Coverage tests check each model against its own claim. To choose between two models that both pass, compare their average **pinball loss** (§5.9) over the same days. The pinball loss is a consistent scoring function for quantiles. Its expected value is smallest when the forecast is the true quantile. The model with the lower average is therefore the better forecaster, and the score rewards both calibration and responsiveness. The daily differences in pinball loss between the two models form a time series. To test whether the difference is significant, use standard errors robust to autocorrelation on that series, in the manner of Diebold and Mariano (1995). In §6.13 the pinball loss ranked filtered historical simulation and the Student t EWMA model first. It ranked the 1,000-day historical simulation and the equal-weighted normal last.

12.7 Backtesting expected shortfall

ES is not elicitable alone (§5.9), but a backtest can still check it. The simplest check averages $L_t / \mathrm{ES}_{\alpha,t}$ over the days on which the loss exceeded the VaR at the same level, $\mathrm{VaR}_{\alpha,t}$. ES forecasts the average loss on exactly those days, so the ratio should average about one if the ES forecast is right. Acerbi and Szekely (2014) turn this and related quantities into formal tests with good power. The practical difficulty is again sample size. An ES test at 97.5% on 250 days uses only six or so tail observations.

12.8 Pitfalls

- **Look-ahead.** The forecast for day t must use only data up to $t-1$, including in parameter choices. Tuning the EWMA decay to minimise breaches over the backtest, and then reporting the backtest, is overfitting.
- **Which P&L.** Hypothetical P&L on frozen positions tests the model. Actual P&L tests the model plus trading. Mixing them makes problems in both harder to detect (§10.2).
- **Overlapping horizons.** Backtest one-day forecasts. Ten-day hits overlap and so cluster by construction.
- **Too short a sample at 99%.** See §12.5. A year of backtesting at 99% cannot distinguish a good model from a bad one.
- **Ignoring the size of breaches.** Coverage tests count hits. They do not distinguish a breach of 1.1 times the VaR from one of 5 times. Track the ratio.
- **Changing positions.** For an individual, the simplest honest backtest replays *today's* positions through history (§14). That tests the model on the book actually held. It does not test how well past forecasts matched past books.

§12 Key takeaways

1. A correct VaR produces hits that are independent coin flips with probability p. The hit rate is right, and hits do not cluster. Test both.
2. Kupiec's test checks the rate. Christoffersen's test checks whether a hit yesterday predicts one today. Duration and regression tests are stronger versions.
3. The Basel traffic light puts a correct model in yellow 11% of the time, and a model with twice the stated breach rate in green 44% of the time.
4. One year of backtesting at 99% detects a doubled breach rate 38% of the time. At 95% it detects it 92% of the time. Backtest at 95% to check the scale, and at 99% only with years of data.
5. Too few breaches is also a failure. The model overstates risk, and positions are too small.
6. Rank models that pass by their average pinball loss on the same days.
7. Never tune the model on the period that you then report as its backtest, and never backtest overlapping multi-day horizons.

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13. VaR for an individual investor

Earlier sections described tools built for institutions. This section identifies what carries over to one person running one or a few accounts. It also places the number in a personal investing or trading process.

13.1 What transfers and what does not

The table sorts the institutional tools by how well they transfer.

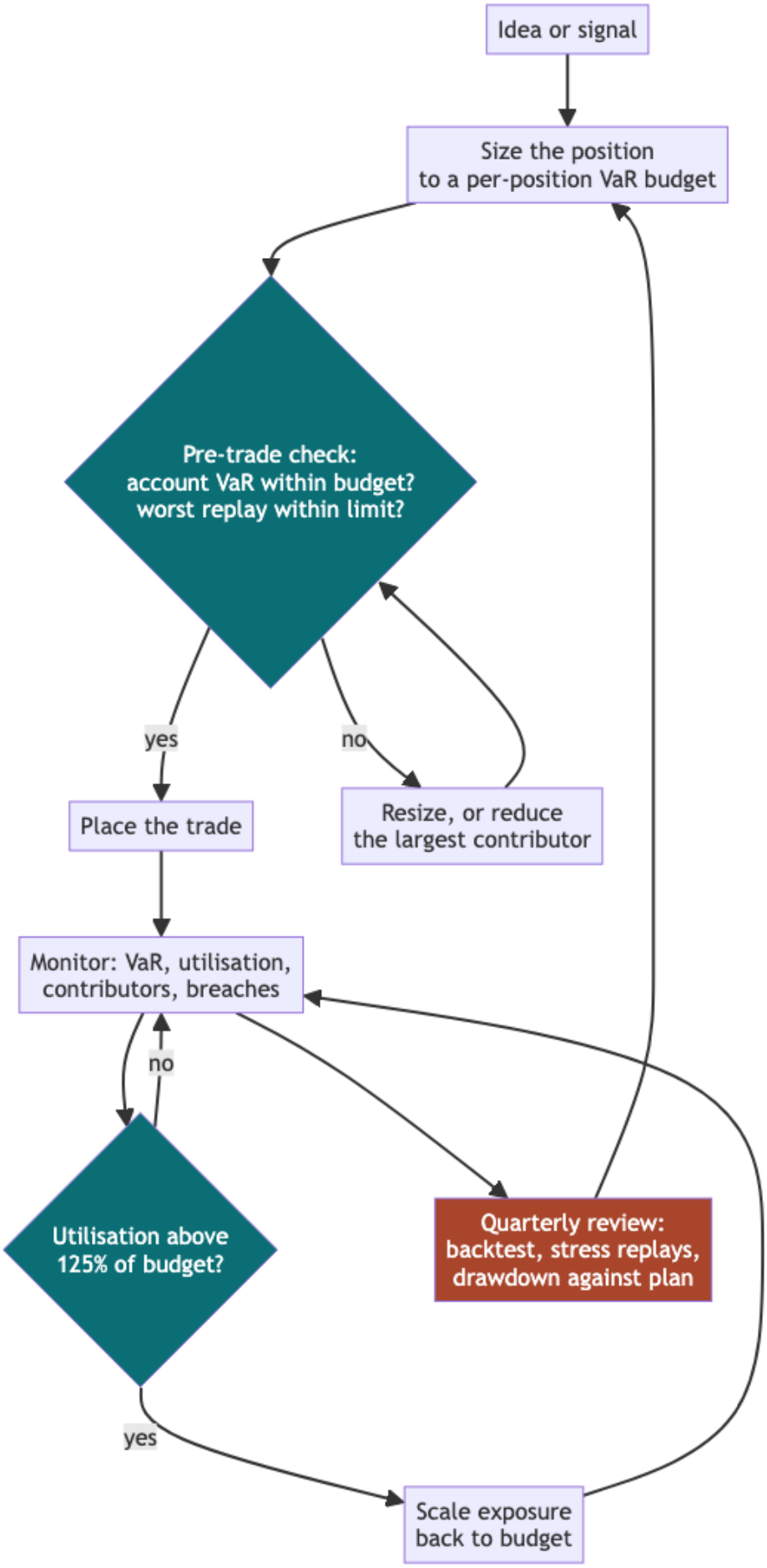
Transfers well	Transfers poorly or not at all
A common unit across stocks, funds, options, currencies and anything else held	The regulatory conventions: 99%, ten days, multipliers of three
Position sizing by risk rather than by dollars	Risk-factor mapping of large derivative books
Decomposition: which holdings carry the risk	Monte Carlo engines
The backtest discipline: count the breaches	Capital allocation across business lines
Historical replays of bad days on today's holdings	Liquidity horizons, for anyone in liquid instruments

An individual also has three structural advantages over an institution. First, an individual is too small to move prices, so the problems of liquidity and endogeneity in §11.6 do not apply. Second, nobody's bonus depends on the individual's VaR, so the gaming of §11.7 is limited to self-deception. Third, an individual chooses the horizon and the conventions. The individual can choose 95% at one day, where VaR is precise and testable, rather than 99% at 10 days, where it is neither.

An individual also has three disadvantages. Personal portfolios are more **concentrated**, so a single position's idiosyncratic risk matters, unlike at a bank. Retail **leverage and options**, such as margin, short puts and leveraged funds, create the non-linear, tail-selling exposures that VaR handles worst. And the person who reads the number is also the person whose behaviour the number should constrain.

13.2 Where it sits in the process

VaR is most useful at four points: when sizing a new position, before placing the trade, in routine monitoring and in periodic review. The flowchart places them in the process.


Idea or signal
Size the position to a per-position VaR budget
Pre-trade check: account VaR within budget? worst replay within limit?
yes
no
Place the trade
Resize, or reduce the largest contributor
Monitor: VaR, utilisation, contributors, breaches
no
Utilisation above 125% of budget?
Quarterly review: backtest, stress replays, drawdown against plan
yes
Scale exposure back to budget

VaR does not generate ideas, time entries and exits, or say whether a position is a good one. It is a constraint and a measurement, not a signal.

13.3 Sizing: three rules that are one rule

The VaR budget. Choose a maximum one-day 95% VaR for the whole account, as a fraction B of the account's equity W, and optionally a smaller one for each position. Take a single position with daily volatility σ_i. Setting its own VaR, $k_{0.95}\,\sigma_i\,x_i$, equal to a per-position budget $B_i W$ gives the position size

$$x_i = \frac{B_i\,W}{k_{0.95}\,\sigma_i}.$$

With B_i = 0.25% of a $100,000 account and a stock whose daily volatility is 2%, the position is $250/(1.645 \times 0.02)$ = $7,600. A broad index fund at 1% daily volatility gets twice as much, $15,200, for the same risk.

The volatility target. As §8.2 showed, an account VaR budget is a volatility target in different units. A budget of B of equity at 95% one-day is a daily volatility target of $B/1.645$. Multiplying by $\sqrt{252} = 15.87$ gives an annual target of $B \times 15.87/1.645 = 9.65B$. A 1% budget is a 9.65% volatility target. A budget of 1.5% is about 14.5%, a little below an all-equity portfolio.

The stop-loss rule. Many traders size by "risking" a fixed fraction of equity per trade, typically 0.5–2%. These traders define the risk as the loss if a stop-loss order executes, which gives $x_i = B_i W/d_i$, with d_i the distance to the stop as a fraction of price. Suppose the trader sets the stop at a multiple of recent volatility, say two times the average true range, a common measure of daily price movement. Then d_i is proportional to σ_i, and this rule is the VaR budget with a different constant in place of $k_{0.95}$. VaR adds information where the rules differ.

- **Gaps.** A stop caps the loss only if the market trades at the stop price. When the price jumps past the stop, the order fills at the next available price. On 4 April 2025, the worked account's technology position fell 6.0% in a day. The 1987 replay costs 20% on the same position. Overnight gaps, halts and fast markets fill stops at prices far beyond the stop level. VaR does not assume that the exit works.
- **Correlation.** If ten positions that each risk 1% at their stops are all technology stocks, they are one 10% bet with 10 exits, not 10 separate 1% risks. Summing per-trade risk, called "portfolio heat", ignores correlation. It charges 10 technology stocks and 10 unrelated positions the same 10%, and it counts a hedge as added risk. Account-level VaR distinguishes these cases.
- **Instruments without stops.** The VaR budget sizes funds held for years, options, and anything not managed with stops, all on the same scale.

When the market, not the trader, breaches the budget. A rise in volatility breaches a VaR budget more often than a trade does (§10.5). Decide in advance what that means. Scaling exposure back to budget every time is a pure volatility target. It sells after volatility rises, generates turnover and, in a taxable account, realises gains. [Practice] A common compromise is a **band**. Act only when utilisation exceeds, say, 125% of budget, and then scale back to 100%. Take the 2025 numbers of §10.3, with the budget set at the year's average VaR. Utilisation would have exceeded the 125% band once, continuously from 4 April to 10 June, two months, and at no other time. §11.6 summarises the evidence that such de-risking has not hurt broad-market returns on average.

How to choose the budget. Derive the budget from the loss that the investor can tolerate, using the relation between daily VaR and drawdowns in §13.5. Roughly, in US equities, a one-day 95% VaR budget of 1% of equity has corresponded to a typical year's worst drawdown of about 9%. It has corresponded to a one-year-in-10 drawdown of about 16%. If a 16% drawdown in a bad year is the limit, 1% is about right. If the limit is 25%, the budget is about 1.5%. Choose the drawdown first, and derive the budget from it.

13.4 Seeing the portfolio

For most individuals, the decomposition of §10.4 is the single most useful output of a VaR system. It shows **which holdings carry the risk**, which brokerage statements do not.

- **Hot spots.** On the worked account, technology was 55% of the risk and autos and durables 21%, on 40% and 10% of the capital. A position whose share of risk is far above its share of capital is either a deliberate high-conviction bet or an accident. The decomposition requires a decision about which.
- **False diversification.** Five funds with different names can be one bet. Component VaR shows it, because highly correlated positions each carry a large share of the total.

- **Real diversification.** Energy was 15% of the capital and 5% of the risk, because its correlation with the rest was near zero in 2025. That is a fact about the recent past, not a law. In the scenario matrix of §10.6, energy lost 2,000–3,000 on three of the five replayed crash days.
- **Before a trade,** compute the incremental VaR (§5.5). Compute the account VaR with and without the proposed position, and compare the difference with the position's stand-alone VaR. Under the normal model the difference can never exceed the stand-alone figure, because VaR is then sub-additive (§5.7). The useful measure is how close the difference comes to that figure. A trade that adds nearly its whole stand-alone VaR is concentrating, because it moves with what the account already holds. One that adds much less is diversifying, and one that lowers the account VaR is a hedge. For a small trade, the ratio of the two is roughly the position's correlation with the account.

13.5 Calibrating expectations

For an individual, VaR's most underrated use is psychological. It states in advance what an ordinary bad day, month and year look like, so the investor does not mistake them for emergencies.

Bad days. At a 95% one-day VaR of $1,100, about 13 days a year should lose more than $1,100, roughly one a month. Over 2025, losses exceeded the worked account's EWMA 95% VaR on 14 days of 250.

The worst day of the year. On US equities from 1930 to 2025, the worst day in a calendar year was typically 2.3 times that year's average 95% VaR. In one year in 10 it was 3.7 times or more. For a $1,100 VaR, plan for a worst day of about $2,500 in a typical year and $4,000 in a bad one.

Drawdowns. The table gives the percentiles, from 1930 to 2025, of the US market's maximum drawdown within each year, as a multiple of that year's average one-day 95% VaR.

Year, ranked by drawdown	Drawdown ÷ that year's average daily 95% VaR
10th percentile	5.3
25th percentile	6.7
Median	9.2
75th percentile	12.4
90th percentile	16.2
Worst year	19.2

An account run at a daily 95% VaR of 1% of equity should therefore expect a worst peak-to-trough decline of about 9% in a typical year. In one year in 10, it should expect about 16%. These figures count only drawdowns within a calendar year. Multi-year declines, such as 2000–2002, are larger. [Hypothesis] The ratio for other assets will differ with their autocorrelation and the persistence of their volatility. For trend-following or mean-reverting strategies it can differ a great deal. Treat the table as a starting point for equity-like risk, not a constant. For a random walk, the theory gives a similar order of magnitude. The expected maximum drawdown of a driftless Brownian motion over a period is about 1.25 times its standard deviation over that period. The exact factor is $\sqrt{\pi/2}$ (Magdon-Ismail and co-authors, 2004). A year's standard deviation is $\sqrt{252} \approx 15.9$ daily ones. The expected drawdown is therefore about $1.25 \times 15.9 \approx 20$ daily standard deviations, or $20/1.645 \approx 12$ daily 95% VaRs. Positive drift lowers it.

Rules from the expectations. Write the rules down. Examples are "a loss of up to 2.5 VaRs in a day is ordinary" and "a drawdown of nine VaRs is a normal year". Another is "eight or more 95% breaches in 60 trading days, against three expected, happens by chance about 1% of the time, so investigate". That 1% applies to one window. If the investor checks the rule every week on a rolling window, it fires on a correct model in about one year in eight. With seven as the threshold, it fires in about one year in three. The rule is therefore a prompt to look, not a verdict. The purpose is to make the response to a loss a decision taken before the loss.

13.6 Leverage, margin and options

Leverage scales VaR linearly. At 1.5 times leverage, the VaR as a fraction of equity is 1.5 times the unlevered figure. Leverage changes the consequence non-linearly, because a margin call forces a sale at the worst moment. The useful measure is **the market move that triggers a margin call**, not the VaR. Compute that move from the broker's rules, express it in units of daily volatility, and compare it with the replays of §7.4. If a day like 1987 or March 2020 would breach margin, the account runs a risk that VaR does not show.

Long options have a loss bounded by the premium, and the delta approximation overstates their risk (§6.10). Value them by full revaluation, or simply count them at premium value as a worst case.

Short options are where VaR is least trustworthy. A short out-of-the-money put can show a VaR of the same order as the premium it collects, and a crash loss dozens of times larger. In §6.10, 10 such contracts collect $71 and show a delta-normal 99% VaR of $91. They lose $3,975 on a day when the stock falls 10% and implied volatility doubles. [Practice] **Recommendation: for any short-option position, limit the *stress loss*, not the VaR.** The stress loss is the loss under a 10% or 20% move in the underlying, with implied volatility doubled or tripled. Size the position so that the stress loss is survivable without a forced liquidation.

13.7 What VaR will not do

- It will not say when to sell. A rising VaR says that risk has risen, not that prices will fall.
- It will not warn of a crash. The first day of every shock in §11.4 was a breach for every model.
- It will not measure long-horizon goal risk. Whether a retirement portfolio will fund 20 years of withdrawals is a different calculation, about expected returns and sequence risk, and a one-day quantile does not address it.
- It will not replace a plan. Its value lies in making the plan's numbers explicit: the budget, the band, the stress limit, and the response to a breach of each.

§13 Key takeaways

1. The common unit, risk-based sizing, decomposition, the backtest habit and stress replays transfer to an individual. The regulatory conventions do not. Use 95% at one day.
2. VaR belongs at four points: sizing, the pre-trade check, monitoring and periodic review. It is a constraint and a measurement, not a signal.
3. A VaR budget, a volatility target and volatility-scaled stop sizing are one rule. VaR adds what stops lack. It does not rely on the exit working, and it accounts for correlation across positions.
4. Decide in advance how to respond when volatility, not a trade, breaches the budget. A band that acts above 125% and scales back to 100% avoids churn.
5. Derive the budget from a tolerable drawdown. In US equities, a 1% daily 95% VaR has gone with a typical-year drawdown of about 9% and a one-in-10-year drawdown of about 16%.
6. The decomposition is the most useful single output. It finds the hot spots and false diversification that a brokerage statement does not show.
7. With leverage, ask what move triggers a margin call. With short options, limit the stress loss, not the VaR.

---

14. The minimal version

This section specifies the smallest VaR system that, in the view taken here, is worth running for a personal account. It gives the code and shows the output on the worked account. The system is the recommendation of §6.13, cut down to what one person can maintain.

14.1 Design decisions

The table records each design decision and its reason.

Decision	Choice	Why
Confidence level	95% primary; 99% and 97.5% ES reported	95% is much more precise: ±18% against ±28% from a year of data (§5.8). It is also far more testable (§12.5)
Horizon	One day	Measured directly, not scaled. Multi-day risk comes from replays (§11.5)
Volatility	EWMA, decay 0.94	Reacts within days. Needs no estimation. The best simple method (§6.3)
Tail shape	Empirical, from 500 days of volatility-standardised P&L	Filtered historical simulation. It performed best in §6.13, there with a 1,000-day window
Positions	Today's dollar holdings, replayed through history	Needs no history of past positions. Tests the book actually held
Decomposition	EWMA covariance, Euler components	Components sum to the total. Identifies the largest risk contributors (§5.5)
Validation	Replay backtest over the last 500 days	Expected 25 breaches at 95%. Any count between about 16 and 33 is consistent with a correct model
Stress	Replays of a few named days and the worst 20-day window	Covers what the VaR ignores (§7.4)

Data. The system needs daily closing prices, adjusted for dividends and splits, for every holding. They must cover at least the 500-day window plus a short warm-up, about two and a half years. Four years is better, so that the backtest has 500 days of its own. Cash counts as a position with zero risk. Short positions are negative dollars. For a holding with too little history, use a close proxy, such as the index that a new fund tracks or a similar fund, and note it.

What it leaves out. Options need full revaluation (§6.10). Either reprice them under each replayed move with a pricing model, or exclude them and cover them with stress limits (§13.6). The system also leaves out anything illiquid.

14.2 The code

The code uses only numpy. The array `R` holds daily simple returns, with one row per day, oldest first, and one column per holding. The array `pos` holds the dollar value of each holding today, the vector x of the notation. The functions `minimal_report` and `minimal_backtest` are in `figures/var_common.py`. They produced every number in §14.3 and all but the replays in §10.7.

```python
import numpy as np

LAM, Z95 = 0.94, 1.645


def ewma_vol(pnl, lam=LAM, seed=60):
    # forecast for day t uses data through day t-1 only
    var = np.empty(len(pnl))
    var[0] = np.mean(pnl[:seed] ** 2)
    for t in range(1, len(pnl)):
        var[t] = lam * var[t - 1] + (1 - lam) * pnl[t - 1] ** 2
    return np.sqrt(var)


def ewma_cov(R, lam=LAM):
    S = np.cov(R[:60], rowvar=False)
    for r in R[60:]:
        S = lam * S + (1 - lam) * np.outer(r, r)
    return S


def minimal_report(R, pos, window=500):
    S = ewma_cov(R)
    sigma = np.sqrt(pos @ S @ pos)             # tomorrow's volatility, dollars
    component = Z95 * pos * (S @ pos) / sigma  # sums to Z95 * sigma
    pnl = R @ pos                              # today's book replayed through history
    z = (pnl / ewma_vol(pnl))[-window:]        # each day in units of its own forecast
    var95, var99 = -np.quantile(z, [0.05, 0.01]) * sigma
    es975 = -z[z <= np.quantile(z, 0.025)].mean() * sigma
    return sigma, var95, var99, es975, component


def minimal_backtest(R, pos, level=0.95, days=500, window=500):
    pnl = R @ pos
    vol = ewma_vol(pnl)
    z = pnl / vol
    return sum(pnl[t] < np.quantile(z[t - window:t], 1 - level) * vol[t]
               for t in range(len(pnl) - days, len(pnl)))
```

The list below describes the code function by function.

- `ewma_vol` applies the RiskMetrics recursion of §6.3 to the replayed P&L. The code lags the recursion by a day, so each forecast uses only past data. The recursion starts from the average squared P&L of the first 60 days, so its earliest values are a warm-up rather than true forecasts.
- `ewma_cov` applies the same recursion to the covariance matrix. The matrix gives tomorrow's volatility and the decomposition. For a fixed book x, its $\sqrt{x'\Sigma x}$ and the univariate `ewma_vol` of the replayed P&L are the same estimator up to their starting values. The reason is that $x'\Sigma_t x = \lambda\, x'\Sigma_{t-1}x + (1-\lambda)(x'r_{t-1})^2$ is the univariate recursion applied to the P&L $x'r$. So `minimal_report` can standardise the past with one and scale tomorrow with the other.
- In `minimal_report`, `component` is the Euler decomposition of the normal 95% VaR (§5.5). `z` is the replayed P&L standardised by its own volatility forecast. This is the filtered historical simulation of §6.8. The VaRs equal minus the 5% and 1% quantiles of `z` over the last `window` days, times tomorrow's volatility `sigma`. The ES equals minus the average of the worst 2.5% of `z`, times `sigma`. The minus signs turn P&L into the positive losses of the convention.
- `minimal_backtest` steps forward through the last `days` days. Each day it recomputes the VaR from data available the evening before, and it counts the breaches. It needs `days + window` days of history after the warm-up. This requirement is the source of the four years in §14.1.
- The replays of §14.1 need no further code. They are entries of `R @ pos` on the named days. The worst 20-day window is the most negative sum of 20 consecutive entries.

The only other code loads the inputs. One way uses pandas:

```python
import pandas as pd
```

```
prices = pd.read_csv("prices.csv", index_col=0, parse_dates=True)  # adjusted closes
R = prices.pct_change().dropna().to_numpy()
pos = shares * prices.iloc[-1].to_numpy()  # shares: array of quantities held
```

14.3 What it says about the worked account

The table gives the output for the five-fund account, with data to the end of 2025.

Output	Value	As % of the account
One-day volatility	664	0.66%
95% VaR	1,116	1.12%
99% VaR	1,991	1.99%
97.5% ES	2,080	2.08%
Components of the 95% VaR (tech, health, energy, utilities, autos)	599, 133, 57, 72, 231	
Backtest breaches at 95%, last 500 days (expected 25)	25	
Backtest breaches at 99%, last 500 days (expected 5)	10	
Backtest breaches at 95% and 99%, last 2,500 days (expected 125 and 25)	118 and 27	

Dollars except where marked. Components sum to 1.645×664. *The filtered 95% VaR is slightly larger because the empirical multiplier, 1.68, exceeds 1.645.*

The multipliers carry the most information. The 95% VaR is 1.68 volatilities, and the 99% VaR is 3.00, well above the normal 2.33. This is the fat tail of §2.5, measured on this book. The ES at 97.5% is only 4.5% above the 99% VaR. The gap is 0.5% for a normal distribution and 6.6% for a Student t with four degrees of freedom (§5.6). The book's tail is therefore fat but not pathological. A short-option book would show a much larger gap.

The 99% backtest row needs a second look. Ten breaches in the last 500 days, against five expected, just reaches the 5% rejection threshold of §12.5. The 2,500-day count, 27 against 25, is in line. §10.7 discusses this row.

14.4 The routine

Weekly, or after any large move: run the report. Log the VaR, the utilisation of the budget and the largest contributor, and record any exceedance of the VaR.

Write the **standing rules** down in advance. An example set follows.

1. The account's 95% one-day VaR is at most 1.25% of equity, derived from the drawdown tolerance as in §13.3.
2. If utilisation exceeds 125% of budget, scale positions back to 100%, starting with the largest component.
3. No single position carries more than 40% of the VaR without a written reason.
4. The worst replayed day costs no more than 20% of equity, and the worst replayed 20-day window no more than 40%.
5. Eight or more 95% breaches in any 60 trading days trigger a review of the model and the book.

Quarterly: run the 500-day backtest at 95%. Between about 16 and 33 breaches is consistent with a correct model. Run it at 99% for information only. Check that the ratio of ES to VaR has not risen sharply. Rerun the replays. Compare the year's drawdown with the expectations of §13.5.

14.5 Before trusting it

Run four checks. Each takes minutes.

1. **A single index fund.** Its 95% VaR should be about 1.6–1.7 times its recent daily volatility times its value. If it is not, the returns are wrong. The causes include unadjusted prices, a missing day, or a percentage where a fraction was expected.
2. **Components add up.** The components should sum to $1.645\sigma_p$ exactly.
3. **Sign conventions.** Add to the book a long position and an equal short position in the same fund. The VaR should not move, and the pair's two components should cancel. On the pair alone, the code would divide zero by zero.

4. **The backtest is in range** at 95% on the last 500 days. If it is not, look at the dates of the breaches before changing anything. A cluster in one episode is a different problem from a steady excess.

14.6 Upgrades, in order of value

1. **Options by full revaluation** under each replayed move, if the account holds any.
2. **Direct multi-day risk:** the same filtered simulation on overlapping five-day or 20-day returns, rather than $\sqrt{h}$ scaling (§11.5).
3. **A longer residual window**, of 1,000 days or more, for the 99% and ES figures. These need more tail observations than the 95% figure (§5.8).
4. **Proxies for short histories,** documented.
5. **A GARCH volatility model** in place of EWMA, if multi-day forecasts matter.

Monte Carlo, extreme value theory and copulas are not on the list. For a personal account of liquid holdings, they add complexity faster than accuracy.

§14 Key takeaways

1. The minimal system runs filtered historical simulation on today's book. It uses EWMA volatility, 500 days of standardised replayed P&L, and one-day 95% as the primary figure.
2. It needs only daily adjusted prices and current holdings, and it fits in about 40 lines of numpy.
3. On the worked account, the 2,500-day backtest is well calibrated. It shows 118 breaches at 95% against 125 expected, and 27 at 99% against 25.
4. The empirical multipliers, 1.68 at 95% and 3.00 at 99%, are themselves diagnostics. They measure how fat the book's tail is.
5. Write the rules down before they are needed: budget, band, concentration, stress limits and breach review.
6. Before trusting the output, validate it with an index fund, the component sum, a long-short pair and the backtest.
7. Add option revaluation and direct multi-day risk before anything else.

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15. Synthesis

15.1 The unifying framework

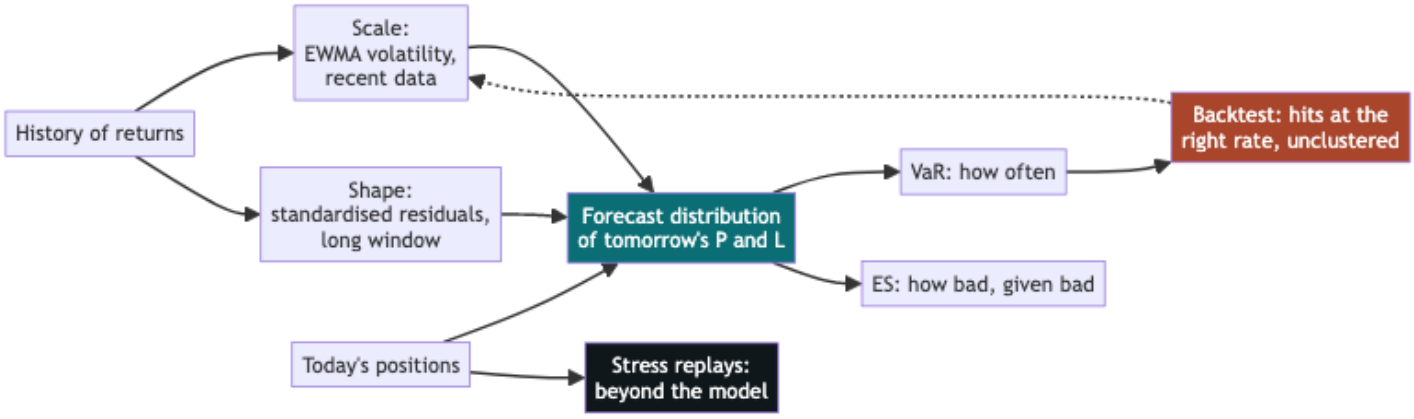
The chapter rests on three central ideas.

VaR is a quantile of a forecast distribution. Every method follows the same pipeline: exposure, distribution, revaluation and quantile. Methods differ only in their forecast of tomorrow's distribution. The difference lies in the scale model, the shape model or the revaluation.

At short horizons it is a volatility forecast times a tail constant. The volatility moves by factors of 10. The constant moves by about 15% at 95% and 25% at 99%. Forecast volatility well from recent data, and take the tail shape from a long history of standardised returns. The method is then nearly as good as any other.

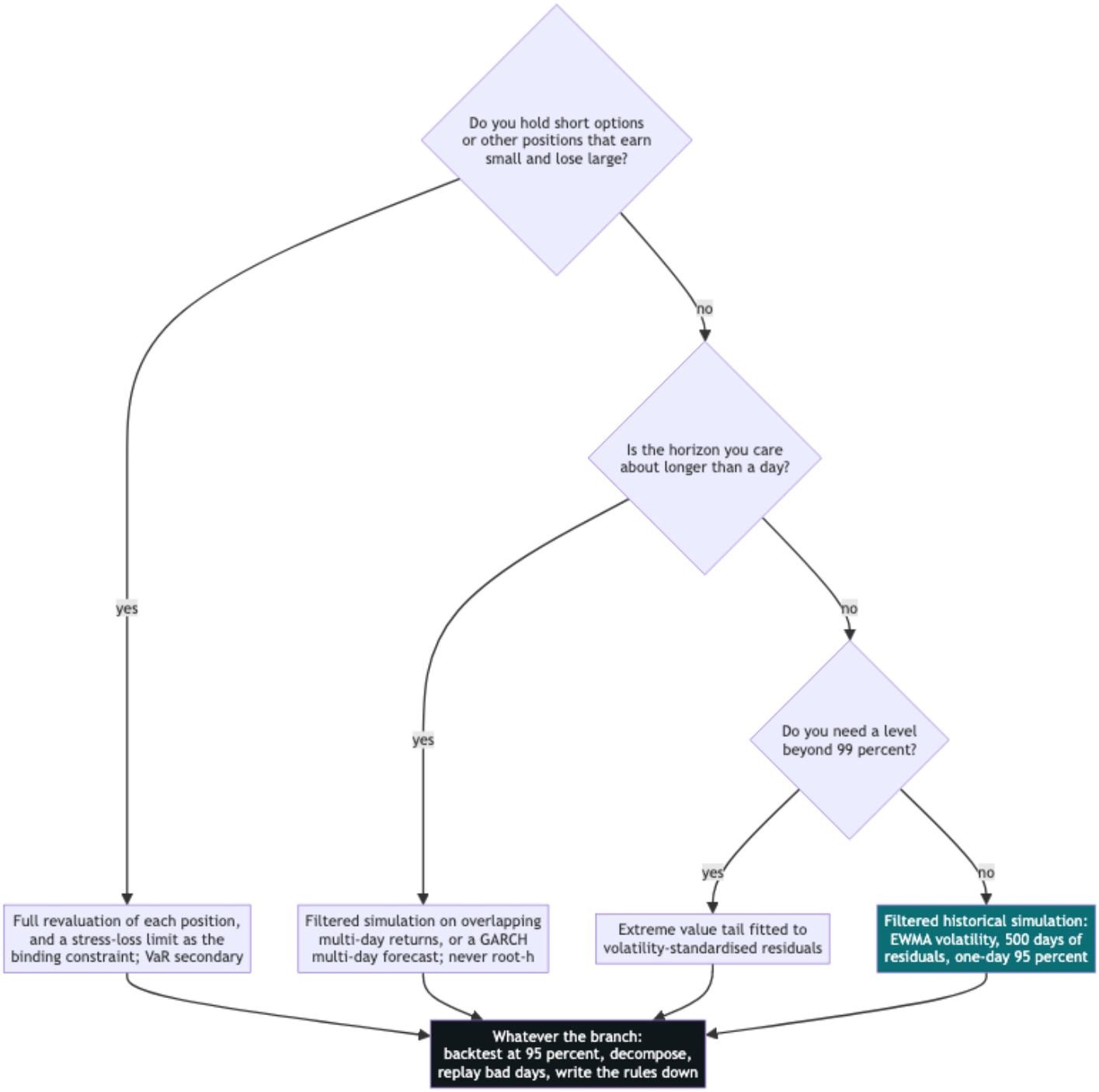
It is a threshold, audited by its hits. VaR says how often losses exceed it, never how much. Its main strength is that anyone can check the "how often". The hits should come at the stated rate and without clustering. The user must add everything beyond the threshold on purpose: ES, stress tests and plans.

The diagram combines the three ideas into one framework.



15.2 Which method

The decision tree ends in a named method for each kind of book.



15.3 Building it, with gates

The table stages the build. Each stage has a gate to pass before moving on.

Stage	Work	Gate before moving on
0. Data	Adjusted daily prices for every holding, current positions in dollars, proxies for short histories	A single index fund's VaR is 1.6–1.7 times its daily volatility times its value
1. Parametric baseline	EWMA covariance, delta-normal VaR, components	Components sum to the total. Adding a long-short pair in one fund leaves the VaR unchanged
2. Filtered historical simulation	Standardised replayed P&L, empirical quantiles, ES	95% backtest over 500 days between 16 and 33 breaches

Stage	Work	Gate before moving on
3. Stress and rules	Replays of named days and the worst 20-day window. Budget, band, concentration and stress limits written down	Every rule says what action follows a breach
4. Routine	Weekly report and log. Quarterly backtest and review	Three months of log entries, and at least one rule exercised or consciously waived
5. Upgrades	Options by full revaluation, direct multi-day risk, longer residual window for 99%	Each upgrade changes a decision, or it is not worth maintaining

Stages 0 and 1 are infrastructure. Most of the errors a VaR system ever produces originate there. Examples are unadjusted prices, positions in the wrong units and a missing account.

15.4 Advice for someone starting today

1. **Read VaR as the minimum loss on a bad day**, never as the maximum possible loss.
2. **Use 95% at one day.** It is precise and testable within a year. It describes a monthly event that the investor will actually experience.
3. **Spend the effort on the volatility forecast.** EWMA with decay 0.94 captures most of the benefit. Filtering history by it captures the rest.
4. **Count the breaches.** Expect 12 or 13 a year at 95%, spread out. If they cluster, the model reacts too slowly. If none occur, the model is too cautious.
5. **Look at the decomposition every time.** It will reveal a concentration that nobody knew existed.
6. **Replay the bad days.** A one-day VaR of about 1% of the account and a replayed 1987 of 18% are both true. Only the second shows whether the account can survive.
7. **Size from the tolerable drawdown.** In a typical year, the worst drawdown has been about nine times the average daily 95% VaR. In one year in 10, it has been about 16 times.
8. **Never scale a calm-period VaR to a longer horizon.** The scaled number is lowest at the moments when it is most wrong.
9. **Distrust a low VaR on a high-yielding position.** A high yield with a low VaR indicates that the position sells tail risk.

15.5 What is and is not known

Several things about VaR are settled. Its definition and its mathematical properties are settled. Conditional volatility models beat unconditional windows. Filtering history by volatility is the best simple method for linear books. [Fact] Researchers have replicated the empirical regularities of this chapter across markets and decades. They include the clustering of breaches in crises, the failure of square-root-of-time scaling from calm states, and the inadequacy of normal tails at 99%.

Other things are not settled. Researchers still disagree on whether the coherence of expected shortfall is worth its loss of robustness. Any realistic sample leaves the tail shape beyond the 99th percentile genuinely uncertain. No method avoids this limit. Above 99%, every estimate is partly assumption. The claim that VaR-based regulation amplifies crises has strong theoretical support and suggestive but not decisive evidence. Whether de-risking when VaR rises improves an individual investor's outcomes depends on the asset and the period. The literature has not resolved how.

VaR and stress analysis have separate jobs. A VaR describes ordinary days honestly and checkably, which is valuable. Scenarios and a plan must examine extraordinary events directly. No better estimate of a quantile will change that.

§15 Key takeaways

1. VaR is a quantile of a forecast distribution. At short horizons it is a volatility forecast times a tail constant. Its hits audit it.
2. For a linear book, filtered historical simulation at 95% over one day is the default. Short options need full revaluation and a stress limit, multi-day horizons need direct computation, and levels beyond 99% need an extreme-value tail.
3. Build in gated stages from clean data, and write the rules down before a breach tests them.

Appendix A. Concepts and prerequisites

This appendix covers every concept that the main text uses without explaining. It targets a reader who is comfortable with mathematics but does not work in finance. Such a reader meets "selling credit protection" in §11.1, "the traffic light" in §12.4 and a likelihood-ratio statistic in §12.2. This reader wants the idea rather than a citation, and prefers to get it without leaving the chapter.

The entries follow **dependency** order, not alphabetical order. Later entries use earlier ones. Five parts group the entries, and the parts also follow dependency order, so the appendix builds up step by step. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. The statistics entries are brief, because most of that material is standard. The finance and regulation entries get more room, because that material is not standard for this reader. The notation follows the main text's notation block exactly. Symbols that an entry introduces are local to it. Where a field's standard symbol conflicts with the notation block, the entry flags the conflict and does not invent new symbols.

The chapter's *argument* does not depend on this appendix. Much of it is needed to check the argument or to implement it.

Index. The table gives the section where each concept is first used.

Concept	First used	Concept	First used
Quantiles and F_L^{-1}	§1.2	Euler allocation	§5.5
Binomial and geometric	§1.3	Expected shortfall	§1.7
Sample quantiles	§1.8	Rockafellar–Uryasev, LPs	§3.4
IID; conditional distributions	§1.5	Elicitability, pinball loss	§5.9
Location-scale families	§1.6	Statistical robustness	§5.7
The normal; sigma counts	§1.5	Copulas, tail dependence	§3.4
Skewness and kurtosis	§2.5	Monte Carlo	§6.9
Student t	§2.2	Bonds and duration	§1.1
Covariance matrix and beta	§1.5	Options, Black–Scholes	§1.1
Elliptical distributions	§5.2	The Greeks	§1.1
Central limit theorem	§5.4	Implied volatility, VIX	§1.5
Sampling error of a quantile	§5.8	Risk-factor mapping	§1.5
Maximum likelihood	§6.4	Credit risk	§1.2
Size, power, p-values	§6.8	Earn small, lose large	§2.3
Likelihood ratios, χ^2	§12.2	Liquidity	§1.2
Two-state Markov chains	§12.3	Leverage and margin	§1.7
Autocorrelation, HAC	§12.1	Stops and sizing rules	§8.2
Simple and log returns	§2.2	Sharpe ratio, vol targeting	§8.2
Volatility clustering	§1.6	Tracking error	§7.3
EWMA	§1.6	Drawdown	§1.7
GARCH and asymmetry	§5.4	Basel internal models	§1.7
Jumps	§5.4	The FRTB	§3.5
Power-law tails	§5.7	Fund and insurance rules	§2.1
Extreme value theory	§6.11	Model risk	§1.3
Cornish–Fisher	§6.5	Backtest hygiene	§6.1
Coherent risk measures	§3.4		

Part I — Probability and statistics. A VaR is a statement about a distribution, and a backtest is a statement about a count. These 17 entries define the terms for both. They cover what a quantile is, how to estimate it, the few distributions the chapter uses, and the statistical tests of §12.

A.1 Distribution functions, quantiles and the generalised inverse

The idea. A distribution function gives, for every loss level ℓ, the probability that the loss is at most ℓ. A quantile reverses this. It gives the loss level that the loss does not exceed with probability α. When the distribution function is continuous and strictly increasing, the two are ordinary inverses of each other. Loss distributions are often neither continuous nor strictly increasing. A default either happens or does not, so the distribution function jumps. Some ranges of outcome are impossible, so the function is flat there. The generalised inverse returns one well-defined number in both cases. It is the smallest loss level at which the probability of not exceeding it reaches α. A jump in a distribution function is an atom of probability on a single value. It differs from a jump in a price path (A.22), although the main text uses the word for both.

Formally. The distribution function of the loss, $F_L(\ell) = P(L \le \ell)$, is non-decreasing and right-continuous, rising from 0 to 1. Its generalised inverse is

$$q_\alpha(L) = F_L^{-1}(\alpha) = \inf\{\ell : F_L(\ell) \ge \alpha\}, \qquad 0 < \alpha < 1.$$

Three properties matter. First, $F_L(q_\alpha) \ge \alpha$, so $P(L > q_\alpha) \le 1 - \alpha$, with equality when F_L is continuous at q_α. At an atom, the probability of exceeding the VaR can be well below p. Second, on a flat stretch the infimum takes the left end. That is why some texts also define the upper quantile $\inf\{\ell : F_L(\ell) > \alpha\}$ and allow any value between the two. Third, quantiles commute with monotone maps. If g is increasing and left-continuous, then $q_\alpha(g(L)) = g(q_\alpha(L))$. So the 5% quantile of log returns is exactly the log of one plus the 5% quantile of simple returns. Their means satisfy no such relation (A.18). A decreasing map reverses the level. For continuous X, $q_\alpha(-X) = -q_{1-\alpha}(X)$.

Why it appears here. §1.2 defines VaR as exactly this infimum. §5.1 shows that the detail matters: a position that loses 100 with probability 4% has a 95% VaR of zero. The same mechanism explains the negative VaR of Position B in §2.3 and the two-bond failure of sub-additivity in §5.7. The step in §5.2 from the quantile of $-\varepsilon$ to $-G^{-1}(1-\alpha)$ is the reflection rule above.

Deeper. Embrechts & Hofert, "A Note on Generalized Inverses," *Mathematical Methods of Operations Research* 77 (2013), 423–432, which proves every property above, edge cases included. McNeil, Frey & Embrechts (2015), in §4.7, for VaR built on it.

A.2 Indicators; Bernoulli, binomial and geometric distributions

The idea. An indicator turns an event into a number: one if it happened, zero if not. With events as numbers, counting them is addition, and their probability is an average. A single yes-or-no outcome with a fixed probability is a Bernoulli trial. The number of successes in a fixed number of independent trials is binomial. The number of trials up to and including the first success is geometric. These three distributions underlie most of §12. A correct VaR model makes every day a Bernoulli trial, with probability p of a hit, independent of the past.

Formally. $\mathbf{1}\{A\}$ is one on the event A and zero off it, so $E[\mathbf{1}\{A\}] = P(A)$. Suppose $I_1, \dots, I_T$ are independent with $P(I_t = 1) = p$. The count $X = \sum_t I_t$ is binomial:

$$P(X = k) = \binom{T}{k} p^k (1-p)^{T-k}, \qquad E[X] = Tp, \qquad \mathrm{Var}(X) = Tp(1-p).$$

The gap D between successive hits is geometric, $P(D = d) = (1-p)^{d-1}p$ for $d = 1, 2, \dots$, with mean $1/p$. Two consequences matter. First, the chance of at least one hit in T days is $1 - (1-p)^T$. A perfect 99% model is therefore exceeded at least once in a 252-day year with probability 92%. Second, the count's standard deviation grows only in proportion to $\sqrt{T}$. At $p = 0.01$ and $T = 250$, the expected count is 2.5 with a standard deviation of 1.6, so one hit and four hits are both unremarkable.

Why it appears here. The 92% of §1.3 and the right-hand column of the table in §2.1 equal $1 - (1-p)^T$. §12.1 states the null hypothesis of every backtest: the I_t are independent Bernoulli trials with probability p. The zone probabilities of §12.4 and the power table of §12.5 are binomial tail sums. The duration tests of §12.3 use the geometric gap, whose mean is 100 days at 99%. The standard error of §5.8 starts from the binomial variance of the count beyond the quantile (A.12).

Deeper. Feller, *An Introduction to Probability Theory and Its Applications*, vol. 1, 3rd ed. (Wiley, 1968), chapter VI.

A.3 Sample quantiles, order statistics and interpolation

The idea. A historical VaR is a quantile of a finite list of numbers, and a finite list has no unique "5% point". Sort 500 daily losses. The 95% quantile should have 25 losses above it. Depending on the convention, the quantile is the 26th largest loss, the 25th, or a point between them. The sorted values are the order statistics. Every package's quantile function is a rule for picking one of them or interpolating between two. None of the rules is wrong. They differ by an amount that is negligible in the middle of a sample and material in its tail, where neighbouring order statistics lie far apart.

Formally. The empirical distribution function of losses $L_1, \dots, L_n$ is the step function $\hat{F}_n(\ell) = n^{-1} \sum_i \mathbf{1}\{L_i \le \ell\}$. Its generalised inverse (A.1) is an order statistic. With the sample sorted as $L_{(1)} \le \dots \le L_{(n)}$,

$$\hat{F}_n^{-1}(\alpha) = L_{(\lceil n\alpha \rceil)}.$$

At $n = 500$ and $\alpha = 0.95$, this is $L_{(475)}$, the 26th largest loss. Interpolating rules instead compute a fractional rank and interpolate linearly between the two order statistics on either side of it. Hyndman and Fan catalogue nine such rules. The default in numpy and R, their type 7, uses the rank $(n-1)\alpha+1$. Their type 6 uses $(n+1)\alpha$. At $n = 250$ and $\alpha = 0.99$, the generalised inverse is the third-largest loss. Type 7 interpolates between the fourth and third largest, and type 6 between the third and second.

Why it appears here. The historical VaR of §1.8 is the loss with 25 worse days, which is the generalised-inverse choice. §5.1 warns that conventions can move a 99% historical VaR by 10% or more on the same data. That is the gap between the ranks above when tail losses lie far apart. The historical simulation of §6.6 and the filtered version of §6.8 use empirical quantiles. The code of §14.2 calls `np.quantile`, whose default is type 7. Check this difference before comparing one system's number with a broker's.

Deeper. Hyndman & Fan, "Sample Quantiles in Statistical Packages," *The American Statistician* 50(4) (1996), 361–365, which sets the nine rules side by side. David & Nagaraja, *Order Statistics*, 3rd ed. (Wiley, 2003), for the distribution theory.

A.4 IID, and conditional versus unconditional distributions

The idea. The phrase "the distribution of returns" can mean two things. One concerns the long run: how returns are spread across all the days of a sample. That is the unconditional distribution. The other concerns tomorrow: how tomorrow's return is spread, given everything known tonight. That is the conditional distribution, and it changes every evening. When returns are IID, independent and identically distributed, the two coincide, because tonight's information carries no information about tomorrow. Daily returns are far from IID. As §2.5 shows, the size of a day's move depends on recent volatility (A.19). The conditional distribution therefore changes while the unconditional one stays fixed. Conditional expectation is the operation that makes "given what is known tonight" precise.

Formally. $r_1, \dots, r_n$ are IID if they are mutually independent and share one distribution. The variance of a sum is then the sum of the variances, which is all that the square-root-of-time rule uses. Write $\mathcal{F}_t$ for the information available on the evening of day t. The conditional distribution of r_{t+1} is its distribution given $\mathcal{F}_t$. $E_t[\cdot] = E[\cdot \mid \mathcal{F}_t]$ is the expectation under it, which is itself a random quantity known at t. Two identities matter. The tower property, $E\big[E_t[Y]\big] = E[Y]$, makes the unconditional mean the average of the conditional means. The variance decomposition reads

$$\mathrm{Var}(Y) = E\big[\,\mathrm{Var}_t(Y)\big] + \mathrm{Var}\,\big(E_t[Y]\big),$$

The unconditional variance is therefore the average conditional variance plus the variance of the conditional mean. For daily returns the second term is negligible. The unconditional variance is therefore the time average of σ_t^2, and the unconditional distribution is a mixture of the day-by-day conditional ones. A.19 computes what mixing does to the tails. Stationarity is a weaker condition than IID. It says that the joint distribution of $(r_t, \dots, r_{t+k})$ does not depend on t. Under stationarity and ergodicity, time averages still converge.

Why it appears here. The $\sigma_p\sqrt{h}$ of §1.5 and the "only if returns are IID" of §2.2 are the variance-addition rule. §2.4 applies the distinction between conditional and unconditional to VaR, and §8.4 warns that "conditional VaR" also names something else. E_t appears in the notation block and is the expectation in the multi-day GARCH forecast of §5.4. The requirement in §12.1 that a correct hit be independent of everything known at $t-1$ is a statement about conditional distributions.

Deeper. Williams, *Probability with Martingales* (CUP, 1991), chapter 9, for conditional expectation done properly. The companion chapter Stochastic Processes covers filtrations, the formal version of "what is known on the evening of day t".

A.5 Location-scale families, standardisation and the tail multiplier

The idea. Many distributions come in families whose members differ only in location and scale. Shifting and stretching a member gives another member. Such a family has one standard shape, and every member is that shape moved and resized. Standardising subtracts the location and divides by the scale, which returns the standard shape. Quantiles transform the same way. The quantile of a shifted, stretched variable equals the shifted, stretched quantile. So in any such model, a VaR is a scale times a number that depends only on the shape. That number is the tail multiplier k_α.

Formally. A family is location-scale if, for a standard member ε with distribution G, every member is the distribution of $\mu + \sigma\varepsilon$ for some μ and some $\sigma > 0$. Normalising G to mean zero and variance one makes μ the mean and σ the standard deviation. Because $u \mapsto \mu + \sigma u$ is increasing, quantiles move with it (A.1), and the loss $L = -(\mu + \sigma\varepsilon)V$ has

$$q_\alpha(L) = (-\mu + \sigma k_\alpha)\,V, \qquad k_\alpha = q_\alpha(-\varepsilon) = -G^{-1}(1-\alpha) \ \text{ for continuous } G.$$

The counterpart of ε in the data is the **standardised residual** $\hat{\varepsilon}_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t$: a return re-expressed in units of the volatility forecast for its own day. If the forecasts are right, these residuals share one distribution even when the raw returns do not.

Why it appears here. This is Idea 2 of §1.6 and the whole of §5.2. The multiplier k_α carries the tail shape and σ the scale, and the table in §2.2 shows how little k_α moves at 95%. §2.5 measures each day's return "in units of the volatility that a forecaster could have known that morning", which is standardisation. The filtered historical simulation of §6.8 takes the empirical quantile of standardised residuals and multiplies it by today's $\hat{\sigma}_t$. §14.3 uses the book's own multipliers, 1.68 and 3.00, as diagnostics.

Deeper. Casella & Berger, *Statistical Inference*, 2nd ed. (Duxbury, 2002), sec. 3.5, on location and scale families.

A.6 The normal distribution, sigma counts and distribution-free bounds

The idea. The normal is the reference shape for the whole chapter, mostly as a contrast with real returns. Its tails thin out extremely fast. Each extra standard deviation divides the tail probability by a larger factor than the last. A "k-sigma day" therefore indicates great rarity if returns are normal, and little otherwise. The opposite extreme assumes only a mean and a variance. Chebyshev's inequality and its one-sided version, Cantelli's, then give the largest probability that a tail can hold. This bound is loose. So a volatility alone cannot give a VaR without an assumption about shape.

Formally. For large k, the normal tail satisfies Mills' approximation,

$$1 - \Phi(k) \;\approx\; \frac{\varphi(k)}{k} = \frac{e^{-k^2/2}}{k\sqrt{2\pi}},$$

The approximation is an upper bound, and it is already within 10% at $k = 3$. The tail probability is about 1.35×10^{-3} at three sigma, 3.2×10^{-5} at four, 2.9×10^{-7} at five, 10^{-9} at six and 3×10^{-138} at 25. For any distribution with mean μ and standard deviation σ, Chebyshev gives $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, and Cantelli gives the one-sided $P(X - \mu \geq k\sigma) \leq 1/(1 + k^2)$. Solving the second for the quantile bounds the tail multiplier of every unit-variance shock:

$$k_\alpha \;\leq\; \sqrt{\frac{\alpha}{1-\alpha}}.$$

The bound is 4.36 at 95% and 9.95 at 99%, against 1.645 and 2.326 for the normal. Two-point distributions approach it. They give a small gain most of the time and a large loss rarely. Position B in §2.3 has this shape.

Why it appears here. Every z_α in the chapter is a normal quantile. The table in §2.5 of days beyond three to six standard deviations, set against the normal's expectation, applies this tail arithmetic. So does its remark that under a normal a 25-sigma day should not happen once in the life of the universe. §3.1 says that Roy had nothing better than Chebyshev bounds in 1952. The Cantelli ceiling shows why these bounds gave no basis for practice. A multiplier that could be anything up to 4.36 at 95% is not an estimate. The narrow band of multipliers in §2.2 is therefore a property of the distributions considered plausible, not of all distributions.

Deeper. Feller, *An Introduction to Probability Theory and Its Applications*, vol. 1, 3rd ed. (Wiley, 1968), chapter VII, for the normal tail estimate. Roy (1952), in §4.1, for the bound used as a decision rule.

A.7 Skewness, kurtosis and fat tails

The idea. The mean and variance give the location and width of a distribution. The next two standardised moments describe its shape. Skewness measures asymmetry, and negative skew means that the long tail is on the loss side. Kurtosis measures how much of the variance comes from rare large deviations rather than many moderate ones. A distribution with high kurtosis and the same variance has more mass at the centre and far out, and less in the "shoulders" between. People use "fat-tailed" loosely, both for high kurtosis and for tails that decay like a power (A.23). The two usually go together but differ. Estimates of both moments depend on the same few extreme days that make them interesting. A sample kurtosis can therefore be very unstable. The formal part quantifies how unstable.

Formally. With mean μ and variance σ^2, skewness and kurtosis are

$$\gamma = \frac{E[(X-\mu)^3]}{\sigma^3}, \qquad K = \frac{E[(X-\mu)^4]}{\sigma^4}.$$

The **excess kurtosis** is $\kappa = K - 3$, which is zero for the normal. The main text uses both conventions. The kurtosis of 19 in §2.5 and the excess kurtosis of 16.1 in §6.5 describe the same data. The sample kurtosis estimates a fourth moment, so its own sampling variance involves the eighth. Suppose the eighth moment is infinite, as under a power-law tail with an exponent below eight (A.23). The sample kurtosis then converges more slowly than the usual $1/\sqrt{n}$. If the fourth moment is infinite, it does not converge at all. The fourth moment also governs the precision of the sample standard deviation. For IID data,

$$\frac{\mathrm{se}(\hat{\sigma})}{\sigma} \approx \sqrt{\frac{K-1}{4n}},$$

This equals $1/\sqrt{2n}$ for the normal and is three times larger at $K = 19$.

Why it appears here. In §2.5, the kurtosis falls from 19 to 8.5 once returns are standardised by their volatility forecast. This fall is the chapter's main evidence for changing volatility as the cause of most fat-tailedness (A.19). The observation in §2.2 that fattening a symmetric tail *lowers* the 95% multiplier reflects the shoulders. The Cornish–Fisher correction of §6.5 uses γ and κ. Its failure there is partly due to instability, because a correction proportional to a sample kurtosis inherits the noise of that kurtosis. The last formula qualifies the remark in §5.8 that the normal formula does better on its own terms. The 4.5% there at $n = 250$ is the case $K = 3$.

Deeper. Cont, "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues," *Quantitative Finance* 1(2) (2001), 223–236, the standard catalogue, including why moment estimates are fragile. Kim & White, "On More Robust Estimation of Skewness and Kurtosis," *Finance Research Letters* 1 (2004), for alternatives that do not depend on a few days.

A.8 The Student t distribution

The idea. The Student t is the simplest one-parameter way to fatten the normal's tails. Its degrees of freedom ν set how heavy they are. Small ν gives tails that decay like a power, and as ν grows the t approaches the normal. The t arises as a normal whose variance is itself drawn at random, which approximates a market whose volatility changes. Two facts matter in practice. First, moments exist only up to an order below ν, so a t with four degrees of freedom has a finite variance and an infinite kurtosis. Second, the raw t has variance $\nu/(\nu-2)$, not one. Analysts must therefore rescale it before using it as a standardised shock. Comparing a raw t quantile with a normal one compares two different scales.

Formally. The t density with ν degrees of freedom is proportional to $(1 + u^2/\nu)^{-(\nu+1)/2}$, so it decays like $|u|^{-(\nu+1)}$. That is a power-law tail with index $\xi = 1/\nu$ (A.23). The moment of order m is finite exactly when $m < \nu$. For $\nu > 2$ the variance is $\nu/(\nu-2)$, and for $\nu > 4$ the excess kurtosis is $6/(\nu-4)$. The unit-variance shock used throughout is $\varepsilon = \sqrt{(\nu-2)/\nu}\, t_\nu$, with tail multiplier

$$k_\alpha = \sqrt{\frac{\nu-2}{\nu}}\, t_\nu^{-1}(\alpha),$$

where t_ν^{-1} is the quantile function of the raw t. Computing the multiplier over all $\nu > 2$ gives two bounds. Every unit-variance t has $k_{0.95}$ below the normal's 1.645. None has $k_{0.99}$ above about 2.66. This maximum occurs near $\nu = 3.6$. The t is also $Z/\sqrt{W/\nu}$, with Z standard normal and W an independent χ^2_ν (A.15). It is a normal with a randomly scaled variance.

Why it appears here. The table of multipliers in §2.2 is the formula above at ν = 10, 5, 4 and 3. §6.4 fits a GARCH model with unit-variance t shocks. Its failure at 95% follows from the first bound, because the data's standardised 95% multiplier, 1.71, lies beyond every t. By the second bound, the same data's 99% multiplier, 2.90 in §2.2, also lies beyond every t. No choice of ν therefore fits these data at either level. The simulation of §5.8 draws from a t with four degrees of freedom, a distribution with infinite kurtosis. §5.6 compares ES with VaR under t tails.

Deeper. Bollerslev, "A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return," *Review of Economics and Statistics* 69(3) (1987), 542–547. This paper introduces GARCH with t shocks, the model of §6.4. Kotz & Nadarajah, *Multivariate t Distributions and Their Applications* (CUP, 2004), for the distribution itself.

A.9 The covariance matrix, quadratic forms and beta

The idea. The variance of a sum equals the sum of the variances plus twice the covariance of every pair of positions. The covariance matrix holds all those pairs. A portfolio's variance is the double sum over the matrix, written compactly as a

quadratic form. Two derived quantities help interpret the matrix. A position's covariance with the whole portfolio, divided by the portfolio's variance, is its beta to the portfolio. The beta measures how far the position tends to move per unit of portfolio movement. Weight times beta is the position's share of the portfolio's variance, and the shares sum to one. The beta quoted for stocks uses the same construction, with a market index in place of the portfolio.

Formally. Take one-day returns $r \in \mathbb{R}^N$ with covariance matrix Σ, where $\Sigma_{ij} = \mathrm{Cov}(r_i, r_j)$. The P&L $x'r$ of dollar positions x has

$$\mathrm{Var}(x'r) = x'\Sigma x = \sum_i \sum_j x_i x_j \Sigma_{ij} = \sigma_p^2.$$

Σ is symmetric and positive semi-definite, because no portfolio has negative variance. With weights $w = x/V$, the beta of asset i to the portfolio is $\beta_i = \mathrm{Cov}(r_i, w'r)/\mathrm{Var}(w'r) = (\Sigma w)_i/(w'\Sigma w)$, and

$$w_i\beta_i = \frac{x_i(\Sigma x)_i}{x'\Sigma x}, \qquad \sum_i w_i\beta_i = 1.$$

Every formula for marginal risk derives from the gradient $\nabla_x \sigma_p = \Sigma x/\sigma_p$ (A.27). A stock's market beta is $\mathrm{Cov}(r_i, r_m)/\mathrm{Var}(r_m)$ for an index return r_m. The β_i here is a regression slope. It is unrelated to the generalised Pareto scale β of §6.11.

Why it appears here. The σ_p of §1.5 and the $z_\alpha\sqrt{x'\Sigma x}$ of §3.1 are this quadratic form, and §5.3 builds the delta-normal VaR on it. The component VaR of §5.5 is the VaR times the variance share above, the position's beta to the portfolio times its weight. This explains how a 10% autos position carries 21% of the risk in §7.2 and §10.4. The equities of §1.1, "measured in beta", use the market beta. The best hedge of §7.2 minimises $x'\Sigma x$ in one coordinate.

Deeper. The companion chapter Portfolio Construction and the Covariance Matrix. Its appendix covers quadratic forms and positive definiteness, and its section 6 covers estimating Σ.

A.10 Elliptical distributions

The idea. Elliptical distributions are the multivariate families whose density contours are ellipses. The multivariate normal and the multivariate t are the cases that matter here. Their defining property is that every linear combination of the components has the same standardised shape. A portfolio of elliptically distributed assets is therefore itself location-scale (A.5), with a scale given by the covariance quadratic form (A.9). Within this family, correlation describes dependence completely and VaR is well behaved. Outside it, neither holds. Cases outside the family include non-linear payoffs, lumpy losses and dependence that strengthens in the tails.

Formally. A random vector $Y \in \mathbb{R}^N$ is elliptical, with location m and dispersion matrix Ω, if

$$x'Y \overset{d}{=} x'm + \sqrt{x'\Omega x}\, Z \qquad \text{for every } x \in \mathbb{R}^N,$$

for one symmetric random variable Z that does not depend on x. When variances are finite, Ω is proportional to the covariance matrix. For the loss $L = -x'Y$ this gives $\mathrm{VaR}_\alpha(x) = -x'm + k\sqrt{x'\Omega x}$, with $k = q_\alpha(Z)$, which is non-negative for $\alpha \geq 1/2$. Because $\sqrt{x'\Omega x}$ is a norm, it obeys the triangle inequality. VaR is therefore sub-additive across positions. A merged book's VaR never exceeds the sum of the parts' VaRs. Combining a Gaussian or t dependence structure with other marginal distributions generally gives a non-elliptical distribution (A.32).

Why it appears here. §5.2 uses ellipticity to say when VaR is a scaled volatility for a whole portfolio. It lists option books and credit as the exceptions, because their P&L is not a linear combination of elliptical factors. The first bullet of §5.7, that VaR is sub-additive whenever the joint distribution is elliptical, is the triangle inequality above. §13.4 relies on it to say that an incremental VaR can never exceed the stand-alone one under the normal model.

Deeper. Embrechts, McNeil & Straumann (2002), in §4.3, which gives the result behind §5.7 and the clearest account of what correlation can and cannot do. Fang, Kotz & Ng, *Symmetric Multivariate and Related Distributions* (Chapman & Hall, 1990), for the theory.

A.11 The central limit theorem, and how slowly it reaches the tails

The idea. The sum of many independent shocks with finite variance looks normal, whatever the shocks looked like. So §5.4 expects multi-day returns to look more normal than daily ones. Two caveats matter for VaR. First, the theorem does not state how many shocks count as many. The number depends on the tails. The fatter the tails, the slower the convergence. Second, the theorem controls absolute errors in probability. These errors are small in the middle of a distribution but large relative to the tiny probabilities of a tail. A sum can be normal to within half a percentage point everywhere and still have a 1% quantile far from the normal one.

Formally. Suppose $r_1, r_2, \ldots$ are IID with mean μ and variance $\sigma^2 < \infty$. Then $(R_h - h\mu)/(\sigma\sqrt{h})$ converges in distribution to a standard normal, with $R_h = \sum_{j=1}^{h} r_j$ as in §5.4. When the third absolute moment is finite, the Berry–Esseen theorem bounds the error:

$$\sup_u \left| P\Big(\frac{R_h - h\mu}{\sigma\sqrt{h}} \le u\Big) - \Phi(u) \right| \;\le\; \frac{C\,E|r_1 - \mu|^3}{\sigma^3\sqrt{h}},$$

with C a universal constant below one half. An error of that size at the point where Φ is 0.01 can be a large fraction of 0.01. The far tail therefore converges last. With a tail index above one half (A.23), the variance is infinite and the theorem fails. Suitably rescaled sums then converge instead to a non-normal stable law.

Why it appears here. §5.4 invokes the theorem for its claim that the h-day multiplier drifts back toward the normal one as h grows. By itself, that drift would make $\sqrt{h}$ scaling overstate the 99% quantile. The tail caveat explains why the drift is slow. It also explains why §5.4 finds the drift offset by volatility clustering and jumps (A.19, A.22), which act in the opposite direction.

Deeper. Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2, 2nd ed. (Wiley, 1971), for Berry–Esseen and the stable laws. The companion chapter Simple and Log Returns covers what aggregation over time does to return distributions.

A.12 The sampling error of a quantile, and simulation studies

The idea. A quantile estimated from n observations is uncertain because the number of observations that land beyond the true quantile is random. If slightly too many land in the tail, the estimate moves outward. If slightly too few land there, it moves inward. The size of the move depends on how densely the data sit near the quantile. In the middle of a distribution the data are dense, and one extra observation barely moves the cut-off. In the tail the data are sparse, and one extra observation moves the cut-off a long way. This is why tail quantiles are hard to estimate. A simulation study checks the large-sample answer directly. It generates many histories from a model with a known true quantile, applies the estimator to each, and measures the spread of the estimates around the true value.

Formally. Let q be the true quantile at tail probability p, with density $f(q) > 0$ there. Of n IID observations, the number beyond q is binomial with variance $np(1-p)$ (A.2). The fraction beyond therefore has standard deviation $\sqrt{p(1-p)/n}$. Moving the cut-off a small distance changes the expected fraction beyond it by about $f(q)$ times that distance. Equating the two gives

$$\sqrt{n}\,(\hat{q} - q) \;\to\; N\Big(0,\; \frac{p(1-p)}{f(q)^2}\Big), \qquad \mathrm{se}(\hat{q}) \approx \frac{1}{f(q)}\sqrt{\frac{p(1-p)}{n}}.$$

The same binomial gives an exact statement. The k-th order statistic is at most ℓ exactly when at least k of the n observations are. $P(L_{(k)} \le \ell)$ is therefore a binomial tail sum in $F_L(\ell)$. That gives distribution-free confidence intervals for a quantile, between two order statistics. Under volatility clustering the observations are not independent, and the true error is larger than either calculation says. A simulation study replaces the algebra with M synthetic histories, and its own Monte Carlo error shrinks in proportion to $1/\sqrt{M}$.

Why it appears here. §5.8 states this standard error and then runs exactly such a study, on 20,000 histories of Student t returns, to get its table. A one-year 99% VaR is uncertain by about ±28%, and a 95% VaR by ±18%. §6.9 applies the formula to Monte Carlo with $n = M$ scenarios. The advice in §11.3 to treat differences smaller than the sampling error as noise follows from it, and so does the preference for 95% in §14.1.

Deeper. Jorion (1996), in §4.4, for standard errors of VaR and the case for reporting them. Van der Vaart, *Asymptotic Statistics* (CUP, 1998), chapter 21, on quantiles and order statistics.

A.13 Maximum likelihood

The idea. Maximum likelihood chooses the parameters of a model under which the observed data would have been most probable. It is the default way to fit a parametric model. Under regularity conditions, no other estimator is more precise in large samples. For a time series, the likelihood factorises into one-step-ahead conditional densities. Fitting a volatility model therefore means choosing the parameters whose daily forecasts make the realised returns most probable. One robustness result matters. If the shocks are not normal, fitting a GARCH model as if they were normal still estimates the volatility parameters consistently, provided that the analyst corrects the standard errors.

Formally. For observations with density $f(\cdot\,;\vartheta)$, the likelihood is $\mathcal{L}(\vartheta) = \prod_t f(r_t;\vartheta)$, and the estimator is $\hat{\vartheta} = \arg\max_\vartheta \ln \mathcal{L}(\vartheta)$. For a model $r_t = \sigma_t(\vartheta)\,\varepsilon_t$ with shock density g,

$$\ln \mathcal{L}(\vartheta) = \sum_t \Big[\ln g(r_t/\sigma_t(\vartheta)) - \ln \sigma_t(\vartheta) \Big],$$

Software maximises it numerically. Under regularity, $\hat{\vartheta}$ is approximately normal with covariance $\mathcal{I}(\vartheta)^{-1}/n$, where $\mathcal{I}$ is the Fisher information per observation. When the fit takes g to be normal but the true shocks are not, the estimator is quasi-maximum likelihood. It is still consistent for the volatility parameters, with a "sandwich" covariance in place of $\mathcal{I}^{-1}$. For Bernoulli hits, the maximum-likelihood estimate of the hit probability is simply $\hat{\pi} = X/T$.

Why it appears here. §6.4 estimates the GARCH parameters and the t degrees of freedom "jointly with the recursion by maximum likelihood". It warns that the optimisation can be unstable on short samples. §6.11 fits the generalised Pareto shape and scale the same way. The $\hat{\pi} = X/T$ of §12.2 is the Bernoulli estimate, and every likelihood-ratio test in §12 compares two maximised likelihoods (A.15).

Deeper. Foundations of Econometrics, section 5.4, on maximum likelihood and quasi-likelihood. Bollerslev & Wooldridge, "Quasi-Maximum Likelihood Estimation and Inference in Dynamic Models with Time-Varying Covariances," *Econometric Reviews* 11(2) (1992), 143–172.

A.14 Hypothesis tests: size, power and p-values

The idea. A test checks whether data are consistent with a claim, the null hypothesis. It measures how surprising the data would be if the claim were true. A test can err in two ways. It can reject a true claim, a Type I error. Or it can fail to reject a false claim, a Type II error. The size of a test is its Type I error rate, fixed in advance at, say, 5%. Its power is the probability of rejecting a false claim. Power depends on how false the claim is and on how much data there is. A p-value is the probability, under the null, of a result at least as extreme as the one observed. A one-sided test looks for departures in one direction only. This gives more power when only that direction matters. For VaR backtests, power is the binding constraint. With a year of data, no test can reliably tell a 1% breach rate from a 2% one.

Formally. A test rejects H_0 when a statistic falls in a rejection region. Its size is $P(\text{reject} \mid H_0)$. Its power against an alternative H_1 is $P(\text{reject} \mid H_1)$, and the Type II error rate is one minus the power. For a backtest, H_0 is $X \sim \text{Binomial}(T, p)$. The one-sided rule "reject if $X \geq c$" has size $P(X \geq c \mid p)$ and power $P(X \geq c \mid p_1)$ against a true rate $p_1 > p$. The one-sided p-value of an observed count x is $P(X \geq x \mid p)$. Because X is discrete, only certain sizes are available. At $T = 250$ and $p = 0.01$, the rule $X \geq 6$ has size 4.1%, and the rule $X \geq 5$ has size 10.8%. A nominal 5% test is therefore really a 4.1% test. Testing repeatedly inflates the size. A rule with a 1% false-alarm rate per look, applied every week, rejects a correct model far more often than 1% of the time over a year.

Why it appears here. §12.2–§12.5 use these pieces. They use Kupiec's statistic and its p-values. They use the traffic light, whose zone probabilities in §12.4 are a size and a Type II error rate. Of correct models, 10.8% leave green. Of models with a doubled breach rate, 44% stay green. They use the power table of §12.5, which is the reason §12 recommends backtesting at 95%. §6.8 and §6.13 report which methods the coverage and independence tests reject. The rule in §13.5 of eight breaches in 60 days is a one-sided test at about 1%. Used weekly, it rejects a correct model in about one year in eight.

Deeper. Foundations of Econometrics, sections 4.6 and 4.7, on tests, p-values and power. Casella & Berger, *Statistical Inference*, 2nd ed. (Duxbury, 2002), chapter 8.

A.15 Likelihood-ratio tests, Wilks' theorem and χ^2

The idea. To test a restriction, fit the model twice, once with the restriction imposed and once without. If the restriction is true, imposing it should cost little likelihood. If it is false, imposing it should cost a lot. The likelihood-ratio statistic measures this cost. Wilks' theorem gives its large-sample distribution under the null, whatever the model. The distribution is chi-squared,

with as many degrees of freedom as there are restrictions. This one result gives the distribution of every test statistic in §12.2 and §12.3.

Formally. Let $\hat{\vartheta}$ be the unrestricted maximum-likelihood estimate and $\hat{\vartheta}_0$ the estimate under H_0. Then

$$\text{LR} = -2 \ln \frac{\mathcal{L}(\hat{\vartheta}_0)}{\mathcal{L}(\hat{\vartheta})} \;\to\; \chi^2_k \quad \text{under } H_0,$$

where k is the number of independent restrictions. χ^2_k is the distribution of a sum of k squared independent standard normals. The 95th percentile is 3.84 for $k = 1$ and 5.99 for $k = 2$. Kupiec's test imposes one restriction: the hit rate equals p. Christoffersen's independence test also imposes one restriction: the two transition probabilities of A.16 are equal. Their sum tests both, with $k = 2$. The approximation needs enough hits. With expected counts of two or three, the exact binomial probabilities of A.14 are more reliable. A zero count enters the likelihood as $0 \ln 0 = 0$.

Why it appears here. The LR_{uc} of §12.2, with its threshold of 3.84, and the LR_{ind} and $\text{LR}_{\text{cc}} \sim \chi^2_2$ of §12.3 are this construction applied to the Bernoulli and Markov likelihoods. §12.2 remarks that the exact binomial test is a different test, whose p-value need not match. That is the gap between the asymptotic and the exact distributions.

Deeper. Casella & Berger, *Statistical Inference*, 2nd ed. (Duxbury, 2002), sec. 8.2.1 and 10.3. Christoffersen (1998), in §4.4, for the application to VaR.

A.16 Two-state Markov chains

The idea. A Markov chain is a sequence whose next state depends on the current state and on nothing earlier. With two states, hit or no hit, two numbers define the chain. One is the probability of a hit after a quiet day. The other is the probability of a hit after a hit. Independence is the special case in which the two are equal. A chain with a much larger probability of a hit after a hit produces clustered hits, even when its long-run hit rate is right. Christoffersen's test detects exactly this failure.

Formally. With $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$, the transition matrix is

$$\begin{pmatrix} 1-\pi_{01} & \pi_{01} \\ 1-\pi_{11} & \pi_{11} \end{pmatrix},$$

and the long-run fraction of hit days is $\pi_{01}/(1 - \pi_{11} + \pi_{01})$. The likelihood of an observed hit sequence is

$$(1-\pi_{01})^{n_{00}}\, \pi_{01}^{n_{01}}\, (1-\pi_{11})^{n_{10}}\, \pi_{11}^{n_{11}},$$

which the transition frequencies $\hat{\pi}_{01}$ and $\hat{\pi}_{11}$ of §12.3 maximise. Under independence, $\pi_{01} = \pi_{11}$, and the gaps between hits are geometric (A.2). With $\pi_{11} > \pi_{01}$, short gaps after a hit are too common. A long-run rate near the target can coexist with this pattern. Transition probabilities of $\pi_{01} = 0.01$ and $\pi_{11} = 0.10$ give a long-run hit rate of 1.1%, close to a 99% model's claim. Yet after a hit, the probability of another hit is 10 times what it should be.

Why it appears here. The independence test of §12.3 is a likelihood ratio (A.15) between this chain and the independent one. Its summary statistic $\hat{\pi}_{11}$ is the "hit after a hit" column of §6.13: 8.8–10.3% for the equal-weighted 99% methods, against 1%. Because the chain looks only one day back, §12.3 adds duration tests for slower clustering.

Deeper. Christoffersen (1998), in §4.4. Norris, *Markov Chains* (CUP, 1997), chapter 1.

A.17 Autocorrelation, HAC standard errors and the Diebold–Mariano test

The idea. Standard errors computed as though observations were independent are too small when the observations are positively autocorrelated. Each new observation partly repeats the last, so the sample holds fewer independent pieces of information than its length suggests. Overlapping windows create exactly this problem. Ten-day returns computed every day share nine of their 10 days with their neighbours, so consecutive 10-day outcomes are nearly the same draw. The problem has two remedies. One avoids the overlap, by backtesting at one day. The other keeps the overlap and widens the standard errors to allow for the autocorrelation. Estimators that are consistent under heteroskedasticity and autocorrelation (HAC), such as Newey and West's, do this. The Diebold–Mariano test applies the second remedy to a comparison of two forecasts.

Formally. The autocorrelation at lag j of a stationary series y_t is $\text{Corr}(y_t, y_{t-j})$. For IID daily returns, overlapping h-day sums have autocorrelation $(h - j)/h$ at lags $j < h$: 0.9 at lag one for 10-day windows. The variance of the mean of T

autocorrelated observations is approximately Ω/T. Here $\Omega = \gamma_0 + 2\sum_{j\geq 1}\gamma_j$ is the long-run variance, and γ_j is the lag-j autocovariance, not the skewness of §6.5. Newey and West estimate it as

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{m}\Big(1 - \frac{j}{m+1}\Big)\hat{\gamma}_j,$$

with a truncation lag m and declining weights that keep $\hat{\Omega}$ non-negative. To compare two forecasts, let d_t be the day-t difference in their losses, for instance in pinball loss (A.30). The Diebold–Mariano statistic is $\bar{d}/\sqrt{\hat{\Omega}/T}$, with $\hat{\Omega}$ computed from the d_t. It is approximately standard normal when the two forecasts are equally accurate.

Why it appears here. §12.1 and §12.8 require one-day backtests, because overlapping 10-day hits are autocorrelated by construction. Autocorrelated hits invalidate the tests of §12, all of which assume independent days. §12.6 tests whether one model's average pinball loss is significantly below another's. It uses standard errors robust to autocorrelation, in the manner of Diebold and Mariano (1995). That is this construction. The same caution applies to the direct multi-day risk from overlapping returns in §14.6. A 500-day window holds only 25 non-overlapping 20-day periods.

Deeper. Foundations of Econometrics, section 4.5, on serial correlation and HAC standard errors. Newey & West, "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica* 55(3) (1987), 703–708. Diebold & Mariano (1995), in §4.6.

Part II — Returns and volatility. Idea 2 of §1.6 says that at short horizons VaR is a volatility forecast times a constant. These five entries concern the forecast. They cover what the forecast measures, why volatility moves, the two recursions the chapter uses to track it, and the moves that no recursion predicts.

A.18 Simple and log returns, annualisation and basis points

The idea. Two measures of return exist, and each is exact for a different job. The simple return, the fractional change in price, is what a position actually earns. It adds across positions. A portfolio's simple return is the weighted sum of its holdings' returns. The log return, the change in the logarithm of price, adds across time. A week's log return is the sum of its days' log returns. No single measure does both. VaR uses each where it is exact: simple returns to turn positions into P&L, and log returns to aggregate over days. At daily horizons the two are close. Over long horizons and in crashes they differ. Annualising a daily volatility by $\sqrt{252}$ uses the same aggregation argument, and it holds only if days are independent.

Formally. For a price S_t adjusted for dividends and splits (A.51), the simple return is $r_t = S_t/S_{t-1} - 1$, and the log return is $\ln(1 + r_t) = \ln S_t - \ln S_{t-1}$. A book's one-day P&L is exactly $\sum_i x_i r_{i,t}$ in simple returns, and an h-day log return is exactly the sum of the daily ones. Since $\ln(1 + r) = r - r^2/2 + \ldots$, the two differ by about half the squared return. The difference is 0.5 basis points on a 1% day and about 2 percentage points on a 20% day. Under IID returns, variances add across days, so an annual volatility is $\sqrt{252} = 15.87$ times a daily one. A **basis point** is 10^{-4}, a hundredth of a percentage point. The r_t of the notation block denotes both kinds of return. §5.3 and §14.2 use simple returns, and §5.4 uses log returns.

Why it appears here. §5.4 sums log returns to derive the square-root-of-time rule. The P&L formula of §5.3, exact "for a buy-and-hold portfolio of simple returns over one period", and the `pct_change` of §14.2 use simple returns. The conversion table of §2.2 divides annual volatility by $\sqrt{252}$, and the $9.65B$ a year of §8.2 multiplies by it. Both inherit the IID caveat. §6.13 reports pinball losses in basis points.

Deeper. The companion chapter Simple and Log Returns covers this choice in full.

A.19 Volatility clustering and conditional heteroskedasticity

The idea. Large moves tend to follow large moves, of either sign, and quiet days follow quiet days. These are the calm and violent periods of §2.5. A volatility model uses this pattern. It forecasts the size of tomorrow's move, not its direction. A series whose conditional variance changes over time in this way is conditionally heteroskedastic. The consequence for tails is the most important fact behind §2.5. Mixing calm normal days with violent normal days produces a sample with far more extreme days, and more near-zero days, than any single normal would. The result is high kurtosis, although no fat-tailed law generated any individual day. Dividing each day by its own volatility removes the mixing.

Formally. Write $r_t = \sigma_t \varepsilon_t$, with the ε_t IID, mean zero and variance one, and with σ_t known on the evening of day $t-1$. Then r_t is uncorrelated over time but r_t^2 is not, and the conditional variance $\mathrm{Var}_{t-1}(r_t) = \sigma_t^2$ changes over time. Unconditionally, r_t is a scale mixture. Because ε_t is independent of σ_t, its kurtosis (A.7) is

$$K(r) = K(\varepsilon)\,\frac{E[\sigma_t^4]}{\left(E[\sigma_t^2]\right)^2} \;\geq\; K(\varepsilon),$$

with equality only if σ_t is constant, by Jensen's inequality. Even with normal shocks, where $K(\varepsilon) = 3$, any variation in volatility therefore raises the unconditional kurtosis above three. If σ_t is 1 or 2 with equal probability, $K(r) = 4.08$.

Why it appears here. In §2.5, the kurtosis falls from 19 to 8.5 when returns are standardised by a volatility forecast. This identity explains the fall: most of the 19 was mixing. It also explains why Idea 2 of §1.6 calls volatility "the part that moves". The same identity explains why §6.1 expects the conditional column to beat the unconditional one. It also explains why §6.13 finds that every method that updates volatility daily has roughly half the breach clustering of its equal-weighted counterpart.

Deeper. Mandelbrot, "The Variation of Certain Speculative Prices," *Journal of Business* 36(4) (1963), 394–419, an early statement of the clustering. Engle (1982), in §4.2, for the model that made it forecastable. The companion chapter Market Regimes and Hidden Markov Models takes the regime view of the same phenomenon.

A.20 Exponentially weighted moving averages

The idea. An equal-weighted window treats a day from a year ago as just as informative as yesterday, then drops it completely when it leaves the window. An exponentially weighted average weights every past day, with weights that shrink geometrically

with age. Recent days dominate, and old days lose weight gradually. One parameter, the decay, sets the memory length. It trades quick reaction, which needs a short memory, against precise estimation, which needs a long one. A few summaries of the weights quantify the trade: how fast they halve, how old the average observation is, and how many equally weighted observations would give the same precision.

Formally. With decay $\lambda \in (0,1)$, the recursion $\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$ puts weight $(1-\lambda)\lambda^j$ on the squared return from $j+1$ days ago. At $\lambda = 0.94$:

- The weights halve every $\ln 0.5 / \ln \lambda = 11.2$ days.
- The weighted average age of the data is $1/(1-\lambda) = 16.7$ days.
- The effective sample size is $(1+\lambda)/(1-\lambda) = 32$. This is the number of equally weighted observations whose average would be as precise. It equals Kish's ratio of the squared sum of the weights to the sum of their squares.
- The starting value's weight after j updates is λ^j, which is 2.4% after 60 days and 0.2% after 100. A warm-up must run this long before the forecast no longer depends on its starting value.

The forecast for every future day equals tomorrow's, because EWMA has no long-run level to revert to (A.21).

Why it appears here. §6.3 states the first three numbers for $\lambda = 0.94$. It concludes that EWMA behaves like a 32-day window, an equivalence that §8.2 marks as approximate. The "effective memory of about 30 observations" of §3.3 is the same calculation. The `ewma_vol` of §14.2 starts the recursion from the first 60 days. §14.2 calls its earliest values "a warm-up rather than true forecasts". The last bullet says how long that lasts. EWMA supplies the scale in the elevenfold move of §1.6, in the standardisation of §2.5, and in the filtered simulation of §6.8 and §14.

Deeper. J.P. Morgan/Reuters (1996), in §4.1: the *RiskMetrics Technical Document*, which chose 0.94. The companion chapter Portfolio Construction and the Covariance Matrix, section 6.3 and its appendix entry on effective sample size, for the same argument applied to a whole covariance matrix.

A.21 GARCH(1,1), persistence and asymmetric volatility

The idea. GARCH adds one ingredient to EWMA: a long-run level of variance toward which the forecast reverts. Today's variance combines three things: the long-run level, yesterday's squared return and yesterday's variance. The sum of the last two weights, the persistence, measures how slowly a shock to volatility decays. Mean reversion gives GARCH sensible multi-day forecasts. From a calm start it predicts rising volatility, and from a turbulent start falling volatility. EWMA is the limiting case with persistence exactly one and no long-run level, called integrated GARCH. Asymmetric variants let falls raise volatility more than rises of the same size, the effect that §6.4 notes in equities. GARCH makes tomorrow's variance a fixed function of past returns. Stochastic-volatility models, the main alternative, give the variance a random shock of its own.

Formally. With $r_t = \sigma_t \varepsilon_t$, GARCH(1,1) is

$$\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\, \sigma_{t-1}^2, \qquad \omega > 0,\ a, b \geq 0.$$

The model is covariance-stationary when $a + b < 1$, with long-run variance $\bar{\sigma}^2 = \omega/(1-a-b)$. The j-step forecast is $E_t[\sigma_{t+j}^2] = \bar{\sigma}^2 + (a+b)^{j-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$, so a volatility shock has a half-life of $\ln 0.5 / \ln(a+b)$ days. Setting $\omega = 0$, $a = 1-\lambda$ and $b = \lambda$ gives EWMA. Any model with $a + b = 1$ is integrated (IGARCH) and has no level to revert to. The GJR model adds a term for down days, $\sigma_t^2 = \omega + (a + g\,\mathbf{1}\{r_{t-1} < 0\})\, r_{t-1}^2 + b\,\sigma_{t-1}^2$ with $g > 0$. Nelson's EGARCH models $\ln \sigma_t^2$ instead. That keeps the variance positive without parameter constraints, and lets the sign of ε_{t-1} enter directly. The asymmetry is called the **leverage effect**, after one proposed explanation. A falling share price raises a firm's debt relative to its equity, which makes the equity riskier. Most texts write α_1 and β_1 for a and b. This chapter already uses α for the confidence level.

Why it appears here. §5.4 uses the multi-step forecast to show that $\sqrt{h}$ scaling understates risk from calm starting points and overstates it from turbulent ones, and §11.5 then measures this effect. §6.4 fits GARCH with t shocks and names GJR and EGARCH. The description of EWMA in §6.3 as an "integrated" GARCH, and the corresponding row of §8.2, are the special case above. §14.6 lists GARCH as the upgrade for when multi-day forecasts matter.

Deeper. Engle (1982) and Bollerslev (1986), in §4.2. Glosten, Jagannathan & Runkle, "On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks," *Journal of Finance* 48(5) (1993), 1779–1801. Nelson, "Conditional Heteroskedasticity in Asset Returns: A New Approach," *Econometrica* 59(2) (1991), 347–370. Both papers give the asymmetric models. Foundations of Econometrics, section 8.6.

A.22 Jumps

The idea. A diffusion moves continuously, so over a short enough interval a large move is vanishingly unlikely. Real prices also jump, for example in a crash, after a surprise announcement or in a gap at the open. A jump does not shrink with the interval, as a diffusion's moves do. Jumps matter for VaR in two ways. First, a volatility model updates its estimate only after a jump. So the first day of a shock is a breach for every model. Second, a jump too rare to show in a one-day quantile can be common enough over 10 days to dominate the 10-day quantile. This is one way square-root-of-time scaling understates long-horizon risk. Here a jump is a jump in a price path. A.1 uses jumps of a distribution function, which are a different object.

Formally. A jump-diffusion adds a compound Poisson term to a continuous part. Over h days the return is $\mu h + \sigma\sqrt{h}\,Z + \sum_{k=1}^{N_h} J_k$. Here Z is standard normal. N_h is Poisson with mean ηh for a jump intensity η per day. The jump sizes J_k are IID. The chance of at least one jump in h days is $1 - e^{-\eta h} \approx \eta h$. Take η = 0.005 and a large negative jump. A single day then contains a jump with probability 0.5%, less than the 1% tail. The one-day 99% VaR therefore depends on how often the jump occurs, not on how large it is. Position B in §2.3 behaves the same way. Over 10 days, the chance of a jump is 4.9%. The jump's size then largely sets the 10-day 99% VaR. The cumulants of the jump term grow in proportion to h, so its excess kurtosis falls in proportion to $1/h$. This is the central limit theorem (A.11), which applies eventually.

Why it appears here. §5.4 says that jumps and volatility clustering push multi-day risk above what $\sqrt{h}$ predicts. It cites Danielsson and Zigrand (2006), whose model is a jump-diffusion. The arithmetic above is one mechanism. §11.4 observes that volatility rises in jumps. §13.7 reminds the reader that the first day of every shock was a breach for every model. In both cases the jump is in the volatility path, and no recursion anticipates it. The gaps of §13.3 are jumps that affect a stop order (A.43).

Deeper. Merton, "Option Pricing When Underlying Stock Returns Are Discontinuous," *Journal of Financial Economics* 3(1–2) (1976), 125–144, for the jump-diffusion model. Danielsson & Zigrand (2006), in §4.5. The companion chapter Stochastic Processes, for the general framework.

Part III — Tails and risk measures. Part I treated VaR as a quantile to estimate. This part treats it as one of many ways to turn a loss distribution into a single number. It covers the shape of the far tail and how to extrapolate into it. It covers the properties such a number should have. It covers what expected shortfall gains and what it costs. These 11 entries supply the mathematics behind §5.5–§5.9, §6.5, §6.9 and §6.11.

A.23 Power-law tails and the tail index

The idea. Normal tails decay faster than exponentially. Power-law tails decay as a power of the loss. The difference is qualitative. Under a power law, the chance that a loss beyond some large level also exceeds twice that level is a fixed fraction. That fraction stays the same however large the level. Big losses do not become relatively rarer as they get bigger, as they do under a normal. One number, the tail index, measures how heavy the tail is and decides which moments exist. A heavy enough tail has no finite variance. A heavier one still has no finite mean. At that point expected shortfall is infinite, and diversifying can raise VaR.

Formally. A loss has a power-law, or regularly varying, right tail with index $\xi > 0$ if

$$\frac{P(L > s\ell)}{P(L > \ell)} \;\to\; s^{-1/\xi} \quad \text{as } \ell \to \infty, \text{ for every } s > 0.$$

Roughly, then, $P(L > \ell) \approx C\ell^{-1/\xi}$ far out, with tail exponent $1/\xi$. Moments $E[|L|^m]$ are finite for $m < 1/\xi$ and infinite for $m > 1/\xi$. A finite mean needs $\xi < 1$, a finite variance $\xi < 1/2$ and a finite kurtosis $\xi < 1/4$. The Student t with ν degrees of freedom has $\xi = 1/\nu$ (A.8). At $\xi = 1/4$, a loss beyond any large level exceeds twice that level with probability $2^{-4} = 1/16$. The classical estimator, Hill's, uses the k largest losses, $L_{[1]} \geq L_{[2]} \geq \ldots$:

$$\hat{\xi} = \frac{1}{k}\sum_{i=1}^{k} \ln \frac{L_{[i]}}{L_{[k+1]}}.$$

Like every tail estimator, it trades bias against variance in its choice of k (A.24).

Why it appears here. The counterexamples of §5.7 need extremely heavy tails, with no finite mean. That is the region $\xi \geq 1$, where the sub-additivity result of Danielsson and co-authors fails. §6.11 reports ξ between about 0.2 and 0.4 for daily equity returns, a tail exponent of 2.5 to 5. Its ES formula requires $\xi < 1$. The warning in A.7 about sample kurtosis says that ξ near or above 1/4 leaves the fourth moment barely finite or infinite.

Deeper. Embrechts, Klüppelberg & Mikosch, *Modelling Extremal Events for Insurance and Finance* (Springer, 1997), for regular variation and the Hill estimator in full. Hill, "A Simple General Approach to Inference About the Tail of a Distribution," *Annals of Statistics* 3(5) (1975), 1163–1174.

A.24 Extreme value theory and peaks over threshold

The idea. The central limit theorem says that sums approach a normal whatever the summands. Extreme value theory has a counterpart for tails. In almost any distribution, the excesses over a high enough threshold follow one family, the generalised Pareto. Its shape parameter is the tail index of A.23. That universality allows extrapolation. Instead of assuming a whole distribution, assume only that the threshold is high enough for the limit to hold. Fit the family's two parameters to the excesses, the peaks over threshold, and read off quantiles beyond the data. The cost is the choice of threshold. Too low a threshold means the limit does not yet apply, which biases the fit. Too high a threshold leaves too few excesses, which makes the fit noisy.

Formally. For a threshold u, the excess distribution is $F_u(y) = P(L - u \leq y \mid L > u)$. The generalised Pareto distribution function is

$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \qquad y \geq 0.$$

The Pickands–Balkema–de Haan theorem covers every F_L in the domain of attraction of an extreme value law. That includes the normal, the Student t and most continuous distributions used in practice. For each, a scale $\beta(u)$ exists with $\sup_y |F_u(y) - G_{\xi,\beta(u)}(y)| \to 0$ as u rises to the top of the support. $\xi > 0$ gives a power-law tail. $\xi = 0$, read as the limit $1 - e^{-y/\beta}$, gives the exponential tail of distributions such as the normal. $\xi < 0$ gives a tail with a finite end point. The

companion result for block maxima, the Fisher–Tippett–Gnedenko theorem, gives the generalised extreme value distribution with the same ξ. One diagnostic for the threshold is the mean excess $E[L - v \mid L > v]$. In a generalised Pareto tail, it rises linearly with v, with slope $\xi/(1-\xi)$. Choose u where the empirical version becomes linear.

Why it appears here. §6.11 builds on this theorem. Its VaR and ES formulas invert and integrate the fitted $G_{\xi,\beta}$ above the threshold. Its weakness is the bias–variance trade-off above. §6.11 says "The choice of threshold is a judgement with a visible effect on the result". It also says the estimates are noisy because the fit uses "the few dozen most extreme observations". McNeil and Frey's method, which §6.11 calls "the right way round", applies the fit to standardised residuals (A.5). Their distribution is closer to fixed than that of raw returns.

Deeper. McNeil & Frey (2000), in §4.2. Coles, *An Introduction to Statistical Modeling of Extreme Values* (Springer, 2001), chapter 4, on threshold models.

A.25 Cumulants and the Cornish–Fisher expansion

The idea. Cumulants are combinations of the moments with two properties. The cumulants of independent summands add, and the normal has none beyond the second. That makes them the natural measures of "how far from normal". After standardising, the third cumulant is the skewness and the fourth is the excess kurtosis. The Cornish–Fisher expansion uses them to correct a normal quantile for a mildly non-normal distribution, as a power series in the departure. Like any truncated series, it is accurate when the correction is small and can give absurd values when it is not. For a large enough kurtosis, the "corrected" quantiles stop increasing with the probability level. No quantile function behaves that way.

Formally. The cumulants κ_m are the coefficients of the cumulant generating function, $\ln E[e^{sX}] = \sum_{m\geq 1} \kappa_m s^m/m!$. Here κ_1 is the mean and κ_2 the variance. For a standardised variable, $\kappa_3 = \gamma$ and $\kappa_4 = \kappa$, the skewness and excess kurtosis of A.7. The main text's unsubscripted κ is this standardised fourth cumulant. Cumulants of independent sums add. For a sum of h IID variables, the standardised third and fourth cumulants therefore shrink in proportion to $h^{-1/2}$ and h^{-1}. This is the central limit theorem (A.11), stated in cumulants. Truncating the Edgeworth expansion of the distribution function and inverting it gives the formula of §6.5:

$$\tilde{z} = z + \frac{(z^2-1)\gamma}{6} + \frac{(z^3-3z)\kappa}{24} - \frac{(2z^3-5z)\gamma^2}{36}.$$

For $\tilde{z}$ to be a valid quantile, it must increase with z. With $\gamma = 0$ its derivative is $1 + (z^2-1)\kappa/8$, which is smallest at $z = 0$. The expansion is therefore monotone only when $\kappa \leq 8$.

Why it appears here. The modified VaR of §6.5 is this expansion with the sample γ and κ. It fails on daily data. At $\kappa = 16.1$, its 99% multiplier is 6.21 against an empirical 2.89. That is twice the largest kurtosis at which the expansion is even monotone. §6.10 lists Cornish–Fisher on the moments of a delta-gamma P&L as one way to find its quantile (A.36). The same check applies there.

Deeper. Cornish & Fisher, "Moments and Cumulants in the Specification of Distributions," *Revue de l'Institut International de Statistique* 5(4) (1938), 307–320. Jaschke, "The Cornish–Fisher Expansion in the Context of Delta-Gamma-Normal Approximations," *Journal of Risk* 4(4) (2002), on the expansion applied to option books and the limits of its accuracy.

A.26 Coherent and convex risk measures

The idea. A risk measure maps a loss distribution to an amount of money. The most useful reading of that amount is as capital, the cash a holder would add to make the position acceptable. Four properties follow naturally from that reading. Artzner and co-authors called a measure with all four coherent. Three are uncontroversial. A position that always loses less is less risky. Adding cash reduces the requirement one for one. Doubling a position doubles the requirement. The fourth, sub-additivity, says that merging two positions cannot need more capital than holding them separately. So diversification never hurts. VaR fails it. A later relaxation, convexity, keeps the idea of diversification but drops proportional scaling. That lets a measure treat a very large position as more than proportionally risky, which is closer to how liquidity behaves.

Formally. For losses L_1 and L_2, a risk measure ρ is coherent if it has four properties. Monotonicity means that $L_1 \leq L_2$ almost surely implies $\rho(L_1) \leq \rho(L_2)$. Translation invariance means that $\rho(L+m) = \rho(L) + m$ for constants m. Positive homogeneity means that $\rho(cL) = c\rho(L)$ for $c > 0$. Sub-additivity means that $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. Papers working with payoffs $X = -L$ write translation invariance as $\rho(X+m) = \rho(X) - m$, which is the same axiom. A measure is **convex** if $\rho(\tau L_1 + (1-\tau)L_2) \leq \tau\rho(L_1) + (1-\tau)\rho(L_2)$ for $\tau \in [0,1]$. Positive homogeneity together with sub-additivity implies convexity, but the converse does not hold. Convexity of ρ in the positions makes minimising risk a convex optimisation problem, with no local minimum other than the global one. VaR is monotone, translation invariant and positively homogeneous, but it is neither sub-additive (§5.7) nor convex. ES is coherent (A.28).

Why it appears here. §3.4 records the arrival of the axioms, and §5.7 states them, with the two-bond counterexample to sub-additivity. §11.2 turns the failure into a practical warning about desk-level limits. Two passages make the point about convexity. §5.6 remarks that VaR "is not convex in the portfolio weights, and it has many local minima". The row of the table in §7.1 on optimisation says the same.

Deeper. Artzner, Delbaen, Eber & Heath (1999), in §4.3. Föllmer & Schied, "Convex Measures of Risk and Trading Constraints," *Finance and Stochastics* 6(4) (2002), 429–447, for the convex relaxation.

A.27 Positive homogeneity, Euler's theorem and risk allocation

The idea. A statement such as "Technology carries 55% of the risk" sounds like an accounting convention. In fact the arithmetic forces it. A measure that doubles when every position doubles has a decomposition into per-position pieces that add up exactly to the total. Each piece is the position's size times the sensitivity of the total to it. This is Euler's theorem on homogeneous functions, and the pieces are component VaR or component ES. The decomposition is also the one consistent with rewarding positions by return on risk. If a unit's return per unit of its own contribution exceeds the firm's, growing the unit raises the firm's ratio. That consistency is why the decomposition underlies risk-adjusted performance measurement, from Bankers Trust's RAROC onward.

Formally. A function f on $\mathbb{R}^N$ is positively homogeneous of degree one if $f(cx) = cf(x)$ for $c > 0$. Differentiating in c at $c = 1$ gives Euler's identity,

$$f(x) = \sum_{i=1}^{N} x_i \, \frac{\partial f}{\partial x_i}(x).$$

For $f = \sigma_p$, the partial derivatives are $(\Sigma x)_i / \sigma_p$ (A.9). Variance is homogeneous of degree two. Its Euler sum is $2x'\Sigma x$. For VaR and ES with a continuous P&L, the marginal contributions are conditional expectations of the position's loss per dollar. They are $\partial\,\mathrm{VaR}_\alpha / \partial x_i = E[-r_i \mid L = \mathrm{VaR}_\alpha]$ and $\partial\,\mathrm{ES}_\alpha / \partial x_i = E[-r_i \mid L \geq \mathrm{VaR}_\alpha]$. RAROC, risk-adjusted return on capital, is expected return divided by allocated risk capital. Tasche shows that only the Euler allocation has this property. Under it, a unit whose RAROC exceeds the firm's always raises the firm's RAROC when the unit grows a little.

Why it appears here. §5.5 derives marginal, component and incremental VaR from this identity, and gives the forms as conditional expectations. §7.2 makes component VaR the basis of the risk budget. The decomposition of §10.4 and the `component` line of §14.2 are the delta-normal case. §3.2 associates Bankers Trust's firm-wide measure with its RAROC system. A component is a derivative, so it is not exact for a large trade. That is why §5.5 and §13.4 require recomputing incremental VaR.

Deeper. Tasche (2008), in §4.2. Litterman (1996) and Garman (1996), also in §4.2, for the practitioner versions.

A.28 Expected shortfall, and its careful definition

The idea. VaR is the threshold of the worst p of outcomes, and expected shortfall is their average. For a continuous distribution the definition is unambiguous. Average the losses beyond the VaR. Now suppose the distribution has an atom at the VaR, as for a bond that defaults or not, or a small sample. "The losses beyond the VaR" can then carry less or more than a fraction p of probability. A naive conditional average then covers the wrong amount of probability. The careful definition averages the quantiles beyond α. That fixes the tail mass at exactly p, taking from the atom whatever share is needed. That version is coherent for every distribution. The naive one, the tail conditional expectation, is not.

Formally. Acerbi and Tasche's expected shortfall is

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 q_u(L)\, du = \frac{1}{1-\alpha} \Big(E[L\, \mathbf{1}\{L > q_\alpha\}] + q_\alpha (P(L \leq q_\alpha) - \alpha) \Big),$$

where the second term adds the share of the atom at q_α needed to make the tail mass exactly $1-\alpha$. The **tail conditional expectation** $E[L \mid L \geq q_\alpha]$ agrees with it whenever F_L is continuous at q_α. In the two-bond example of §5.7, the difference is large. One bond has $q_{0.95} = 0$, and every outcome satisfies $L \geq 0$. Its tail conditional expectation therefore averages everything and gives 4. The pair has $q_{0.95} = 100$ and a tail conditional expectation of 102.0. That measure, like VaR, fails sub-additivity there, with 102.0 against 4 + 4. ES gives 80 per bond and 103.2 for the pair. For a long sample, where each atom is a single observation, the two definitions are close. For distributions with large atoms they are not. Under a normal loss with standard deviation σ, $\mathrm{ES}_\alpha = \sigma\varphi(z_\alpha)/(1-\alpha)$.

Why it appears here. §1.7 introduces ES as the average loss in the worst p of outcomes. §5.6 gives the integral definition and the normal closed form. It attributes to Acerbi and Tasche the definition that "remains coherent when the distribution has jumps". Here jumps mean jumps of F_L in the sense of A.1. §7.1 lists ES's many names, and says that the careful definitions differ for discrete distributions.

Deeper. Acerbi & Tasche (2002), in §4.3, which proves coherence and shows where the naive version fails. Rockafellar & Uryasev (2002), also in §4.3, for the same object reached through optimisation.

A.29 The Rockafellar–Uryasev formula and linear programming

The idea. Expected shortfall equals the minimum, over a single number c, of a simple expression, and the minimising c is the VaR. That has two consequences. Computationally, one optimisation yields both VaR and ES. Practically, the expression involves only an average of the positive parts of losses that are linear in the positions. Minimising a portfolio's ES therefore becomes a linear program once each scenario's shortfall has an auxiliary variable of its own. Linear programs are among the most thoroughly solved problems in numerical mathematics. Minimising VaR has no such structure, and it is a hard non-convex problem.

Formally. With $y^+ = \max(y, 0)$, let $H(c) = c + E\big[(L-c)^+\big]/(1-\alpha)$. H is convex in c, and

$$\mathrm{ES}_\alpha(L) = \min_{c \in \mathbb{R}} H(c), \qquad \mathrm{VaR}_\alpha(L) \in \arg\min_c H(c).$$

Take M equally likely scenarios of asset returns $r^{(1)}, \dots, r^{(M)}$. The loss in scenario m is $-x'r^{(m)}$. Minimising ES over positions x subject to linear constraints is then the linear program

$$\min_{x,\,c,\,\zeta} \; c + \frac{1}{(1-\alpha)M} \sum_{m=1}^{M} \zeta_m \quad \text{s.t.} \quad \zeta_m \geq -x'r^{(m)} - c, \;\; \zeta_m \geq 0,$$

plus the constraints on x. At the optimum, each ζ_m equals $(-x'r^{(m)} - c)^+$, because the objective drives it down to the larger of its two lower bounds.

Why it appears here. §3.4 credits Rockafellar and Uryasev with showing that ES is minimisable "by linear programming". §5.6 derives the formula, sketches why the minimiser is the VaR and introduces the auxiliary variables. The row of §7.1, "Convex in positions, so optimisable by linear programming", states the consequence. The same construction optimises the conditional drawdown at risk of §7.7.

Deeper. Rockafellar & Uryasev (2000, 2002), in §4.3. Boyd & Vandenberghe, *Convex Optimization* (CUP, 2004), chapter 4, for linear programs and the auxiliary-variable trick, free at web.stanford.edu/~boyd/cvxbook.

A.30 Elicitability, the pinball loss and quantile regression

The idea. Ranking two forecasters needs a score that rewards honest reports. Its expected value is lowest when the forecaster reports the true value of the statistic being forecast. Such a score is consistent for that statistic, and a statistic that has one is elicitable. Squared error is consistent for the mean, and absolute error for the median. The pinball loss, an asymmetric absolute error, is consistent for any quantile. Expected shortfall has no consistent score of its own, but the pair (VaR, ES) does. Minimising the average pinball loss over the parameters of a model, rather than merely scoring its forecasts, is quantile regression. It fits a conditional quantile directly, with no model of the rest of the distribution.

Formally. A scoring function $\psi(\hat{y}, y)$ is consistent for a statistic T if $E_F[\psi(T(F), Y)] \leq E_F[\psi(\hat{y}, Y)]$ for every forecast $\hat{y}$ and every distribution F in a class. T is elicitable if a strictly consistent ψ exists. For the p-quantile of returns, the pinball loss of §5.9 is

$$\psi(\hat{q}, r) = \big(p - \mathbf{1}\{r < \hat{q}\}\big)(r - \hat{q}).$$

Its expectation has derivative $F_r(\hat{q}) - p$ in $\hat{q}$, with F_r the distribution function of the return. The derivative is zero at the quantile. A necessary condition for elicitability is that the distributions sharing a value of T form a convex set. Mixing two distributions with the same quantile leaves the quantile unchanged. Mixing two with the same ES can change the ES, so ES fails the condition. Fissler and Ziegel give scoring functions of a pair of forecasts whose expectation the true (VaR, ES) pair minimises jointly. Quantile regression chooses the coefficients ϑ of a model $\hat{q}_t(\vartheta)$ for the conditional quantile by minimising

$\sum_t \psi(\hat{q}_t(\vartheta), r_t)$. When the model is linear in ϑ, this is a linear program (A.29), as Koenker and Bassett showed. Separate fits at different levels can cross, so they need not respect the order of the quantiles.

Why it appears here. §5.9 sets out elicitability and the pinball loss. §6.13 and §12.6 rank models by average pinball loss. §5.9 and §12.7 explain why one can still backtest ES and compare it jointly with VaR, although ES is not elicitable alone. The CAViaR of §6.12 is quantile regression in which the fitted quantile follows its own autoregression. The growth-at-risk of §7.6 applies the same method to forecasts of GDP growth.

Deeper. Gneiting (2011) and Fissler & Ziegel (2016), in §4.3. Koenker, *Quantile Regression* (CUP, 2005), and Koenker & Bassett, "Regression Quantiles," *Econometrica* 46(1) (1978), 33–50.

A.31 Statistical robustness

The idea. An estimate is robust if small changes in the data produce only small changes in its value. The changes might be a mis-recorded price, one unusual day or a slightly different window. A sample quantile is robust in this sense. If the single worst day moves anywhere beyond the quantile, the quantile does not move at all. A tail average is not robust. It moves with every observation it averages. One bad data point beyond the VaR shifts the ES by that point's error divided by the number of tail observations, without limit. Robustness and sensitivity to the tail are the same property with opposite signs. So one sentence can state both the case for VaR and the case against it.

Formally. Hampel's qualitative robustness asks that an estimator's distribution change continuously with the distribution generating the data. Two quantitative summaries go with it. The **influence function** of a statistic T at a point y,

$$\mathrm{IF}(y) = \lim_{\eta \downarrow 0} \frac{T((1-\eta)F + \eta\,\delta_y) - T(F)}{\eta},$$

with δ_y a point mass at y, measures the effect of a small contamination there. The **breakdown point** is the smallest fraction of arbitrarily corrupted observations that can carry the estimate arbitrarily far. For the α-quantile, the influence function is bounded, $\mathrm{IF}(y) = (\alpha - \mathbf{1}\{y \le q_\alpha\})/f(q_\alpha)$, and the breakdown point is $\min(\alpha, 1-\alpha)$. For ES, the influence function grows linearly in y beyond the VaR, so a single corrupted point can move it without bound. Cont, Deguest and Scandolo show that for historical estimators this makes VaR robust and ES not robust. They also show that robustness and sub-additivity cannot hold together.

Why it appears here. §5.7 and §7.1 weigh the coherence of ES against its loss of robustness. They cite Cont, Deguest and Scandolo, and Kou, Peng and Heyde. The table row of §7.1, "Robust to a few bad data points", states a property of the influence function. §2.3 remarks that moving the worst day to $50,000 would leave the 95% VaR unchanged. That is the same bounded influence, which §2.3 describes as a blind spot.

Deeper. Cont, Deguest & Scandolo (2010), in §4.5, and Kou, Peng & Heyde (2013), in §4.6. Huber & Ronchetti, *Robust Statistics*, 2nd ed. (Wiley, 2009), for influence functions and breakdown points.

A.32 Copulas, tail dependence and correlation in a crash

The idea. A joint distribution combines two things. One is how each variable behaves alone. The other is how the variables move together. A copula is the second part with the first removed. It is the joint distribution of the variables after each passes through its own distribution function, which gives it a uniform scale. Separating the two lets a model pair fat-tailed marginals with any dependence structure. The dependence that matters for risk is tail dependence. Tail dependence is the probability that one asset has a very bad day, given that another asset has one. Correlation does not measure tail dependence, and the Gaussian copula models it badly. For any correlation below one, joint extremes become negligible relative to single ones as the extremes grow. A statistical artefact pushes the other way. Measured over a volatile period only, a sample correlation rises even when the underlying correlation is unchanged. "Correlations go to one in a crash" therefore mixes a real change in dependence with an artefact of conditioning on large moves.

Formally. Sklar's theorem says that any joint distribution with continuous marginals $F_1, \dots, F_N$ has a unique copula C. The copula is a distribution function on $[0,1]^N$ with uniform marginals, and $P(Y_1 \le y_1, \dots, Y_N \le y_N) = C(F_1(y_1), \dots, F_N(y_N))$. With $U_i = F_i(Y_i)$, the lower tail-dependence coefficient of two variables is

$$\lim_{s \downarrow 0} P(U_2 \le s \mid U_1 \le s) = \lim_{s \downarrow 0} \frac{C(s,s)}{s}.$$

It is zero for the Gaussian copula at any correlation below one. It is positive for the Student t copula, and it grows as the degrees of freedom fall. The conditioning effect holds exactly under joint normality. Let Y_1, Y_2 be bivariate normal with correlation ϱ (not the risk measure ρ). Restrict the sample to days on which Y_1 falls in a set that multiplies its variance by v. The correlation on those days is

$$\varrho_v = \frac{\varrho}{\sqrt{\varrho^2 + (1-\varrho^2)/v}},$$

which exceeds ϱ whenever $v > 1$. Selecting the volatile days raises the correlation, though nothing about the dependence has changed.

Why it appears here. §3.4 attributes part of LTCM's losses to positions "highly correlated in a crisis that its data did not contain". §11.8 lists correlation in a stress among its lessons. §6.9 lists the copula as part of a Monte Carlo model. It also warns that "a normal copula has no tail dependence at all". In the scenario matrix of §10.6, energy carried 5% of the VaR yet lost 2,000–3,000 on three of the five replayed days. That is why analysts should examine joint extremes directly rather than through a correlation.

Deeper. Embrechts, McNeil & Straumann (2002), in §4.3. Nelsen, *An Introduction to Copulas*, 2nd ed. (Springer, 2006). Forbes & Rigobon, "No Contagion, Only Interdependence: Measuring Stock Market Comovements," *Journal of Finance* 57(5) (2002), 2223–2261, for the conditioning effect.

A.33 Monte Carlo estimation and variance reduction

The idea. When a model is too complicated for a formula, simulate it. Examples are options repriced in full, defaults and multi-day dynamics. Draw many scenarios from the model, compute the loss in each, and treat the simulated losses as a sample. The VaR is their quantile. The answer is exact for the model in the limit, and noisy for any finite number of draws. A simulation has the same tail sparsity as a historical sample (A.12). Variance reduction makes each draw more informative. Importance sampling draws more scenarios from the tail than the model would. It reweights them, so the region that decides the quantile is well populated. A control variate uses a cheap approximation whose answer is known exactly, such as the delta-gamma P&L of A.36 for an option book. It then simulates only the approximation's error, which is far smaller than the P&L itself.

Formally. With M IID scenario losses $L_1, \dots, L_M$, the Monte Carlo VaR is their empirical α-quantile, with standard error $\sqrt{p(1-p)/M}/f(q)$ (A.12). Importance sampling draws scenarios from a density g instead of the model's density f. It estimates $P(L > \ell)$ by $M^{-1}\sum_m \mathbf{1}\{L_m > \ell\}\,(f/g)_m$, with $(f/g)_m$ the density ratio at scenario m. The estimate is unbiased for any g that covers the tail, and a g tilted toward large losses cuts its variance sharply. A control variate subtracts from the estimate a multiple of the simulation error in a quantity whose true value is known. At the best multiple, the variance falls by the factor one minus the squared correlation between the two errors.

Why it appears here. §6.9 defines Monte Carlo VaR and its sampling error: "With 10,000 scenarios, a 99% VaR has 100 of them in the tail." It reports that importance sampling and control variates "from the delta-gamma approximation" cut the cost by one to two orders of magnitude (Glasserman, Heidelberger and Shahabuddin, 2000). The table of §5.8 is itself a Monte Carlo experiment on an estimator (A.12).

Deeper. Glasserman, *Monte Carlo Methods in Financial Engineering* (Springer, 2004), chapter 4 on variance reduction and chapter 9 on risk management. Glasserman, Heidelberger & Shahabuddin (2000), in §4.2.

Part IV — Instruments and markets. The main text applies VaR to bonds, options, credit and leveraged accounts without defining them. These nine entries define the terms, ordered so that each builds on the earlier ones. They cover two instruments, the sensitivities that describe them and the mapping that turns any book into factor exposures. They then cover the products and arrangements whose risk VaR handles worst.

A.34 Bonds: yield, duration and modified duration

The idea. A bond is a schedule of fixed payments, coupons and then the principal. Its price is those payments discounted at market interest rates, so when rates rise its price falls. The yield is the single discount rate that reproduces the price. It compares bonds with different schedules by one number. Duration measures how sensitive the price is to the yield. It equals the average time until the payments arrive, weighted by present value. A 10-year zero-coupon bond has a duration of 10 years. It loses roughly 10% of its value when its yield rises by one percentage point. That sensitivity matters for risk, not the bond's face value. That is why a bond desk in the 1980s measured its risk in duration and could not add it to an options desk's delta.

Formally. For cash flows CF_k at times t_k years and an annually compounded yield y, the price is $V(y) = \sum_k \mathrm{CF}_k (1+y)^{-t_k}$. Macaulay duration is $D = \sum_k t_k \,\mathrm{CF}_k (1+y)^{-t_k}/V$, and modified duration is

$$D^\star = -\frac{1}{V}\frac{dV}{dy} = \frac{D}{1+y}.$$

To first order, therefore, $\Delta V \approx -D^\star V\,\Delta y$, which is the mapping that §5.3 uses. Convexity, $V^{-1}\,d^2V/dy^2$, is the second-order term, the bond's analogue of an option's gamma. The yield curve is the set of yields across maturities. In a VaR model, its points at standard maturities are the risk factors. The duration tests of §12.3 measure the time between VaR breaches. They are unrelated to this duration.

Why it appears here. §1.1 names duration as the bond desk's unit of risk. §1.5 lists points on yield curves as the risk factors for a bond book. §5.3 maps a bond to a position in its yield of $-D^\star$ times its value. The sensitivity limits of §9.1 on interest-rate duration are limits on this number.

Deeper. The companion chapter Bonds and Bond Markets, sections 6 and 7, on valuation, duration and convexity.

A.35 Options: calls, puts and the Black–Scholes price

The idea. An option is a right without an obligation. A call gives the right to buy an asset at a fixed price, the strike, at or before an expiry date. A put gives the right to sell. The buyer pays the seller a price up front, the premium, and the seller takes on the obligation. The buyer uses the right only when it pays, so the premium caps the buyer's loss and the seller's gain. The seller's loss can be many times the premium. Moneyness says where the strike sits relative to today's price. A put is out of the money when the price is above the strike, so that exercising now would be worthless. It is deep out of the money when the price is far above the strike. Selling deep out-of-the-money puts is a standard way to collect small, frequent premiums against a rare, large loss. Options are the leading example of derivatives, contracts whose value derives from another price.

Formally. At expiry, a call pays $\max(S-K,0)$ and a put pays $\max(K-S,0)$, where S is the underlying's price then and K the strike. A long position holds the option, and a short position has sold it. Listed US equity options trade in contracts of 100 shares. Black–Scholes prices options by assuming that the log price follows a Brownian motion with constant volatility. The main text assumes a zero interest rate. At that rate, an option with τ years to expiry and volatility σ_{imp} is worth

$$\text{call} = S\,\Phi(d_1) - K\,\Phi(d_2), \qquad \text{put} = K\,\Phi(-d_2) - S\,\Phi(-d_1),$$

$$d_{1,2} = \frac{\ln(S/K) \pm \tfrac{1}{2}\sigma_{\text{imp}}^2 \tau}{\sigma_{\text{imp}}\sqrt{\tau}}.$$

With $S = 100$, $K = 90$, $\sigma_{\text{imp}} = 0.20$ and $\tau = 30/365$, the put is worth 7 cents a share. Some options pay off according to the whole price path rather than its final value, such as barrier and average-price options. They are path-dependent and generally need simulation.

Why it appears here. §2.3 introduces the short out-of-the-money put as the real-world version of Position B. §6.10 prices 10 such contracts at Black–Scholes values with a zero interest rate. Seven cents a share is the $71 premium on 1,000 shares.

The separate rules for long and short options in §13.6 rest on the asymmetric payoffs above. §6.9 names path-dependent instruments as a reason for Monte Carlo.

Deeper. The companion chapters Dealer Hedging and Gamma Exposure, sections 2–3, which introduce options from first principles, and Implied Volatility, section 2, which gives just enough Black–Scholes. Black & Scholes, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy* 81(3) (1973), 637–654.

A.36 The Greeks, and the Taylor expansion of option P&L

The idea. An option's value depends on several market variables at once. The Greeks are its partial derivatives with respect to each. Delta is the derivative with respect to the underlying price, and gamma the second derivative with respect to that price. Vega is the derivative with respect to implied volatility, and theta with respect to the passage of time. Together they give a Taylor expansion of the P&L for small moves. A risk system uses it to turn an option into factor exposures. The expansion works for small moves and fails for large ones, in a predictable direction. A short option position has negative gamma. It loses more on a large move than delta predicts, in either direction. It also has positive theta. It earns a little every day the market stays still. Under Black–Scholes an exact equation links the two. That makes "earn small, lose large" a mathematical property of the position rather than an observation about markets.

Formally. For a position worth $V(S, \sigma_{\text{imp}}, t)$, write $\delta = \partial V/\partial S$, $\Gamma = \partial^2 V/\partial S^2$, $\mathcal{V} = \partial V/\partial \sigma_{\text{imp}}$ and $\Theta = \partial V/\partial t$. The expansion of §6.10 is then

$$\Delta V \approx \delta\,\Delta S + \tfrac{1}{2}\Gamma\,(\Delta S)^2 + \mathcal{V}\,\Delta\sigma_{\text{imp}} + \Theta\,\Delta t.$$

For a long Black–Scholes put at a zero interest rate, $\delta = \Phi(d_1) - 1$ and $\Gamma = \varphi(d_1)/(S\sigma_{\text{imp}}\sqrt{\tau})$, the same gamma as a call. At a zero rate, the Black–Scholes equation is $\Theta = -\frac{1}{2}\sigma_{\text{imp}}^2 S^2\,\Gamma$. A position's time decay is minus half its gamma times the expected squared move per unit of time. Collecting theta and being short gamma are therefore the same thing. When ΔS is normal, the delta-gamma P&L is a quadratic function of a normal variable. Its moments and characteristic function are known in closed form. Three methods find its quantile: Cornish–Fisher on the moments (A.25), numerical Fourier inversion of the characteristic function, and simulation (A.33).

Why it appears here. The options of §1.1, "measured in delta and vega", and the limits on option vega in §9.1 refer to these derivatives. §5.3 maps an option to a stock position of δS dollars per share of the underlying stock. §6.10 builds on the expansion and on the sign of gamma. It uses Britten-Jones and Schaefer's delta-gamma VaR, with its Cornish–Fisher and Fourier routes. The control variates of §6.9 use the delta-gamma P&L.

Deeper. Dealer Hedging and Gamma Exposure, sections 3–5, on the Greeks and the trade-off between gamma and time decay. Britten-Jones & Schaefer (1999), in §4.2.

A.37 Implied volatility, the VIX and short-volatility products

The idea. Black–Scholes turns a volatility into a price. Implied volatility is the reverse calculation. It turns a market price into the volatility that justifies that price. It quotes options in a common unit. It is also the market's price of future movement, which rises sharply when markets fall. The VIX index combines the implied volatilities of a strip of S&P 500 options into one number. That number is the market's price of 30-day volatility, quoted in annual percentage points. Some products profit while volatility stays low, such as exchange-traded products that are short VIX futures. They collect that price as a steady income, and they can lose most of their value when volatility spikes. They have sold what buyers want most when volatility spikes.

Formally. Implied volatility solves $\text{BS}(S, K, \tau, \sigma_{\text{imp}}) =$ market price, where BS is the Black–Scholes formula of A.35. The solution is unique because that formula increases with volatility. A daily equivalent is $\sigma_{\text{imp}}/\sqrt{252}$, so 20% implied is 1.26% a day, as in §6.10. The VIX comes from out-of-the-money option prices across strikes and needs no pricing model. The VIX is the square root of a weighted sum that approximates the market price of 30-day variance. Markets quote it like an annualised volatility. A VIX of 20 is about 1.26% a day. An inverse product that delivers minus the daily return of a volatility index loses its whole value on a day when that index doubles.

Why it appears here. §1.5 lists implied volatility among the risk factors for an option book. §2.3 and §6.10 shock it together with the price. In §6.10, a five-point rise raises the loss of the at-the-money puts at the 99% move from 1,695 to 2,191. §8.1 lists implied volatility as a possible scale model. The February 2018 case of §11.8 is short-volatility products on a day when the VIX more than doubled.

Deeper. Implied Volatility, section 1 on the number itself and section 5 on variance swaps, the VIX and the term structure. Whaley, "Understanding the VIX," *Journal of Portfolio Management* 35(3) (2009).

A.38 Risk factors and risk-factor mapping

The idea. A bank holds hundreds of thousands of positions, but their values depend on far fewer market variables. These are stock prices and indices, points on yield curves, exchange rates, credit spreads and implied volatilities. They are the risk factors. Mapping expresses each position as a function of the factors, so that a forecast of factor moves becomes a forecast of P&L. Two methods do the conversion. Sensitivity-based revaluation multiplies factor moves by first-order or second-order sensitivities (A.34, A.36) and adds them up. It is fast and decomposable, and wrong for large moves in non-linear positions. Full revaluation reprices every instrument in every scenario. It is slow and exact for the pricing model, and it is the only accurate method for short options. A foreign stock maps to two factors, its local price and the currency. A bond maps to a few yield-curve points, and an option to its underlying and its implied volatility.

Formally. Write the book's value as $V(f)$ for a vector of factor levels f. Sensitivity-based revaluation uses $\Delta V \approx \nabla V' \Delta f$, or adds $\frac{1}{2}\,\Delta f'\,\nabla^2 V\,\Delta f$ for delta-gamma. Full revaluation computes $V(f+\Delta f) - V(f)$ exactly. Under linear mapping, the dollar exposures to the factors' returns take the place of the positions x. The delta-normal VaR is then $z_\alpha \sqrt{x'\Sigma x}$, with Σ now the covariance matrix of factor returns. **Notional** is the face amount of a contract, such as the number of shares an option contract covers times their price. It is not a measure of risk. A contract's sensitivity can be a small or a large fraction of its notional.

Why it appears here. The incomparable units of §1.1, which are duration, delta and vega, notional, and beta, are the problem that mapping solves. §1.5 makes mapping the first stage of the pipeline. §5.3 says that risk-factor mapping absorbs most of a bank's VaR engineering effort. §6.10 compares delta, delta-gamma and full revaluation. Together with the revaluation step of §8.1, they are the two methods of turning factor moves into P&L. §11.7 shows the danger of positions "mapped to proxies that do not move like them".

Deeper. J.P. Morgan/Reuters (1996), in §4.1, which standardised the mapping, and Holton (2014), in §4.7, for the engineering.

A.39 Credit risk, credit default swaps and the one-factor model

The idea. Credit risk is the risk that a borrower fails to pay. For a single bond the loss is lumpy. Most of the time nothing happens. Occasionally the investor loses a large part of the principal. Investors receive compensation for bearing this risk through a **credit spread**, the extra yield over a risk-free bond. They lose value on a default, and also on a downgrade or a widening spread that makes a default look likelier. A **credit default swap** isolates the default risk. The protection buyer pays a regular premium, and the protection seller pays the loss if the borrower defaults. Selling protection is economically like owning the bond without lending the money. It earns steady income and risks a rare large loss. A synthetic credit portfolio takes its exposure through such swaps, or through indices of them, rather than through bonds. **Tranching** divides the losses of a pool of loans or swaps into layers, paid in order. An equity tranche absorbs the first losses. A senior tranche absorbs only those beyond a high attachment point, so it loses only if many borrowers default together. Its risk depends on default correlation, which a history without stress cannot reveal.

Formally. The standard model of correlated defaults is Vasicek's one-factor model, the basis of the Basel internal-ratings capital formula. Borrower i defaults within the horizon if $\sqrt{\varrho}\,Y + \sqrt{1-\varrho}\,Y_i$ falls below $\Phi^{-1}(\mathrm{PD})$. Here the common factor Y and the borrower-specific factors Y_i are independent standard normals. PD is the default probability. ϱ is the asset correlation (not the risk measure ρ). Given $Y = y$, defaults are independent with probability $\Phi\big((\Phi^{-1}(\mathrm{PD}) - \sqrt{\varrho}\,y)/\sqrt{1-\varrho}\big)$. In a large portfolio, the loss fraction is therefore a decreasing function of Y alone, and its quantile follows from that of Y:

$$\mathrm{VaR}_\alpha \approx \mathrm{LGD} \cdot \Phi\left(\frac{\Phi^{-1}(\mathrm{PD}) + \sqrt{\varrho}\,\Phi^{-1}(\alpha)}{\sqrt{1-\varrho}}\right)$$

as a fraction of exposure, with LGD the loss given default. The Basel formula evaluates this at $\alpha = 0.999$, subtracts the expected loss and applies a maturity adjustment.

Why it appears here. §1.2 and §5.7 use the defaulting bond as the standard lumpy distribution. §7.6 describes credit VaR at 99.9% over a year, which the Basel internal-ratings capital formula computes in closed form from a one-factor model. §11.1 lists selling credit protection among the strategies whose losses sit in the tail. §3.5 and §11.7 describe the London Whale's synthetic credit portfolio. §11.7 also names highly rated structured credit as the canonical position whose calm history gave no sign of its risk.

Deeper. Bonds and Bond Markets, section 9 on credit risk, and its appendix entries on default correlation and credit default swaps. Gordy, "A Risk-Factor Model Foundation for Ratings-Based Bank Capital Rules," *Journal of Financial Intermediation* 12(3) (2003), 199–232. Vasicek, "The Distribution of Loan Portfolio Value," *Risk* 15(12) (2002).

A.40 Earn small, lose large: carry and other tail-selling trades

The idea. A family of strategies shares one payoff shape. It is a small, steady gain on most days and a rare, large loss. Selling out-of-the-money options and selling credit protection have it (A.35, A.39). So does the currency carry trade. It borrows in a low-interest currency and invests in a high-interest one. It keeps the difference for as long as the exchange rate does not move against it. For a pegged or managed currency, the rate mostly does not move, until the peg breaks. Short-volatility products (A.37) share the shape. So do relative-value or convergence trades. These expect a small price gap between similar securities to close, and they lose when such gaps widen together in a stress. In ordinary times, all of them look like excellent risk-adjusted investments. Their returns are negatively skewed, and the loss that defines them sits beyond any ordinary confidence level.

Formally. The stylised version has two outcomes. One is a gain g with probability $1-\pi$. The other is a loss G with probability π. If $\pi < p = 1-\alpha$, the α-VaR is $-g$, a gain, whatever the size of G. The expected P&L $(1-\pi)g - \pi G$ can be positive or negative without the VaR registering either. The ES (A.28) is $\big(\pi G - (p-\pi)g\big)/p$, which does depend on G. Position B of §2.3 has $g = 200$, $G = 20{,}000$ and $\pi = 0.03$. Its expected daily P&L is -406. Yet its 95% VaR is a gain of 200, and only its 95% ES, 11,920, registers the loss. A carry trade's expected excess return is the interest-rate differential minus the expected depreciation of the high-rate currency. It earns while realised depreciation falls short of the differential, as it does in the long stretches between breaks.

Why it appears here. This payoff shape appears in several places. Position B of §2.3 and its short-put counterpart have it. So do the credit and option books of §5.7 where sub-additivity fails. So do the strategies of §11.1 that traders can design to have a low or even negative VaR. The short-volatility products of §11.8 and the advice of §15.4 to distrust a low VaR on a high-yielding position also concern it.

Deeper. Brunnermeier, Nagel & Pedersen, "Carry Trades and Currency Crashes," *NBER Macroeconomics Annual* 23 (2008), 313–347. Basak & Shapiro (2001), in §4.5, for why a VaR limit rewards this shape.

A.41 Liquidity: spreads, market impact and liquidation horizons

The idea. A quote contains two prices. One is the highest bid that a buyer will pay. The other is the lowest offer that a seller will accept. The gap between them is the bid-ask spread. The midpoint is the mid price, the number that most valuations and risk systems use. Turning a position into cash costs at least half the spread. A large position costs more, because selling it pushes the price down as it goes. That is market impact. Both costs rise in a crisis, when spreads widen and buyers withdraw. The liquidation horizon is how long it would take to exit a position without moving the price much. It is the natural choice of VaR horizon. A position that takes a month to sell carries a month's risk, regardless of its one-day VaR.

Formally. With bid b and ask a, the mid is $(a+b)/2$, and the proportional spread is $s = (a-b)/\text{mid}$. Selling at the bid costs $s/2$ of the position's mid value. Summed over positions, that is the add-on $\frac{1}{2}\sum_i |x_i|\, s_i$ of §7.5. The most common model of impact makes the cost per share grow with the square root of the fraction of daily volume traded:

$$\text{cost per share} \;\approx\; \text{const} \times \sigma \times \sqrt{Q/\text{ADV}},$$

with Q the quantity traded, ADV the average daily volume and σ the daily volatility. Total cost then grows as $Q^{3/2}$. Doubling a position therefore more than doubles the cost of exiting it at a given speed.

Why it appears here. §1.2 says that the horizon should reflect how long exiting the position would take. §7.5 adds the spread cost and the liquidity horizons of the 2019 Basel rules (A.48). §11.6 attributes LTCM's losses to positions so large that "the act of reducing them was itself the loss". §13.1 notes that an individual is too small to move prices, which removes most of the problem.

Deeper. Almgren & Chriss, "Optimal Execution of Portfolio Transactions," *Journal of Risk* 3(2) (2000), 5–39. Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (OUP, 2003), for spreads and order books.

A.42 Leverage and margin

The idea. Leverage is holding more exposure than there is equity, financed by borrowing or by derivatives. It scales gains and losses alike, and it adds a new risk. A lender may force the holder out at the worst moment. Margin is the lender's protection. A broker that lends against securities requires the account's equity to stay above a set fraction of the position, the maintenance margin. When a fall pushes equity below it, the broker issues a margin call. The holder must post more cash or have positions sold. Clearing houses do the same job for derivatives. A clearing house stands between buyer and seller and requires initial margin. That is collateral sized to cover the loss on a defaulting member's positions over the days it would take to close them out. Setting initial margin is therefore a VaR problem by construction. Short selling borrows

shares and sells them, to buy them back later. It is leverage with an unbounded loss, because a price can rise without limit. A leveraged fund delivers a multiple of an index's daily return. A haircut is the discount that a lender or regulator applies to a security's value when counting it as collateral or capital.

Formally. Leverage is gross exposure over equity, $\sum_i |x_i|/W$. A levered book's VaR as a fraction of equity is the leverage times the unlevered fraction. Consider a long position bought with an initial equity fraction e_0 and a maintenance requirement m on equity over market value. A margin call comes after a fall in price of

$$1 - \frac{1 - e_0}{1 - m}.$$

That is one third for the common US combination of half the purchase borrowed, the Regulation T initial requirement, and FINRA's 25% maintenance minimum. Divided by the daily volatility, it answers the question of §13.6, which asks what market move triggers a margin call. A clearing house's initial margin is a VaR or ES of the account's loss over a close-out period of several days. The confidence level is 99% or more. Variation margin settles each day's P&L in cash. CME's SPAN instead charges the worst loss over a fixed grid of price and volatility scenarios. ISDA's SIMM, for uncleared trades between dealers, aggregates sensitivities with prescribed risk weights and correlations, calibrated to a 99% 10-day loss. A clearing house's members pool its default fund. It covers losses beyond a defaulter's margin. Stress scenarios, not a VaR, set its size.

Why it appears here. §1.7 distinguishes margin from VaR and notes that the loss on a short position has no upper bound. §3.1 describes the SEC's move in 1980 to statistically based haircuts. In §9.3, prime brokers, the brokers that finance hedge funds, set margin from their own models. §9.4 describes clearing-house margin, SPAN and SIMM, and Regulation T and portfolio-margin accounts for individuals. §13.6 asks what move triggers a margin call. §13.1 lists margin and leveraged funds among the retail exposures that VaR handles worst. §6.11 names clearing-house default funds among the uses of extreme value theory.

Deeper. CPMI–IOSCO, *Principles for Financial Market Infrastructures* (Bank for International Settlements, 2012), principles 4 and 6, on credit risk and margin at clearing houses. The Federal Reserve's Regulation T (12 CFR Part 220) and FINRA Rule 4210, for the retail rules.

Part V — Practice, regulation and evaluation. The last part covers how practitioners use and evaluate VaR. It defines the terms of sizing and performance in §8 and §13, the rules of §9, and the practices that make a backtest reliable. Several entries concern institutions, not mathematics, so their formal parts state rules, not theorems.

A.43 Stops, gaps, average true range and sizing rules

The idea. Many traders control risk trade by trade. They decide in advance the price at which they will exit a losing position. They then size the position so that the loss at that price is a fixed fraction of equity. The exit is usually a stop order, an instruction to sell once the market trades at or through a given price. A stop does not guarantee that price. Once the market reaches the stop, it becomes an order to sell at the next available price. A market gaps when it opens far below the previous close or trades through the level in a halt or a rush. Then the fill can land well beyond the stop. The difference is slippage. Volatility-scaled stops set the exit at a multiple of a recent range measure, most often the average true range. That makes the rule equivalent to a volatility-based sizing rule. A rebalancing band is the analogous device for a risk budget. It acts only when a quantity drifts outside a band, and then trades back toward the target.

Formally. Fixed-fractional sizing sets the position at $x_i = B_i W / d_i$, with d_i the distance to the stop as a fraction of the price. The planned loss at the stop is then $B_i W$. The true range of a day is

$$\max\left(\text{high}_t - \text{low}_t,\ |\text{high}_t - \text{close}_{t-1}|,\ |\text{low}_t - \text{close}_{t-1}|\right),$$

which counts an overnight gap as part of the day's range. The average true range (ATR) is its moving average, conventionally over 14 days. Suppose the trader sets the stop at c ATRs, and the ATR is roughly proportional to σ_i. Then $d_i \propto \sigma_i$, and the rule becomes $x_i \propto B_i W / \sigma_i$. That is the VaR budget of §13.3, with a different constant in place of $k_{0.95}$. Summing the B_i over open positions, sometimes called portfolio heat, adds risks while ignoring correlation. A band rule acts when utilisation exceeds an upper limit such as 1.25. Utilisation is the VaR divided by its budget. The rule then trades back to 1.

Why it appears here. §13.3 shows that a VaR budget, a volatility target and volatility-scaled stop sizing are one rule. It lists three things VaR adds to stops: gaps, correlation and holdings managed without stops. §8.2 marks as a heuristic the claim that a fixed-fraction stop caps the loss, because "gaps and halts go through it". The flowchart of §13.2, §13.3 and §14.4 all use a 125% band.

Deeper. Trend-Following in Financial Markets, section 8.5 on position sizing and section 11.3 on stops. Wilder, *New Concepts in Technical Trading Systems* (Trend Research, 1978), where the average true range originates.

A.44 The Sharpe ratio, factor portfolios and volatility management

The idea. The Sharpe ratio is return per unit of risk. It is the average excess return over the risk-free rate, divided by volatility. Leverage does not change it, so it ranks strategies rather than position sizes. An analyst builds a factor portfolio to isolate one source of return. It may be the market as a whole, or a long-short position in a characteristic such as value or momentum. The Kenneth French data library publishes the returns of many such portfolios, and of industry portfolios, for US stocks from 1926. A value-weighted portfolio holds each stock in proportion to its market capitalisation. Volatility targeting scales exposure so that forecast volatility stays constant. It holds more when markets are calm and less when they are volatile. Moreira and Muir's volatility-managed portfolios scale by the inverse of recent variance rather than of volatility, a more aggressive version of the same idea. Both are forms of a VaR budget.

Formally. For a risk-free rate r_f, the Sharpe ratio is $\text{SR} = E[r - r_f] / \text{sd}(r - r_f)$, annualised from daily data by $\sqrt{252}$ under IID returns. Volatility targeting holds the exposure multiple $\sigma^\star / \sigma_t$ for a target $\sigma^\star$. The scaled position's forecast volatility is then $\sigma^\star$ every day. A one-day 95% VaR budget B is the daily target $\sigma^\star = B / k_{0.95}$ (§8.2). In the normal case that is $\sqrt{252}\, B / 1.645 = 9.65 B$ a year. Moreira and Muir's managed factor return is the factor's return times $c / \hat{\sigma}_t^2$. Here $\hat{\sigma}_t^2$ is the previous month's realised variance, and c is a constant that matches the unmanaged factor's volatility. Exposure therefore falls in proportion to the rise in variance, rather than to the rise in volatility.

Why it appears here. The value-weighted market and industry portfolios of the front matter come from the French library. §8.2 shows that a VaR budget and a volatility target are the same rule. A 1% budget is a 9.65% annual target. §13.3 uses the rule for sizing. §11.6 cites Moreira and Muir's finding that inverse-variance scaling raised Sharpe ratios for the market and many factor portfolios. It also cites the out-of-sample challenge from Cederburg and co-authors.

Deeper. Moreira & Muir (2017) and Cederburg, O'Doherty, Wang & Yan (2020), in §4.6. Sharpe, "The Sharpe Ratio," *Journal of Portfolio Management* 21(1) (1994), 49–58. The Kenneth R. French data library, and Trend-Following in Financial Markets, section 11.2, on volatility targeting.

A.45 Benchmarks, active return and tracking error

The idea. Investors judge an active manager against a benchmark, usually an index. The relevant risk is trailing the benchmark, not losing money. Active return is the portfolio's return minus the benchmark's, and tracking error is its volatility. Take a manager with 3% tracking error and normal active returns. The manager can expect to finish within three percentage points of the index in about two years out of three. Because tracking error is a volatility, a VaR of the active return is a multiple of it. An active manager's risk budget is therefore a tracking-error budget. The information ratio, mean active return divided by tracking error, is the active manager's Sharpe ratio.

Formally. With portfolio weights w and benchmark weights w_b, the active weights are $w - w_b$. The ex-ante tracking error is $\sqrt{(w-w_b)'\Sigma(w-w_b)}$, annualised by $\sqrt{252}$ from a daily Σ. Under normality, with the mean dropped, the relative VaR of §7.3 is z_α times the tracking error times V. The information ratio is $E[r_p - r_b]/\operatorname{sd}(r_p - r_b)$ for portfolio and benchmark returns r_p and r_b.

Why it appears here. §7.3 defines relative and tracking-error VaR. §9.2 describes the tracking-error budget as a relative VaR in volatility units. §8.4 separates the three meanings of "relative VaR", of which this is one.

Deeper. Grinold & Kahn, *Active Portfolio Management*, 2nd ed. (McGraw-Hill, 1999). Roll, "A Mean/Variance Analysis of Tracking Error," *Journal of Portfolio Management* 18(4) (1992), 13–22.

A.46 Drawdowns, the drawdown of a random walk, and sequence risk

The idea. A drawdown is the fall from the highest value reached so far to the current value. The maximum drawdown is the worst such fall over a period. It is a path property. It depends on the order in which gains and losses arrive, and it accumulates over many steps. Investors therefore feel it, and a one-step measure like VaR does not describe it. A random walk has drawdowns too, and probability theory gives their typical size. The expected maximum drawdown grows with the square root of time, with a constant that relates it to the daily volatility. Sequence-of-returns risk is the same dependence on order, as it affects someone adding or withdrawing money. With withdrawals, losses early in a retirement do more damage than the same losses late, even when the average return is identical.

Formally. For an account's equity W_t, the drawdown at t is $1 - W_t/\max_{s\le t} W_s$. The maximum drawdown over a period is its largest value. For a Brownian motion with no drift and volatility σ per day, the expected maximum drawdown over h days is

$$\sqrt{\pi/2}\,\sigma\sqrt{h} \;\approx\; 1.25\,\sigma\sqrt{h},$$

and positive drift lowers it. Without cash flows, terminal wealth $W_0\prod_t(1+r_t)$ does not depend on the order of the returns. With a withdrawal c each period, $W_t = (W_{t-1} - c)(1 + r_t)$, and the order does matter. The same returns in a different order give a different ending balance, because an early loss shrinks the base that every later gain applies to.

Why it appears here. §1.7 contrasts drawdown with VaR, and §7.7 defines conditional drawdown at risk, the ES of drawdowns. The multi-manager platforms of §9.3 cut capital at drawdown thresholds. §13.5 tabulates US drawdowns as multiples of daily VaR and checks them against this result, about 12 daily 95% VaRs over a year. §13.7 names sequence risk as something that a one-day quantile cannot capture.

Deeper. Magdon-Ismail, Atiya, Pratap & Abu-Mostafa (2004) and Chekhlov, Uryasev & Zabarankin (2005), in §4.6. Bengen, "Determining Withdrawal Rates Using Historical Data," *Journal of Financial Planning* 7(4) (1994), on sequence risk in retirement. Trend-Following in Financial Markets, section 11.4, on drawdown control.

A.47 Bank capital and the Basel internal-models approach

The idea. Regulatory capital is the equity that a bank must hold so that it can absorb losses without failing. Market risk covers the losses on the trading book, the positions held to trade rather than to keep. For market risk, the 1996 Basel amendment let large banks compute the charge from their own VaR models, subject to standards. Banks no longer had to use fixed regulatory percentages. The rules had two parts. First, they multiplied the model's number by at least three, a safety factor for everything the model leaves out. Second, they required daily backtests of the model against the bank's P&L, with a penalty for too many breaches. The traffic light raises the multiplier from three toward four as exceptions accumulate. After the crisis, Basel 2.5 added a second VaR computed on a stressed year, so that a calm period could not drive capital down. A bank's own estimate of the capital it needs, usually at a higher confidence level, is economic capital. It is a separate number from the regulatory one.

Formally. Under the 1996 rules, the daily charge is the first term below. VaR is at 99% over 10 days, and $\overline{\mathrm{VaR}}_{60}$ is its average over the last 60 business days. Basel 2.5 adds the second term, the same expression for stressed VaR:

$$\max\left(\mathrm{VaR}_{t-1},\ m_c\,\overline{\mathrm{VaR}}_{60}\right)+\max\left(\mathrm{sVaR}_{t-1},\ m_s\,\overline{\mathrm{sVaR}}_{60}\right).$$

m_c is three plus a plus factor set by the traffic light (§12.4). The plus factor is 0 for four or fewer exceptions in 250 days, and rises to 1 for 10 or more. m_s is also at least three. The backtest compares each day's VaR with that day's P&L in two forms. **Hypothetical** P&L is the change in value of the previous evening's positions held fixed, which is what the VaR forecasts. **Actual** P&L is the result actually booked, which also reflects intraday trading and new positions.

Why it appears here. §1.7 says that regulators set capital "as a multiple of VaR". The 10-day 99% row of §2.1 is this rule's convention. §3.3 and §9.1 describe the 1996 amendment and its capital formula. §12.4 sets out the traffic light and its error rates. §3.5, §7.4 and §9.1 describe stressed VaR. §10.2 and §12.8 distinguish hypothetical from actual P&L. The economic capital of §6.11 is the internal counterpart.

Deeper. Basel Committee on Banking Supervision (1996a, 1996b, 2009), in §4.1. The 1996b backtesting framework is short and models how to write a statistical rule. Jorion (2006), in §4.7, for how the rules work in practice.

A.48 The Fundamental Review of the Trading Book

The idea. The FRTB is the post-crisis rewrite of bank market-risk capital, finalised in 2019. It keeps the internal-model architecture of A.47 but changes most of its parts. The capital measure moves from 99% VaR to 97.5% expected shortfall, so that the size of tail losses counts. Regulators calibrate the measure to a period of stress rather than to recent data. Horizons move from a uniform 10 days to between 10 and 120 days. The horizon depends on how quickly a bank could trade out of each risk factor. Approval moves from the bank to the trading desk. Each desk keeps its model only while it passes two tests. One is a backtest of its one-day VaR. The other is a profit-and-loss attribution test, which checks that the risk model's view of the desk's P&L matches the desk's own pricing.

Formally. With a base horizon of 10 days and liquidity horizons $\mathrm{LH}_j \in \{10, 20, 40, 60, 120\}$ days, the liquidity-adjusted ES is a nested square-root-of-time rule:

$$\mathrm{ES}=\sqrt{\mathrm{ES}_{10}(P)^2+\sum_{j\geq 2}\left(\mathrm{ES}_{10}(P,j)\sqrt{\tfrac{\mathrm{LH}_j-\mathrm{LH}_{j-1}}{10}}\right)^2}.$$

Here, at the 97.5% level, $\mathrm{ES}_{10}(P)$ is the 10-day ES of the portfolio P with all its risk factors shocked. $\mathrm{ES}_{10}(P,j)$ is the same with only the factors whose liquidity horizon is at least LH_j shocked. A bank computes the ES on a reduced set of factors over the worst 12-month stress period. It then scales the result by the ratio of the full-set ES to the reduced-set ES in current data. Desk-level backtesting counts exceptions of one-day VaR at 99% and 97.5% over the last 250 days. The attribution test compares the risk model's P&L with hypothetical P&L, using a rank correlation and a distributional distance.

Why it appears here. §3.5 and §9.1 describe the move to 97.5% ES, the stress calibration and the desk-level tests. §5.6 explains the choice of 97.5%, which matches 99% VaR for a normal distribution. §7.5 lists the liquidity horizons. Read the formula alongside §5.4 and §11.5. The regulation that retired VaR still builds its long horizons by square-root-of-time scaling.

Deeper. Basel Committee on Banking Supervision (2019), in §4.1: the standard itself, which contains the formula above.

A.49 Fund and insurance rules: UCITS, Rule 18f-4 and Solvency II

The idea. VaR has become a standard way to cap the leverage of investment funds that use derivatives. One rule can then apply to very different funds. UCITS is the EU framework for funds sold to the public. A European UCITS fund that uses derivatives in a sophisticated way must measure its global exposure by VaR. It must keep the VaR under one of two caps. The absolute cap is a percentage of its net asset value. The relative cap is a multiple of the VaR of a reference portfolio without derivatives. The US adopted the same design for registered funds in Rule 18f-4. European insurers face a different VaR rule. Under Solvency II, their capital must cover a one-in-200-year fall over one year in their own funds, the excess of assets over liabilities.

Formally. UCITS (CESR, 2010) uses 99% confidence, a 20-day horizon and at least one year of data. Absolute VaR may be at most 20% of net asset value, or relative VaR at most twice the reference portfolio's. Funds with simpler use of derivatives apply the commitment approach instead, which converts each derivative into an equivalent position in its underlying. Rule 18f-4 (Securities and Exchange Commission, 2020, in force from 2022) uses 99% over 20 trading days and at least three years of data. VaR may be at most 200% of a designated reference portfolio's VaR, or 20% of net assets where no suitable

index exists. The reference portfolio is an unleveraged index reflecting the fund's markets. Funds with limited derivatives exposure are exempt from the VaR test. Under Solvency II, the solvency capital requirement is the 99.5% one-year VaR of basic own funds. A fund's net asset value is its assets minus its liabilities.

Why it appears here. The 20-day and one-year rows of §2.1 are these conventions, and §3.6 dates them. §7.3 and §8.4 contrast the regulatory "relative VaR", a ratio to a reference portfolio's VaR, with the asset manager's. §9.2 sets out all three regimes.

Deeper. CESR (2010) and Securities and Exchange Commission (2020), in §4.1. Directive 2009/138/EC (Solvency II), article 101, for the 99.5% calibration.

A.50 Model risk and independent model validation

The idea. Model risk is the risk of loss from decisions based on a model that is wrong or misused. A model can be wrong in its specification or in its implementation, or right but used outside the conditions its builders designed it for. VaR has all three. Reasonable methods disagree by large factors on the same book. Implementations of the same method disagree. Users apply a model calibrated on calm data to a crisis. Once a number drives limits and pay, a further form appears. People have an incentive to change the model rather than the risk. That is an instance of Goodhart's law, which says that a measure made into a target stops measuring what it measured. The institutional answer is independent validation. People with no stake in the model's output review its design, test its implementation and its outcomes, and can stop its use.

Formally. US supervisory guidance (SR 11-7) defines a model as a quantitative method that turns input data into estimates, using statistical, economic, financial or mathematical theory. It defines model risk as the potential for adverse consequences from decisions based on incorrect or misused model outputs. It asks for effective challenge, and sets out three elements of validation: evaluation of conceptual soundness, ongoing monitoring, and outcomes analysis. Backtesting is the main instance of outcomes analysis for VaR. Implementation risk, the same specification coded differently, is part of model risk. Independent reimplementation tests it, and so do invariance checks such as those of §14.5.

Why it appears here. §1.3 warns that two competent teams routinely differ by tens of percent. §11.3 quantifies the disagreement with the results of Beder, and of Marshall and Siegel. §3.4 makes LTCM "the standard case study for model risk". §11.7 documents the London Whale's model change and names independent validation as the defence. The four checks of §14.5 are a minimal validation for a personal system.

Deeper. Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, *Supervisory Guidance on Model Risk Management*, SR 11-7 (2011). Beder (1995) and Marshall & Siegel (1997), in §4.5.

A.51 Backtest hygiene: look-ahead, overfitting and data

The idea. A backtest is evidence only if the forecaster could have made each forecast at the time. The forecast must use only data available then and a method chosen before the test. Look-ahead bias breaks the first condition. A forecast that uses later data, even indirectly through a parameter fitted on the whole sample, looks better than any real forecast could. Overfitting, or data snooping, breaks the second. If a researcher tries enough variants on the same data, the best will look good by luck. The remedies are two. Choose the method on one period and judge it on another that it never saw, which is out-of-sample testing. Count every variant tried. Two data problems are as common as the statistical ones. First, analysts must adjust prices for dividends and splits. Otherwise the return series contains false crashes at every split and misses the income from every payout. Second, a holding with a short history needs a proxy, and the proxy's differences from the real thing become an unmeasured risk.

Formally. A forecast for day t may depend only on $\mathcal{F}_{t-1}$ (A.4). That includes its parameters, its window and the choice of model. If a researcher tries M variants on the same data, the best of their test statistics is biased upward. For independent standard normal statistics, the expected maximum is about 2.5 for 100 variants, and it grows as $\sqrt{2 \ln M}$. The bar for significance should rise with it. A split-adjusted series divides every price before an n-for-one split by n. A total-return series reinvests each dividend on the day the stock starts trading without it. Then $S_t / S_{t-1} - 1$ computed from the series is the holder's actual return. An unadjusted two-for-one split appears as a return of -50%.

Why it appears here. §6.1 requires every backtested forecast to use "only data available the evening before". §12.1 and §12.8 forbid refitting on the full sample, and tuning the EWMA decay on the period then reported. §14.1 asks for prices adjusted for dividends and splits, and a documented proxy for short histories. The first check of §14.5 exists to catch unadjusted prices, and §11.7 shows proxies leaving risk unmeasured.

Deeper. White, "A Reality Check for Data Snooping," *Econometrica* 68(5) (2000), 1097–1126. Foundations of Econometrics, section 9.7, "Backtests are econometrics". Bailey, Borwein, López de Prado & Zhu, "Pseudo-Mathematics and Financial Charlatanism," *Notices of the AMS* 61(5) (2014), 458–471, on backtest overfitting.

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